BASE24 Release 6.0 Version 5
Implementation Guide

Version Record

11/2004, Version 5.0 First Printing

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Preface

Document Control

A description of the document revisions is shown below. Any changes of text from one version to another are indicated with the symbol "|" in the outside margin.

Version 5.0   (D. Barton, J. Dargy, S.Echilman, R. Schieuer, P. Spanke, J. Steffes, C. Rink, J. Urbanec,L. Watson, J. Wishart)-11/2004

Fifth Version

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Section 1

Introduction

The ***BASE24 Release 6.0 Version 5 Implementation Guide*** is the primary reference document for information needed to upgrade existing BASE24 release 6.0 version 4 systems to BASE24 release 6.0 version 5.

This section provides an overview of the planning requirements for BASE24 users migrating to release 6.0 version 5. It also provides guidance for using this manual.

Also included in this section is a listing of other documents that a BASE24 user should obtain for implementation planning, the general software requirements for release 6.0 version 5, and a glossary of terms used throughout the guide.

Implementation Planning

BASE24 users must develop their own project plans for migrating their BASE24 systems to release 6.0 version 5. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

ACI has provided BASE24 users with the ***BASE24 Release 6.0 Version 5 Implementation Guide*** to assist in preparing project plans. This guide includes a list of services available to the BASE24 user and information that existing BASE24 users will need to upgrade to release 6.0 version 5.

BASE24 Release 6.0 products include the following:

- BASE24-atm Release 6.0
- BASE24-pos Release 6.0
- BASE24-teller Release 6.0
- BASE24 Kernel/IEP Release 6.0
- BASE24-telebanking Release 6.0
- BASE24-billpay Release 6.0
- BASE24-customer service Release 6.0
- BASE24 Interchange Interfaces Release 6.0
- BASE24 Value-Added Products:
    - BASE24-Infobase
    - BASE24 Retail Products
    - BASE24-card
    - ACI Host Link
    - ACI Plastic Card Control System
    - ACI Claims Manager
    - ACI Proactive Risk Manager
    - ACI Settlement Manager
    - ACI Payments Manager

Who should use this guide

There are two audiences for this guide: those who plan and manage the migration process and those who implement the migration to release 6.0 version 5. In either case, the reader should be experienced with BASE24 software, HP NSK products, and the BASE24 user's current configuration. This guide is not directed at new BASE24 users who will be installing a BASE24 application for the first time. New users should refer to the formal BASE24 publications listed under the Publications heading in Section 3.

To migrate to release 6.0 version 5, users must be on release 6.0 version 4. Refer to the ***BASE24 Release 6.0 Implementation Guide (Version 4)*** to assist in planning the tasks necessary to get to release 6.0 version 4.

Using this guide, you can do the following

- Determine what ACI services are available to assist in the migration to release 6.0 version 5.
- Develop a project plan to migrate to release 6.0 version 5.
- Determine staffing and training needs.
- Gain a detailed understanding of the release 6.0 version 5 enhancements.

Software Requirements

HP NSK Software Requirements

ACI recommends that the BASE24 user be on a current HP NSK-supported operating system to obtain maximum performance and support. However, BASE24 release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 release 6.0 requires TMF for several files.

XPNET

ACI requires that the BASE24 user be on the XPNET 3.0 system.

Conversion Planning

BASE24 users must develop their own project plans for upgrading their systems to release 6.0 version 5. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

In general, all changes for release 6.0 version 5 can be obtained by requesting the latest fixes from ACI and then retrofitting all fixes to the user's release 6.0 version 4 modules.

Conversion Program Guidelines

Users of previous releases of BASE24 and previous versions of BASE24 Release 6.0 must run file conversion programs in order to migrate to the latest version of release 6.0. Information about specific conversion programs can be found in subsequent sections of this document under the applicable enhancements. Following are general conversion program guidelines.

The *UC04 subvolumes (e.g., AT60UC04, BA60UC04, PS60UC04) contain the conversion programs for users migrating from release 5.3 to the latest version of release 6.0. The conversion programs on these subvolumes are kept up-to-date with each new version of release 6.0.

The *UC05 subvolumes (e.g., AT60UC05, BA60UC05, PS60UC05) contain the conversion programs for users migrating from release 6.0 version 1 to release 6.0 version 2. No changes are made to these programs other than necessary fixes. Also residing on these subvolumes are the Transaction Security Services (TSS) conversion programs, which are used to move key data to the new TSS database. The TSS conversion programs are required only if the user is implementing TSS.

The *UC06 subvolumes (e.g., AT60UC06, BA60UC06, PS60UC06) contain the conversion program for users migrating from release 6.0 version 2 to release 6.0 version 3. No changes are made to these programs other than necessary fixes.

The *UC07 subvolumes (e.g., AT60UC07, BA60UC07, PS60UC07) contain the conversion program for users migrating from release 6.0 version 3 to release 6.0 version 4. No changes are made to these programs other than necessary fixes.

The *UC08 subvolumes (e.g., AT60UC08, BA60UC08, PS60UC08) contain the conversion program for users migrating from release 6.0 version 4 to release 6.0 version 5. No changes are made to these programs other than necessary fixes.

If a release 5.3 user wants to convert to the latest version of release 6.0, the conversion programs on *UC04 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 1 user wants to convert to release 6.0 version 2, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 2 user wants to convert to release 6.0 version 3, the conversion programs on *UC06 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 3 user wants to convert to release 6.0 version 4, the conversion programs on *UC07 should be run.

If a release 6.0 version 4 user wants to convert to release 6.0 version 5, the conversion programs on *UC08 should be run.

If a release 6.0 version 1 user wants to convert to release 6.0 version 5, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run. Following this, the user should run the conversion programs on *UC06 and on *UC07. Finally, the user should run the conversion programs on *UC08.

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Other Documents

In addition to this guide, the reader can obtain other BASE24 and NET24-XPNET publications that cover operational and product-specific information required to fully understand and implement the release 6.0 version 5 software. Such manuals include the formal publications mentioned under the Publications heading in Section 3, the event message documentation located on the BA60LOGM and OK60LOGM subvolumes, and the following documents:

BASE24 Classic 6.0 Installation Guide (Informal) - BA-AE000-60

A primary reference document for information about the installation of a BASE24 release 6.0 system from scratch. This document is included as a Microsoft Word document and is shipped on the BASE24 6.0 CD-ROM.

The BASE24 Classic 6.0 Installation guide is also provided on the software tape with the code. The document is located on the BA60UD04 subvolume as a file called BASE24. On this subvolume there is also a readme file called AAREAD04, which details how to the move the install document to your hard drive and open it with Microsoft Word.

BASE24 Release 6.0 Interchange Interface Implementation Guide (Informal) - SW-DS0000-60

Describes the function and purpose of ACI's Switch 6.0 interfaces. It explains how customers can implement Switch 6.0 interfaces in their existing networks as well as in a BASE24 6.0 network. It also explains how customers can implement their Fortify.1/Fortify.3 Interfaces in a 6.0 network. This document is intended to assist customers in configuring and controlling their Switch 6.0 Interchange Interfaces in all BASE24 release networks and their Fortify.1/Fortify.3 Interchange Interfaces in a BASE24 release 6.0 network. This document is included as a HP NSK TGAL document and is shipped with the release 6.0 Interchange Interface software. This document is named NSTLSW60 and is located on the SW60UD04 subvolume.

Note: This guide is to be used with BASE24 releases 4.0, 5.0, 5.1.2, 5.3, and 6.0.

User Bulletin for BASE24 Release 6.0 Version 5 (Informal)

A primary reference for information about all changes made to each BASE24 release 6.0 version 5 source module. This documentation is included as a set of HP NSK TGAL documents and is shipped with BASE24 release 6.0 version 5 software. This documentation consists of a number of files located on the BA60UB06 subvolume. All files ending in FH contain fix history sections for all changes made. All files ending in FP contain a list of all fixed Procedures for each change made to all modules, except those files that have been resequenced.

Release 6.0 Data File Impacts Document (Informal)

A primary reference document for information about the data file impacts and the methods that can be used to upgrade to release 6.0 version 5. This document is included as a HP NSK TGAL document and is shipped with the release 6.0 version 5 software. This document is named AAREAD06 and is located on the BA60UC06 subvolume.

Definition of Terminology

The following list has been defined to assist the reader in understanding the terms referred to within this document.

6.0 Interchange Interfaces

The BASE24 Interchange Interfaces developed in conjunction with release 6.0. These replace the Fortify.3 Interchange Interfaces and can be used with releases 4.0, 5.0, 5.1, 5.1.2, 5.3 and 6.0 of BASE24.

Accountholder

A person for whom a financial account has been established for some purpose. Accountholders may or may not use cards to initiate transactions on their accounts.

BASE24 User

An entity that has licensed BASE24 software products.

Cardholder

A person who has been issued a plastic card for the purpose of initiating transactions.

Consumer

A person who utilizes services offered by a BASE24 user who may or may not require a plastic card.

Conversion Program

A utility that reads the records from an input file, adds new fields and initializes appropriate fields and then writes the record to an output file. The input file (existing file) and output file (converted/initialized file) are different files. This method (versus the initialization program and online conversion) is required when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.). This method can also be used to initialize existing user fields in a record.

This method requires system down time to convert the file (also requires an operator to manually rename the file that has been converted), move in the new objects of modules that access the file (e.g., device handler, auth, etc.) and then restart the objects. This method impacts all BASE24 users who use the file being converted even if they are not taking advantage of the enhancement for which the modifications are being made.

Dynamic Key Management (DKM)

The automated distribution of encryption keys between BASE24 and an outside entity.

Extended Memory Table file (EMT)

A data file produced by the EMT Build utility program. Once the EMT has been built with the EMT Build utility, processes that use the EMT must be warmbooted (or started if not running).

Extended Memory Table Build utility (EMT Build)

The EMT Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom designed tables. Refer to the ***BASE24-atm Transaction Processing Manual***, ***BASE24-pos Transaction Processing Manual***, or the ***BASE24 ITS Kernel Configuration Manual*** for additional information about this utility.

Online (Inline) Conversion

A utility that reads and updates each record in a file to initialize user fields that have been redefined as new fields for use by an enhancement. The file is converted in place (i.e., a new file is not created). The record is read, updated and then rewritten to the same file. This method (versus the conversion program and online conversion) cannot be used when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.).

The online conversion code will be executed each time a record is accessed (versus the conversion and initialization programs that are run once). When a record is read, the process will determine if the value of a new field is valid. If the value is not valid, the field will be initialized to the default value. The database is gradually converted as records are read and updated.

This method requires an operator to stop a process, roll in the new object and start processes that use the files that are impacted. This method does not require any additional steps to be performed by an operator (e.g., running conversion/initialization programs, renaming files).

Operator

A person who carries out network operations functions, such as starting and stopping the system, entering ONCF commands, controlling logging, dealing with event messages, performing refreshes and extracts. The term *operator* also can refer to persons using BASE24 screens for such purposes as file maintenance, perusal, etc.

Section 2

Functionality

This section contains an overview of the BASE24 release 6.0 verison 5 enhancements.

Functionality has been organized by product, and within each product, according to whether the enhancement is a general enhancement or an optional enhancement.

- General enhancements are those enhancements that are inherent in the release 6.0 version 5 offerings. These enhancements affect all BASE24 users. BASE24 users must review the features that are part of these enhancements to ensure an understanding of the new processing and ensure proper configuration within the BASE24 environment.
- Optional enhancements are those enhancements that can be enabled or disabled in release 6.0 version 5. BASE24 users can implement these enhancements based on their own BASE24 processing and marketing requirements.

Enhancements are described in this section by product as follows:

- BASE24 Base (applicable to all products)
- BASE24-atm
- BASE24-pos
- BASE24-teller
- BASE24 Kernel/IEP (applicable to all ITS based products)
- BASE24-telebanking
- BASE24-billpay
- BASE24 Interchange Interfaces

BASE24 users should acquaint themselves with the BASE24 Base section (Section 4) as well as the applicable product-specific sections in order to obtain an understanding of the overall release 6.0 version 5 offering.

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BASE24 Base Enhancements

The BASE24 Base product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Base product, refer to Section 4.

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General Enhancements

The following enhancements are common to all products or provide the base for product-specific enhancements. As these enhancements are inherent in release 6.0 version 5, BASE24 users must review them to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- Changes for all BASE24 release 6.0 version 5 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes were done under one SCN per file as well.
- Changes were made to a Base DDL file (BA60DDL.DDLGTSS) and the security utilities (BA60SRC.SECUTILS) to provide compatibility with BASE24-es version 04.4 TSS module, available in early January 2005.
- In BASE24-atm the AT50 and AT-FLG1 tokens have been used to hold disparate data and flags. Since the last remaining byte in these generic ATM tokens has been used, a new token has been added to allow for future ATM changes that require additional data/flags.
- Updates have been made to the currency code and country code table in BAUTILS to include the latest ISO codes for Ecuador, East Timor, Madagascar and Turkey.
- A new token has been added for integration with the new IFX ATM Manager.
- Modifications have been made to the Transaction Based Pricing reports to support mobile top-up transactions. All customers, regardless of whether they process mobile top-up transactions, will need to apply the TBP report changes.

Reopen TSS After Restart

BASE24 applications have been enhanced to re-establish communications with a TSS process once a TSS process fails and restarts. Prior to this enhancement, operator intervention was required to warmboot the applications whenever a TSS process failed. This enhancement enables the application's security utilities (SECUTILS) to automatically re-establish communications with TSS. Based upon a failure count and the "notify count" thresholds set in the LCONF, the application will periodically attempt to connect with TSS until communication is re-established.

Optional Enhancements

The following BASE24 Base release 6.0 version 5 enhancements are optional. BASE24 users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Absent ICC Application PAN Sequence Number

BASE24-atm and BASE24-pos EMV transaction processing was modified to use spaces rather than zeros as the default value for the Application PAN Sequence Number (APSN) in the EMV Status Token. This allows BASE24-atm and BASE24-pos EMV device extensions and BASE24 EMV interface extensions to differentiate between an APSN of “00” and an absent APSN.

American Express Card Security Code

This enhancement added support for validation of the American Express Card Security Code (CSC) values in BASE24. American Express cards support three card verification values in their fight against fraud. One CSC is a value embedded on the magnetic stripe (5-digit value), another is imprinted on the front of an American Express card (4-digit value) and the third is imprinted on the signature panel on the back of an American Express card (3-digit value). On any given magnetic stripe-read transaction, it is possible that the transaction will include the CSC from the magnetic stripe as well as one of the other CSC values imprinted on the card (could be either one).

With this enhancement, BASE24 has the ability to contain all three values on the BASE24 database, verify the 5-digit value from the magnetic stripe and perform a second card verification if an authorization message includes both the CSC value from the magnetic stripe and one of the two available CSC values printed on the card.

Prerequisite: Transaction Security Services (TSS) version 04.4 is required.

Automated Hot Card Updates

From Host Maintenance Host ISO Interface has been enhanced to allow the host to initiate a Card Status Update message to as many as three interchanges using the Card Status (0300) Update Request message. When a Card Status Update Request message is received from the host, the From Host Maintenance Host ISO Interface will build a request message and send it to the corresponding interchange interface, so that the interchange databases can be updated. It will also send the Card Status Update Request message to the BASE24 From Host Maintenance process, which will update the CAF or NEG files on BASE24. Based upon the values in various data elements in the external Card Status Update Request message, the From Host

Maintenance Host ISO Interface will send a message to the interchange interface process(es) only, to the BASE24 From Host Maintenance process only, or to both the interchange interface process(es) and the BASE24 From Host Maintenance process. The host will have the ability to send a hot card update request to as many as three interchange interface processes in a single message.

EMV 2000 Session Key Derivation

This enhancement supports new cryptographic enhancements defined in the EMV 2000 (version 4.0) specifications for EMV cryptographic processing. The enhancement allows issuers to utilize session key derivation, which offers additional flexibility and security in managing session keys.

This enhancement is invoked based on the security scheme supported by the card. For cards compliant with VIS 1.4.0 (Visa) and M/Chip 4.0 (MasterCard), the Cryptogram Version Number within the Issuer Application Data received in an EMV request message determines the type of key derivation used. The enhancement is backward compatible with current EMV 1996 key derivation.

Enhanced Second Card Data Processing

BASE24-atm and BASE24-pos currently support (with BASE24 release 6.0 version 4) second card processing, which enables an institution to issue two cards with the same primary account number (PAN) and member number. The two cards are differentiated by comparing the expiration date on the card with two expiration dates held in the CAF – one for the primary (original) card and another for the second (reissued) card.

By setting a new LCONF parameter appropriately, this enhancement enables an institution to automatically remove the current primary card from the system and make the second card the new primary card. When the first transaction is received from the second card, this feature, if enabled, will automatically replace the primary card's data in the CAF with the second card's data and then clear the second card fields.

This enhancement also maintains another ATC value in the CAF for the second card, for those institutions that utilize EMV. When a transaction is received from the second card and ATC checking is enabled, the ATC from the card's microchip will be checked against a new field in the CAF. Additionally, if the system is configured to automatically make the second card the new primary card (as described above), the second card's ATC is included in the data that is moved to the primary card's data.

Forward Declined Advices to Host

BASE24-atm Authorization and BASE24-pos Router/Authorization have been enhanced to send all advices, if configured, to the host system using a new field on the Issuer Processing Control File. This would be done regardless of the transaction being approved, declined, or referred (POS only). This enhancement enables host systems to better track consumer activity, thereby increasing customer service and potentially lowering fraud and credit losses.

Inhibit UI Display of POFST/PVV

There are potential cryptographic attacks that can be attempted when provided with card numbers and their associated PIN offset. The current CAF screen 1 displays both these data elements when enabled in the users' security profile. To further secure the offsets from users, an enhancement has been made to mask the PIN offset/PVV using asterisks. If the user is granted add/update access, asterisks will obscure the offsets after the record has been added/updated. This enhancement can be enabled or disabled via an LCONF parameter.

PBF Refresh Early Notification

A new LCONF parameter was introduced to provide notification to BASE24 authorization processes that a new PBF is available and to reduce the number of records impacted during the PBF Refresh process. If the new parameter is set to "yes", the PBF Refresh process will perform the following:

- Update the IDF with a new refresh indicator.
- Notify the authorization process that a new PBF is available.
- After the first authorization process acknowledges the notification, it will set a new global to the current time.
- Read the transaction log file until the new global time value is reached.

When the new parameter is set to "no", the PBF Refresh process will perform as it did prior to this performance improvement. In other words, Refresh will read the corresponding transaction log file until the end of file is reached.

Round Robin Access to TSS

An enhancement has been made to "round robin" to multiple TSS processes in a TSS environment when multiple security devices are in a logical network. SECUTILS has been enhanced to support up to 20 security devices or TSS processes. Currently, applications are limited to two primary/alternate devices or TSS processes through the SECURE-DEV-1 and SECURE-DEV-2 LCONF assigns. The new capability is optional and can be enabled via LCONF parameter.

This enhancement greatly simplifies configuration of larger networks using TSS. Rather than defining a unique pair of SECURE-DEV-1 and SECURE-DEV-2 assigns for each process defined in the network, this enhancement allows one set of assigns to define a common pool of TSS processes for the entire logical network.

The enhancement will track which process and/or device to use for the next transaction, the number of HSMs and/or TSS processes open, and the time to delay before attempting to reinstate a device and/or process. If a particular device or process is marked down, an attempt to restart will be made after the specified reinstatement time.

This enhancement is a compliment to the Reopen TSS After Restart enhancement described under Base General Enhancements.

Visa iCVV Support

BASE24-atm and BASE24-pos have been enhanced to allow issuers to validate the CVD (card verification data) during EMV authorization processing, when the CVD on the track-2 equivalent data is different from the CVD on the magnetic stripe. To provide complete flexibility, BASE24-atm and BASE24-pos have been enhanced to allow issuers optionally to validate the CVD during EMV authorization processing, even when the ARQC has not been verified successfully or whenever ARQC verification has been performed.

Four new fields were added to the EMV segment of the Card Prefix File (CPF):

- An EMV card verification check type field to specify the circumstances (if any) under which card verification should be performed on EMV transactions.
- An EMV card verification method field to specify the algorithm and data used when card verification is performed on EMV transactions.
- An EMV card verification effective date field to specify the earliest card expiration date for which card verification should be performed on EMV transactions.
- An EMV card verification data field to contain any additional data required when card verification is performed on EMV transactions.

BASE24-atm Enhancements

The BASE24-atm product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-atm product, refer to Section 5.

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General Enhancements

The following BASE24-atm enhancements are inherent in the release 6.0 version 5 offerings. All BASE24-atm users, to ensure an understanding of the new processing and proper configuration within the BASE24 environment, must review these enhancements.

General Changes

- BASE24-atm Authorization was revised to not perform CVD validation when an interchange has already performed this validation. If the CVD was successfully checked by an interchange, the card verification will be bypassed. If the CVD was checked by an interchange and failed, then ATM Authorization will check the BAD CV ACTION on the Card Prefix File (CPF) to determine the appropriate authorization action.

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Optional Enhancements

The following BASE24-atm release 6.0 verison 5 enhancements are optional. BASE24-atm users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Absent ICC Application PAN Sequence Number

Refer to **Absent ICC Application PAN Sequence Number** under Base Optional Enhancements for a description of this enhancement.

AKDS Capability Check and Load All Keys for Diebold and NCR

The Diebold 10XX/478X and NCR 5XXX device handlers were enhanced to support the process of sending the ATM an inquiry as to its Automated Key Distribution System (AKDS) capabilities. When the device handler receives an AKDS DCT command, it will close the ATM and send it a configuration request message. When the device handler receives the ATM's configuration response message back, it will determine if the ATM is AKDS capable. If the ATM is determined to be AKDS capable, the DCT command will be allowed to proceed. If not, the DCT command will not be performed and an EMS event will be displayed and the device handler will re-open the ATM.

In addition, a new DCT command, LOADALLKEYS, has been developed. When the device handler receives this command, it will check the AKDS capability of the ATM as described above. If the ATM supports AKDS, the device handler will sequentially download RSA and DES keys in a specified order and then open the ATM. If the ATM does not support AKDS, the device handler will sequentially download only DES keys and then open the ATM.

Prerequisite: Transaction Security Services (TSS), Automated Key Distribution System (AKDS), and either the Diebold 10XX/478X device handler and/or the NCR 5XXX device handler are required.

American Express Card Security Code

This enhancement added support for validation of the American Express Card Security Code (CSC) value in BASE24-atm. American Express cards support a new 5-digit CSC value imbedded in the magnetic stripe.

With this enhancement, BASE24-atm has the ability to verify the new 5-digit CSC value in an authorization request.

Prerequisite: Transaction Security Services (TSS) version 04.4 is required.

BASE24-es ATM IFX ATM Manager and BASE24-atm Integration

BASE24-atm has been enhanced to integrate the new BASE24-es IFX ATM Manager into a standard BASE24-atm system. The IFX ATM Manager is a new device handler that uses the IFX Business Message Specification as its ATM-to-host message format. The BASE24-es IFX ATM Manager interfaces with the BASE24atm Authorization process for authorization, routing and processing. The IFX ATM Manager supports the following standard ATM transactions:

- Withdrawal from checking account
- Withdrawal from savings account
- Fast cash
- Fast cash from checking account
- Fast cash from savings account
- Fast cash from credit account
- Cash advance from credit account
- Deposit to checking account
- Deposit to savings account
- Transfer from checking to savings
- Transfer from savings to checking
- Transfer from credit account to checking
- Transfer from checking to checking
- Transfer from savings to savings
- Payment Enclosed
- Payment from checking to credit account
- Payment from savings to credit account
- PIN change
- Balance inquiry from checking
- Balance inquiry from savings
- Inquiry into credit account availability
- Message enclosed from cardholder to financial institution
- Statement print request
- Log only #1
- Log only #2
- Log only #3
- Log only #4

Note: The BASE24-es IFX ATM Manager user interface is not integrated with BASE24-atm.

Prerequisite: The Diebold 10XX/478X device handler is required.

Diebold Cash Source Plus (CSP) 200 Triple DES

The BASE24-atm Diebold CSP device handler has been modified to support triple DES encryption. Triple DES encryption uses two or three keys for encryption, whereby single DES only used one key. Because of the additional keys used in the algorithm, triple DES is inherently more secure.

For more information on Transaction Security Services (TSS) and triple DES, please refer to the *BASE24 Release 6.0 Version 1 and 2 Implementation Guide.*

Prerequisite: The Diebold CSP 200 device handler, TSS and the related BASE24 enhancements from earlier versions must be installed prior to the configuration of the Diebold CSP 200 device handler for triple DES.

Diebold EMV

The BASE24-atm Diebold 10XX/478X device handler has been enhanced to support the Europay, MasterCard, and Visa (EMV) scheme for chip cards. A new EMV extension module for the Diebold has been developed to handle EMV processing. The Diebold EMV extension module is responsible for scanning request messages for EMV data, checking the validity if it does exist, handling any EMV data received in responses and formatting the appropriate message returned to the ATM. It will also handle the printing of the application identifier and the application PAN sequence number on a receipt.

In addition, the Diebold EMV extension module processes the EMV configuration data load messages, EMV data in solicited and unsolicited status messages and any special reversal processing needed because of a failure from the device's EMV Smart Card Application.

For more information on EMV processing, please refer to "EMV Support" in the BASE24 Base Optional Enhancements of the *BASE24 Release 6.0 Version 1 and 2 Implementation Guide.*

Prerequisite: The Diebold 10XX/478X device handler is required.

Diebold Text to Speech (ADA) and Agilis 91x (Opteva) Features

The BASE24-atm Diebold 10XX/478X device handler has been enhanced to support Diebold's latest technology, as it relates to supporting the new American Disabilities Act (ADA) guidelines. This enhancement includes adding support for the new Text-to-Speech state (AU) to the Diebold Entry Program and modifying the Diebold Entry Program to allow additional types of voice prompting files to be embedded in consumer display screens. (This includes text (.TXT) files that contain a text string to be spoken or an actual text string.) It should be noted that both of these enhancements require ATM software of TCS Plus 1.3 or higher or Agilis 91x.

The BASE24-atm Diebold 10XX/478X device handler has also been enhanced to support new features offered on Diebold's Opteva line of ATMs. Support has been added for automatic envelope dispensing, multi-colored LED Lights, and two-colored text printing for thermal receipt and thermal statement printers. The enhancement also includes the recognition of the associated solicited and unsolicited status messages from the envelope dispenser, the LED lights, the receipt printer and the statement printer.

Prerequisite: The Diebold 10XX/478X device handler is required.

EMV 2000 Session Key Derivation

Refer to **EMV 2000 Session Key Derivation** under Base Optional Enhancements for a description of this enhancement.

Enhanced ATM Status Reporting

BASE24-atm has been enhanced to provide additional information to ATM monitoring systems to aid in problem resolution. This new information will allow customers to better utilize their ATM support by being able to better interrogate problem situations at ATMs. In order to accomplish this, the Diebold 10XX/478X and NCR 5XXX device handlers have been modified to provide to EMS the native data the device handler received in a status message. The display of the device's native message data is controlled by a new LCONF parameter.

Prerequisite: The BASE24-atm NCR 5XXX device handler and/or the Diebold 10XX/478X device handler are required.

Forward Declined Advices to Host

Refer to **Forward Declined Advices to Host** under Base Optional Enhancements for a description of this enhancement.

Mobile Top-Up Integration

With this version, BASE24-atm has been enhanced to support mobile top-up functionality. This was accomplished in part by integrating the ACI EMEA ATM Mobile Top-Up product into standard product. In combination with a phone time provider, BASE24-atm systems are now capable of providing an avenue for mobile consumers to purchase phone time.

In the US, ACI has completed development of an interface to a third party phone time provider, Boston Communications Group, Inc. (*bcgi*). This new interface and its associated interface specific documentation are now available.

Typically, replenish (or top-up) transactions are initiated by the consumer at the ATM. The consumer inserts either a debit or credit card and selects a "phone top-up" transaction. The consumer specifies the amount of phone time to be replenished (either a dollar amount or a unit of time) and uses the ATM keypad to enter the Mobile Directory Number (MDN) or phone number to be replenished.

After the consumer enters all necessary data, the transaction is sent from the ATM to the ATM device handler. The device handler formats the transaction and sends the transaction to a new BASE24 module, the Transaction Context Manager (TCM). Mobile top-up extensions are available for the NCR 5XXX and Diebold 10XX/478X device handlers.

The TCM routes the transaction to the applicable mobile phone carrier interface. The mobile phone carrier (or third party provider) validates whether the mobile number supplied in the request can accept a replenishment, calculates any sales tax associated, determines if a voucher must be dispensed, and sends a response back to the TCM.

The TCM sends the response to the ATM device handler, which sends the information back to the ATM. The consumer then chooses to accept or decline the transaction. If the consumer declines the transaction, the TCM logs a record of the transaction to the Transaction Log File (TLF). If the consumer accepts the transaction, TCM routes the funds portion of the transaction to BASE24-atm for authorization. Depending upon the identity of the funds issuer, BASE24-atm Auth may or may not route the funds portion of the transaction to an interchange. Once the funds portion of the transaction is approved or denied, BASE24-atm Auth logs a record to the TLF and sends a response to the TCM.

If the funds portion of the transaction was approved, the TCM will forward the transaction to the applicable mobile phone carrier, if applicable, to update their files. Upon receiving a response from the mobile phone carrier, a record is logged to the Interchange Log File (ILF) and the response is sent to the TCM.

The TCM completes the transaction by logging the transaction to the TLF and forwarding the response to the device handler.

BASE24 Mobile Top-Up Integration supports two models for obtaining information about different carriers: definitions-based and message-based. The definitions-based model is used when carrier information (e.g., name, voucher values, etc.) is kept on the BASE24 database. The message-based model is when carrier information is obtained from a third party (e.g., bcgi) via request/response message processing. A Transaction Context Manager (TCM) process and device handler process can be configured to use only one model, so separate TCM and DH processes would be necessary to support both models.

Prerequisites: The NCR 5XXX device handler and/or the Diebold 10XX/478X device handler, the respective device handler mobile top-up extension, and one mobile phone carrier interface are required for mobile top-up functionality. Non-Currency Dispense is also required. Additional requirements may exist to support individual mobile phone carriers.

NCR NDC+ CAM II (EMV) Upgrade and EMV Dip Card Reader

The NCR 5XXX device handler and its EMV extension have been upgraded to support the NCR's latest CAM II requirements. The NCR 5XXX device handler and its EMV extension have been upgraded to support the following NCR CAM II features: a language indicator, a default application label, multiple transaction currency codes, transaction category codes, hexadecimal lowest and highest version numbering, deleting entries from the terminal's acceptable AIDS table and for clearing the application name screen once the name has been displayed to the cardholder.

The NCR 5XXX EMV device handler has also been enhanced to support NCR's new EMV dip card reader. The EMV dip card reader may not always read the card stripe, and therefore may not send the mag-stripe track 2 information into BASE24. This enhancement checks for the EMV track 2 equivalent data (and PIN) when the mag-stripe Track 2 information is not sent. If the EMV track 2 equivalent data (and PIN) is sent, the transaction is processed normally. If the EMV track 2 equivalent data (and PIN) is not sent and the mag-stripe track 2 buffer has not been sent, the transaction is treated as an intermediate leg of a multi-legged transaction.

For more information on EMV processing, please refer to "EMV Support" in the BASE24 Base Optional Enhancements of the *BASE24 Release 6.0 Version 1 and 2 Implementation Guide.*

NOTE: The NCR 5XXX device handler has also been enhanced to support a new EMV extension for devices that support Wincor's shared NDC+ EMV specifications. Please refer to the Shared (Wincor) NDC+ EMV Support enhancement in this section for more details.

Prerequisite: The NCR 5XXX device handler and the NCR 5XXX EMV extension are required.

Offline PIN Change Notification

BASE24-atm processing supports the changing of the offline PIN on an EMV card with BASE24 release 6.0 version 4. However, a card management system may require knowledge of the new PIN whenever a PIN change transaction is performed successfully at an ATM. The card management system may have to include the new PIN on the card if or when the card is re-issued. The PIN block will be translated using a new key and stored on the TLF for extract by the card management system using the new key.

PBF Refresh Early Notification

Refer to **PBF Refresh Early Notification** under Base Optional Enhancements for a description of this enhancement.

Shared (Wincor) NDC+ EMV Support

BASE24-atm has been enhanced to support Wincor's version of NDC+ EMV. Because Wincor has agreed to share their EMV specification with other ATM manufacturers, in the future this enhancement will be referred to as "Shared EMV".

A new Shared EMV extension module for the NCR 5XXX device handler has been developed to handle all of the Shared NDC+ EMV processing. The Shared NDC+ EMV extension module is responsible for scanning request messages for EMV data, checking the validity if it does exist, handling any EMV data received in responses and formatting the appropriate message sent back to the ATM. It also handles the printing of the application identifier and the application PAN sequence number on a receipt.

In addition, the Shared NDC+ EMV extension module processes the EMV configuration data load messages, EMV data in solicited and unsolicited status messages and any special reversal processing needed because of a failure from the device's EMV Smart Card Application.

For more information on EMV processing, please refer to "EMV Support" in the BASE24 Base Optional Enhancements of the *BASE24 Release 6.0 Version 1 and 2 Implementation Guide.*

Prerequisite: The NCR 5XXX device handler is required.

Triton EMV

BASE24-atm has been enhanced to support the Triton implementation of EMV.

A new Triton EMV extension module for the Triton device handler has been developed to handle all EMV processing. The Triton EMV extension module is responsible for scanning request messages for EMV data, checking the validity if it does exist, handling any EMV data received in responses and formatting the appropriate message sent back to the ATM.

In addition, the Triton EMV extension module processes the EMV data in solicited and unsolicited status messages and any special reversal processing needed because of a failure from the device's EMV Smart Card Application.

For more information on EMV processing, please refer to "EMV Support" in the BASE24 Base Optional Enhancements of the *BASE24 Release 6.0 Version 1 and 2 Implementation Guide.*

Prerequisite: The Triton device handler is required.

Triton TCP/IP

The BASE24-atm Triton device handler has been enhanced to support TCP/IP. Before this enhancement, the only protocol that could be configured was Visa II Dial-Up. Operating Triton ATMs in a TCP/IP mode can make customer ATM networks more efficient and more reliable.

Customers wishing to use Triton ATMs in a TCP/IP mode must ensure that their ATMs have been upgraded to the proper software and hardware levels that are TCP/IP enabled. For additional TCP/IP configuration assistance, refer to the XPNET TCP/IP Protocol Implementation Guide.

Prerequisite: The BASE24-atm Triton device handler is required.

Triton Triple DES MACing

The Triton device handler has also been enhanced to support the triple DES method of computing the Message Authentication Code (MAC). MACing is used to verify whether or not critical pieces of information in a transaction have been tampered with. Triple DES MACing offers a more secure method of computing the Message Authentication Code than the current single DES method.

For more information pertaining to triple DES, refer to the *BASE24 Release 6.0 Version 1 and 2 Implementation Guide*.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Triton Triple DES MACing enhancement.

Visa iCVV Support

Refer to **Visa iCVV Support** under Base Optional Enhancements for a description of this enhancement.

BASE24-pos Enhancements

The BASE24-pos product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-pos product, refer to Section 6.

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General Enhancements

The following BASE24-pos enhancements are inherent in the release 6.0 version 5 offerings. These enhancements must be reviewed by all BASE24-pos users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- Hooks were added to several modules in POS to support future enhancements related to mobile top-up support.

- BASE24-pos Router/Authorization was revised to not perform CVD/CVD2 validation when an interchange has already performed this validation. If the CVD/CVD2 were successfully checked by an interchange, the card verification will be bypassed. If the CVD/CVD2 were checked by an interchange and failed, then POS Router/Authorization will check the BAD CV ACTION on the Card Prefix File (CPF) to determine the appropriate authorization action.

Router/Auth/DH Data Stack Relief

This enhancement is in response to an increasing number of process abends due to excessive memory allocation, which occurs more frequently as custom modifications increase in POS device handlers. The enhancement targets various data elements in Router/Authorization for removal from the data stack and placed in a flat, extended memory segment, which is always available to a process. This enhancement also impacts the entire range of BASE24-pos device handlers.

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Optional Enhancements

The following BASE24-pos release 6.0 verison 5 enhancements are optional. BASE24-pos users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Absent ICC Application PAN Sequence Number

Refer to **Absent ICC Application PAN Sequence Number** under Base Optional Enhancements for a description of this enhancement.

American Express Card Security Code

This enhancement added support for validation of the American Express Card Security Code (CSC) values in BASE24-pos. American Express cards support three card verification values in their fight against fraud. One CSC is a value embedded on the magnetic stripe (5-digit value), another is imprinted on the front of an American Express card (4-digit value) and the third is imprinted on the signature panel on the back of an American Express card (3-digit value). On any given magnetic stripe-read transaction, it is possible that the transaction will include the CSC from the magnetic stripe as well as one of the other CSC values imprinted on the card (could be either one).

With this enhancement, BASE24-pos Router/Authorization is able to verify the new 5-digit CSC value embedded in the magnetic stripe, and perform a second card verification if an authorization message includes both the CSC value from the magnetic stripe and one of the two available CSC values printed on the card. When an authorization message contains a 5-digit electronically entered CSC and a manually entered 3 or 4-digit CSC, Router/Authorization will first check the results code of the electronic CSC validation. If the results code indicates the electronic CSC verification was not successful, BASE24-pos Router/Authorization will take the appropriate action as indicated by the BAD CV ACTION flag from the CPF. If the results code indicate that CSC verification was successful, and the authorization included a CSC manually entered value, Router/Authorization will check the results code for the manually entered CSC. If the CSC results code indicates the card verification was not successful, Router/Authorization will take the appropriate action as indicated by the BAD CV ACTION flag from the CPF.

Prerequisite: Transaction Security Services (TSS) version 04.4 is required.

EMV 2000 Session Key Derivation

Refer to **EMV 2000 Session Key Derivation** under Base Optional Enhancements for a description of this enhancement.

Forward Declined Advices to Host

Refer to **Forward Declined Advices to Host** under Base Optional Enhancements for a description of this enhancement.

Hypercom Triple DES Master/Session and DUKPT

The Hypercom device handler (HPDH) for BASE24-pos has been enhanced to support triple DES PIN encryption in coordination with the Transaction Security Services (TSS) module. Support for double-length PIN encryption keys of 32 hex characters was added, enabling HPDH to generate either single or double-length PIN encryption keys.

This enhancement includes both Master/Session and Derived Unique Key Per Transaction (DUKPT) Key Management protocols.

Prerequisite: Single DES Master/Session PIN encryption and Single DES DUKPT do not require the use of Transaction Security Services. However, if triple DES Master/Session PIN encryption or triple DES DUKPT is required, the Transaction Security Services enhancement must be installed. For more information about Transaction Security Services refer to the *BASE24 Release 6.0 Version 1 and 2 Implementation Guide.*

PBF Refresh Early Notification

Refer to **PBF Refresh Early Notification** under Base Optional Enhancements for a description of this enhancement.

Stored Value Dormancy and Escheatment

Stored Value processing within BASE24-pos has been enhanced to support the processing and assessing of dormancy fees. The dormancy process has been enhanced to produce escheatment report files that can be used to report reversion of funds to the appropriate state agency.

A POS new file, called the Stored Value Dormancy File (SVDF), has been created to contain the various dormancy configuration parameters. The dormancy process reads the SVDF, assesses applicable fees to the corresponding PBF record, logs the fees to the Stored Value History File (SVHF), and creates a summary report for the fees applied. Access to the dormancy process is via a TACL prompt. All AFT maintenance changes are recorded in the Online Maintenance File (OMF).

The Stored Value segment of the CPF was modified to include an escheatment flag. The escheatment flag is set based on the FIID to indicate if escheatment is permitted.

Stored Value Offline Card

Stored Value processing within BASE24-pos has also been enhanced to provide offline transaction capabilities. With this enhancement, card activation, replenishment, and redemption transactions can be processed while in offline mode, and then forwarded to BASE24 via the SPDH as a 0200 request. This processing was designed to improve the customer experience.

Upon receipt of the offline card transaction, BASE24 will process the transaction as if it were online. However, a new value of "S" will be included in the message subtype field, and subsequently stored in the PS51 Token. A response message is sent by BASE24 to the terminal/store controller to bring both systems back in sync.

Visa iCVV Support

Refer to **Visa iCVV Support** under Base Optional Enhancements for a description of this enhancement.

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BASE24-teller Enhancements

The BASE24-teller product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-teller product, refer to Section 7.

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General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

The following BASE24-teller release 6.0 verison 5 enhancement is optional. BASE24-teller users should review this change to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of its changes.

PBF Refresh Early Notification

Refer to **PBF Refresh Early Notification** under Base Optional Enhancements for a description of this enhancement.

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BASE24 Kernel/IEP Enhancements

The BASE24 ITS Kernel/IEP product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 ITS Kernel/IEP product, refer to Section 8.

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General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

There are no optional enhancements inherent to this product.

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BASE24-telebanking Enhancements

The BASE24-telebanking product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-telebanking product, refer to Section 9.

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General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

The following BASE24-telebanking release 6.0 verison 5 enhancement is optional. BASE24-telebanking users should review this change to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of its changes.

PBF Refresh Early Notification

Refer to **PBF Refresh Early Notification** under Base Optional Enhancements for a description of this enhancement.

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BASE24-billpay Enhancements

The BASE24-billpay product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-billpay product, refer to Section 10.

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General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

There are no optional enhancements inherent to this product.

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BASE24 Interchange Interface Enhancements

All BASE24 Interchange Interface product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Interchange Interfaces, refer to Section 12.

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General Enhancements

The following enhancements are common to the applicable interface products or provide the base for product-specific enhancements. As these enhancements are inherent in release 6.0 version 5, BASE24 users must review them to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- The BIC ISO Interface and ISO Host Interface were enhanced to include MAKE flags for the Boston Communications Group Interface (US Based interface used for mobile top-up transactions).
- The BIC ISO Interface and ISO Host Interface were enhanced to include support of a new Reversal Reason code of "40", where 40 is equal to “Split Routing Enabled; Secondary Service Not Approved”.
- Software fix to HISWUTLS procedure hiswilf^add^tkn to restore the original extended memory segment before returning.

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Optional Enhancements

The following BASE24 Interchange Interface release 6.0 verison 5 enhancements are optional. BASE24 Interchange Interface users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Absent ICC Application PAN Sequence Number

Refer to **Absent ICC Application PAN Sequence Number** under Base Optional Enhancements for a description of this enhancement.

American Express Card Security Code

This enhancement added support for validation of the American Express Card Security Code (CSC) values in the BASE24 AEGN (American Express Global Network) Interchange Interface. American Express cards support three card verification values in their fight against fraud. One CSC is a value embedded on the magnetic stripe (5-digit value), another is imprinted on the front of an American Express card (4-digit value) and the third is imprinted on the signature panel on the back of an American Express card (3-digit value). On any given magnetic stripe-read transaction, it is possible that the transaction will include the CSC from the magnetic stripe as well as one of the other CSC values imprinted on the card (could be either one).

With this enhancement, the BASE24 AEGN I/F will correctly pass all CSC values correctly between the AEGN Network and BASE24-atm/BASE24-pos Authorization modules for verification.

Prerequisite: Transaction Security Services (TSS) version 04.4 is required.

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Section 3

Support Services

This section summarizes the various support services available from ACI. These support services are described as follows:

- Product consulting and project management
- Publications
- Education
- HELP24
- ASTech services

Other services, besides the traditional Installation service, such as go-live support, onsite conversion assistance, and software methodology and utility training are available or are being planned; please contact your ACI Customer Manager for more details of the services ACI can offer.

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Product Consulting and Project Management

No formal service programs have been created by ACI for upgrading customers to BASE24 release 6.0 version 5. Customers need to review this entire document and any referenced manuals, to determine what they will need to do to upgrade their BASE24 system.

After reviewing, customers should contact their ACI Customer Manager to discuss any services, such as project management, custom code upgrades, onsite staff, etc., required for upgrading, after reviewing all materials.

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Project Planning Guide

The following Project Planning Guide is designed to assist in the successful upgrade of a BASE24 system to release 6.0 version 5. Information contained in this guide provides insight into required planning and control of the entire project.

Following this project guide helps ensure the realization of the goals and business objectives of the BASE24 user.

1. IDENTIFY PROJECT TEAM

   One of the initial tasks for the BASE24 upgrade is for the user to assign a project team who will be responsible for the success of the project. A project manager and team members will need to be assigned and specific responsibilities established.

2. DETERMINE HARDWARE REQUIREMENTS

   An ACI sales representative can assist in determining whether additional hardware will be required for the BASE24 upgrade. It is the BASE24 user's responsibility to order the hardware and establish the delivery date with HP NSK. HP NSK will perform diagnostic testing on the hardware when it is installed at the site. Should hardware encryption be required, it will be necessary to determine the hardware requirements. Additional hardware should be ordered for installation before testing can occur.

   If applicable, the BASE24 user should order any new device firmware/software required to implement the enhancements they have chosen to support. The delivery date for the device firmware/software should be agreed upon with the vendor, as this time frame will affect the entire project.

3. IDENTIFY COMMUNICATION REQUIREMENTS

   If the BASE24 user has identified additional protocol requirements, the communications lines and modems should be ordered. The customer must coordinate this effort with the local telephone company to ensure that the lines are configured in the least costly manner. The lines and modems must be installed and tested prior to the scheduled software installation date.

4. UPDATE HP NSK SOFTWARE REQUIREMENTS

   ACI recommends that the BASE24 user be on a current HP NSK supported operating system to obtain maximum performance and support. BASE24 release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 release 6.0 requires TMF for several files.

5. UPGRADE TO XPNET 3.0

   Customers should note that XPNET 3.0 is the minimum requirement for BASE24 release 6.0.

6. PROJECT SCOPE MEETING

   ACI does not believe this step is necessary for customers upgrading to BASE24 release 6.0 version 5.

7. MIGRATION PLANNING

   Users will need to use this document, and any references within it, for detailed analysis when creating their migration plan.

8. DEVELOP PROJECT SCOPE DOCUMENT

   ACI does not believe this step is necessary for customers upgrading to BASE24 release 6.0 version 5.

9. DEFINE REQUIREMENTS DEFINITION

   The customer will be required to review existing CSMs to determine which ones will require retrofitting to the new version. With ACI's Fix Release methodology, BASE24 users should evaluate their CSMs to determine if they are handled as standard product in release 6.0 version 5 or if they are specific to the customer's environment and will need to be upgraded to the new version.

10. DEVELOP PROJECT PLAN

    The customer's project manager is responsible for developing an internal project plan. A review of the ACI project guidelines will help this person identify job responsibilities and establish time frames. A tracking mechanism should be included in the plan to ensure that the project is proceeding on schedule.

11. IDENTIFY CUSTOMER SPECIFIC MODIFICATION DEVELOPMENT

    If the customer elects to have ACI develop modifications identified as required for the new version, a request will be submitted to ACI Software Development for a cost estimate. The functional descriptions and estimates will be sent to the customer for review and approval. Resources will be assigned and the final delivery dates established upon receipt of the signed contract.

    Should the customer decide to develop any modifications in-house, resources will need to be assigned and scheduled. Once delivery dates are established, impacts to system testing can be determined.

12. DEFINE CERTIFICATION REQUIREMENTS

Customers currently on Fortify.3 will need to verify with each of the interchanges they connect to, to determine if they need to re-certify with the Interchanges with release 6.0 version 5 code.

If re-certification is required, preliminary test times should be determined and a copy of the interchange test script secured.

The customer needs to set up test time with the interchange well in advance in order to insure time frames are met.

13. ESTABLISH TEST FACILITY

It is recommended that an area be designated specifically for testing the BASE24 release 6.0 version 5 software. The following items are recommended for the test area:

- 3 PCs
- Test devices
- Hard copy printer
- Test cards

14. INSTALL PRODUCTION HARDWARE

Installation of any additional computer hardware or peripherals is generally performed by the local hardware service organization or customer staff.

15. ATTEND BASE24 EDUCATION CLASSES

ACI offers a full suite of instructor led courses for BASE24. An enhancement course is also available upon request. The course descriptions can be found in the BASE24 Education Catalog available in hardcopy through your ACI Customer Manager or viewed on the ACI Worldwide web site at www.aciworldwide.com.

16. DEVELOP INTERNAL PROCEDURES

There are no new internal procedures that ACI is aware of for BASE24 release 6.0 version 5 enhancements. User/Operational impacts are noted in the detailed sections for each enhancement described within this document.

17. DEVELOP SYSTEM TEST PLAN

The customer is responsible for developing a system test plan. Included in the test plan should be a description of each step specifying the test card to be used, the account to be used, the transaction to be executed, and the expected results. A procedure for retaining the results of the transaction processing should also be specified. Controlled testing and result verification procedures are key elements for successful system testing.

The execution of the system test plan will occur over a period of weeks to fully test all aspects of the system. BASE24 product functionality should be the first application tested. Any system modifications should then be thoroughly tested. The interface between the host and BASE24 should also be tested. This includes testing on-line authorization and the refresh and extract programs.

18. DEVELOP HOST SOFTWARE

During the course of a business day, BASE24 logs all completed and declined transactions to a transaction log file. After network settlement, the transaction log file can be extracted to tape and used as input into the customer's nightly processing cycle on the host. If the customer would like to take advantage of the new features in release 6.0 version 5, a host program will need to be developed to interpret the new token data that can be extracted with release 6.0 version 5.

After the nightly processing cycle on the host or periodically during the business day the host may need to update card and account files used by BASE24 for pre-authorization or stand-in processing. If the customer would like to take advantage of the new features in release 6.0 version 5 in these files, a host program will need to be developed to initialize the new data fields in BASE24 files.

19. INSTALL BASE24

Release 6.0 version 5 software will need to be installed at the customer site. A tape containing the licensed release 6.0 version 5 software modules will be sent to the customer upon request for installation on their system.

20. CONVERT DATA FILES

ACI provides database conversion programs for selected BASE24 files. These files are discussed in the detail sections for each enhancement described within this document.

21. CERTIFY INTERCHANGE

This task involves implementing the interchange test script and receiving notice of certification from the interchange. Current BASE24 users should not have to re-certify their Interchange Interfaces, unless they are implementing new functionality not previously tested.

22. PERFORM SYSTEM TESTING

Full business day cycle testing of the system as determined by the customer testing plan and procedures.

23. DEVELOP CONVERSION PLAN

The customer should develop an implementation plan for converting existing devices and interchange links to release 6.0 version 5. The installation of new devices and interchanges should also be included in this plan. A method of informing users of any changes to existing procedures should also be developed. The conversion schedule should be routed to all departments.

24. GO-LIVE

At the customer's request, an operational support analyst will visit the site one day prior to the "go-live" date to review the current software status and the BASE24 facilities. This individual will remain onsite to assist the customer with production cutover. The migration plan developed by the customer will be utilized in production cutover. On-site go-live support is a billable service.

25. COMPLETE POST IMPLEMENTATION REVIEW

In general, the purpose of this meeting is to ensure overall satisfaction of the customer’s project.

26. TURN OVER TO CUSTOMER MANAGEMENT

If an ACI Project Manager has handled the BASE24 upgrade, a formal turnover meeting is held between the ACI Project Manager and the ACI Customer Manager to ensure a smooth transition in turnover of responsibilities for on-going customer support.

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Publications

The BASE24 formal publications set has been updated to include the release 6.0 version 5 enhancements. The updated BASE24 documentation is available via CD-ROM.

The BASE24 6.0 CD-ROM also contains the previous release (5.3/1.3) documentation. The 5.3/1.3 version of a manual can be considered the most current if the manual is not listed under the 6.0 documentation menus. The XPNET 3.0 documentation is available on a separate CD-ROM titled NET24-XPNET.

BASE24 customers can order a copy of the BASE24 6.0 CD-ROM and/or XPNET 3.0 CD-ROM from their ACI Customer Manager.

ACI has discontinued hard copy distribution of manuals that are available on CD-ROM. Hard copies, however, can be printed at your site from the CD-ROM as necessary and in accordance with your license agreement.

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Education

BASE24 Technical Education offers a complete set of one- to four-day modules covering the components of the BASE24 system. BASE24 courses are structured into a modular format that eliminates repetitious material, provides information specific to certain job functions, offers hands-on lab experience, and enhances retention of information taught.

A BASE24 Enhancement course is also available upon request for existing customers upgrading to BASE24 6.0 version 5.

The modular format enables BASE24 users to choose modules applicable to their employees' job responsibilities. This provides consolidated training time and offers opportunities for additional employees to receive education.

For a current education schedule, see the BASE24 Education Catalog available in hardcopy through your ACI Customer Manager or on the ACI Worldwide web site at \\www.aciworldwide.com.

ACI Worldwide Inc.

HELP24

Please use ACI Worldwide's eSupport site to report problems or ask questions about BASE24 release 6.0 version 5. eSupport is an Internet Web based application that interfaces with the HELP24 Clarify database. Using eSupport, you can search our KnowledgeBase Solutions for information and resolutions to problems you are experiencing. You can also directly open Clarify cases. An eSupport option, InfoLink, is your link to a wealth of information that can help you in your day to day business. It allows you on-line access to ACI Publications, Fix History Information, HELP24 Hotlines and much more. Contact your local Help24 office for an eSupport logon.

As a result of reviewing new troubleshooting methods, HELP24 has acquired a tool called WebEx. This is a state of the art web-based technology that allows us to troubleshoot issues in a secure, online environment. Compared to the dial-up environment we have used in the past, WebEx provides convenient access, a more stable connection and most importantly, it provides a collaborative environment which also allows you to actively participate in the session by viewing and asking questions about everything that goes on.

To address audit security requirements, WebEx provides the option to record the session so all steps can be documented in a file that can be accessed after the fact if necessary. WebEx has a file transfer option which allows the ability to transfer files from your PC to ours, or vice versa. This feature can be used in the event the file(s) are too large to email.

For more information, please refer to the WebEx User's Guide located under Product Info on Infolink. Contact your local HELP24 office to schedule your WebEx demonstration.

ACI Worldwide Inc.

ASTech Services

ACI offers customer support services through our ASTech Division which allows the selection of a la carte support services to meet individual customer requirements. ASTech, ACI Support and Technical Services, is a service support team which provides the customer with a variety of on-site and remote technical support. ASTech provides customers with supplemental staffing, system performance analysis, review and/or upgrading of previous release customer-specific software development, and detailed operational technical training.

The following are example descriptions of the various services ASTech could undertake to assist you during the BASE24 upgrade. The number of staff weeks identified to accomplish these tasks are estimates and will vary depending on the amount of customer specific code to be developed and delivered by ACI.

Your ACI Customer Manager can provide you with details of services available in your region that have been tailored to the special needs of different areas of the world.

Customer Software Analysis

Effort: Two to four staff weeks

ASTech personnel will perform a complete review of the customer's BASE24 software system and determine what current modifications (CSM code) exist at the customer location. ASTech personnel will perform a detailed comparison of existing customer specific software modifications to release 6.0 version 5 functions. Upon completion of the review, the customer will be provided with a document that identifies all modules that contain code unique to their BASE24 system (file names, custom functions, custom function applicability, etc.).

Conversion Planning

Effort: One and one-half to three staff weeks depending on the size and current system configuration.

This service option provides the customer with comprehensive support during the system migration to the new version. ASTech assigns the customer a Technical Project Lead, or Tech Lead from ACI. The Tech Lead will be responsible for ensuring the success of the BASE24 upgrade project.

The Tech Lead will:

- Review the migration options and develop a detailed migration plan for the conversions project
- Review the Project Scope Document on behalf of the customer (if applicable)
- Review the customer's internal procedures to accommodate release 6.0 version 5 functions.

Upgrade Custom Code

Effort: The amount of time necessary for this task varies depending on the number of modules identified as having modifications unique to the customer. Systems with a great number of customizations may take up to sixteen staff weeks. Actual estimates would be provided upon completion of the Customer Software Analysis.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

With this service, ASTech will upgrade all custom code to release 6.0 version 5. This includes unit testing of the code. Customizations that are no longer necessary will be removed. Customizations that must be carried forward can either be upgraded in their current form or be reengineered to take advantage of new functionality in release 6.0 version 5. Reengineering customizations will add time to the estimate.

System Test Planning

Effort: One to four staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products to be tested. Some of this time will be on-site to perform the System Test Plan.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

ASTech personnel will develop and execute a BASE24 System test plan designed for the customer's specific needs. This service also includes the performance and documentation of the system acceptance testing.

Switch and Host Certification Support

Effort: Two to five staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products and interfaces to be certified. All of this time will be on-site time.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

This service option allows ASTech personnel to provide complete support of the various switch and/or host communication certification testing. This includes reviewing and refining the test scripts, performance of certification testing functions, evaluation of the testing results, and documentation and presentation of the results obtained during the certification testing.

Go-Live Support

Effort: One week (on-site).

ASTech personnel will be at the customer site assisting in all conversion tasks and being present for go-live support. This includes remaining on-site for a couple of days to ensure system stability.

Post Go-Live Support

Effort: Varies.

After a successful go-live, a Tech Lead will provide on-going support which may be required by the customer. Tech Lead support can be provided at the customer location and/or remotely through ACI.

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Section 4

BASE24 Base

Section 4 discusses the changes and enhancements made to the BASE24 Base modules in release 6.0 version 5, along with the resulting conversion and implementation requirements that existing BASE24 users must consider when upgrading from release 6.0 version 4.

Changes and enhancements to the BASE24 Base modules impact all users. Consequently, all users should review the material in this section to determine how it applies to them.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the Base modules changes that affect all BASE24 users, along with the associated conversion and implementation requirements that all BASE24 users must address.
- Optional enhancements describe the Base modules changes that affect only those BASE24 users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 4 must be used in conjunction with Sections 5 through 13 to identify the changes that BASE24 users need to consider when upgrading to release 6.0 version 5.

Standard Information Format

A standard format is used in Sections 4 through 13 to highlight and describe the various changes, conversion requirements, and implementation requirements that BASE24 users need to consider when upgrading to release 6.0 version 5. This standard format is defined below.

In situations where the following information does not apply, the information is labeled as “Not applicable”

Software Impacts

Describes the software modules impacted for a given change or enhancement and the nature of the impacts. This information includes a high level list of the modules impacted, how the modules are impacted (i.e., inline changes, hooks, dummy procs, extensions) and which customers are affected (i.e., all, only those implementing the enhancement).

CUSTMACS Changes

Describes the CUSTMACS changes required to implement a given change or enhancement. This information includes Make defines that have been added, deleted, or modified.

LCONF Assign Changes

Describes the LCONF assign changes required to implement a given change or enhancement. This information includes assigns that have been added, deleted, or modified, along with an explanation of how these assigns can or must be set.

LCONF Param Changes

Describes the LCONF param changes required to implement a given change or enhancement. This information includes params that have been added, deleted, or modified, along with an explanation of how these params can or must be set.

Database Configuration

Describes the data file changes required to implement a given change or enhancement. This information includes file fields that have been added, deleted, or modified, along with an explanation of how these fields can or must be set. It also identifies file records and record segments that must be added, deleted, or modified.

New Data Files

Identifies new data files that must be created in order to implement a given change or enhancement.

Template Changes

Identifies file templates that must be created or modified in order to implement a given change or enhancement. It also identifies COBNAMES and Device Handler Names (xxxxNAMS) file modifications that must be made.

Device Configuration

Identifies device configuration changes that are required in order to implement a given change or enhancement.

Mega Table (MEGATBL) Changes

Identifies Mega Table changes that are required in order to implement a given change or enhancement.

PI-Table Changes

Identifies Product Indicator (PI) Table changes that are required in order to implement a given change or enhancement.

SEG-Table Changes

Identifies Segment Indicator (SEG) Table changes that are required in order to implement a given change or enhancement.

Security Table (SECTBL) Changes

Identifies Security Table changes that are required in order to implement a given change or enhancement.

PATHCONF Changes

Identifies PATHCONF changes that are required in order to implement a given change or enhancement.

CRT Access Security Changes

Identifies CRT Access security changes that are required in order to implement a given change or enhancement. This includes changes that need to be made to BASE24 operator security records.

Routing Changes

Identifies routing and configuration changes that are required in order to implement a given change or enhancement. This includes required changes to files such as SPROUTE and the BASE24-pos Authorization Selection Table (AST).

Network Configuration

Identifies configuration changes that are required in order to implement a given change or enhancement. This includes NEF changes.

File Conversions

Identifies the files that need to be converted in order to implement a given change or enhancement. This implies that a given file could be impacted by multiple changes and/or enhancements. BASE24 users need to consider all of the impacts to any one specific file before the actual conversion of a file.

Host/Co-Network Impacts

Identifies impacts to a user’s host or co-network as a result of a given change or enhancement. This includes changes to the external messages used to communicate with hosts and co-networks as well as changes to refresh and extract tape formats.

User/Operational Impacts

Identifies potential changes to a user’s internal BASE24 operating procedures. This includes changes to operations and file maintenance screens, operator-initiated commands, and reports.

Vendor Impacts

Identifies any issues that need to be addressed with vendors in order to implement a given change or enhancement. This includes hardware, firmware, and software changes that vendors need to make in order for their products to function with the given change or enhancement.

Reference Documents

Lists any outside sources used in the design of a given change or enhancement. BASE24 users can refer to these documents for additional information.

General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 users must address in order to upgrade to release 6.0 version 5

- General Changes
- Reopen TSS After Restart

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

Changes for all BASE24 release 6.0 version 5 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes were done under one SCN per file as well.

The TSS Message DDL (DDLGTSS) was modified for compatibility with the 04.4 version of TSS. This enhancement also resulted in changes to the Security Utilities (SECUTILS) and the Card Authentication Method Utilities (CAMUTILS).

Since all of the available space in the two generic ATM tokens (AT50 Token and AT-FLG1 Token) has been taken, a new generic ATM token was added for future use. This change impacted the ATM Token DDL (DDLATTKN), the Product Token table (COBTKN), the ATM Token Conversion utilities (ATTKNCVS) and the ATM Token Ids (ATTKNID).

The currency code and country code table in BAUTILS was updated with the latest ISO codes for Ecuador, East Timor, Madagascar and Turkey. The spelling in the comment fields for Raroe Islands and Guadeloupe was also corrected.

A new token has been added for integration with IFX. This change impacted the Base Token DDL (DDLBATKN), the Product Token table (COBTKN), the Base Token Conversion utilities (BATKNCVS) and the Base Token Ids (BATKNID).

The Transaction Based Pricing (TBP) reports have been modified to support Mobile Top-Up transactions. These changes impact all BASE24 TBP customers. All BASE24 TBP customers must implement the updated version of the TBP report objects (TBPUPDT) and convert their existing Monthly Transaction Accumulation Files to the new file format.

Software Impacts

The list of files impacted by items #2 - #6 include:

All BASE24 customers:

- Inline changes to BA60DDL.DDLATTKN, DDLBATKN and DDLGTSS
- Inline changes to BA60SRC.ATTKNCVS, ATTKNID, BATKNCVS, BATKNID, BAUTILS and SECUTILS

- Inline changes to BA60AFT.COBTKN

All BASE24 TBP customers:

- Inline changes to BA60TBP.ATDRPTS, BAMRPTS, DDLFMTAF, DRPTG, RPTLIBS
- A new TBP reports object BA60TBP.TBPUPDT

All BASE24 EMV customers:

- Inline changes to BA60IEMV.CAMUTILS

The list of files impacted by item #1 is included in the **Software Impacts** section of the applicable enhancement(s).

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

The BASE24 ATM Data1 Token has been added for future use.

If the new token is not to be logged to the TLF or ILF, then the appropriate token records must be configured.

If the new token is to be extracted with the TLF, or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Running the conversion program is required for any EMV user. Users must create the CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF records from the release 5.3 format to the release 6.0 version 5 format. The CPF conversion program on the BA60UC08 subvolume can be used to convert the CPF records from the release 6.0 version 4 format to the release 6.0 version 5 format. If modifications to the conversion program are necessary, the program can be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/ cnvcpfm –force –list –listreplace

Release 5.3 users should update the CNVCPFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 4 users should update the CNVCPFR obey file on the BA60UC08 subvolume with the appropriate information. The CNVCPFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and to the AAREAD08 on the BA60UC08 subvolume for release 6.0 version 4 customers.

Release 6.0 version 1, release 6.0 version 2 and release 6.0 version 3 customers should refer to the ***Conversion Program Guidelines*** within Section 1 of this document.

Monthly Transaction Accumulation Files (MTAF)

Changes to the Transaction Based Pricing (TBP) reports necessitates that all BASE24 TBP customers implement the updated version of the TBP report objects into their system and convert their existing Monthly Transaction Accumulation Files (MTAFs) to the new format by performing the following steps:

1. All programs must be installed on the proper subvolume.

2. All file assigns must point to the new TBP programs and MTAFs.

3. The RIP process must be stopped.

4. Existing MTAFs must be converted to the new format by performing the following steps for each MTAF.

    a. Rename the MTAF.

    b. Dup the MTAF template to the new MTAF name: FUP DUP <subvolume>.XXMTYYMM, <MTAF name>.

    c. Copy the records from the old MTAF to the new MTAF: FUP COPY <old renamed MTAF>, <new MTAF>, PAD 0.

5. The RIP process must be started.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

If a customer specific token is required to be displayed and processed by the TKN AFT, the file COBTKN must be updated to contain the customer specific token information.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Reopen TSS After Restart

This section describes the implementation requirements associated with the Reopen TSS After Restart enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Reopen TSS After Restart enhancement.

This change impacts all customers who perform security processing using TSS. Changes have been made to the security utilities to periodically attempt to reopen the TSS process when it has been marked down, instead of raising a warning message saying it was down.

Software Impacts

All BASE24 customers:

- Inline changes to BA60SRC.SECUTILS
- Updated event (log) message manual BA60LOGM.BASECUTL

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNTSSRE.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

SECURE-DEV-FAIL-NOTIFY

The meaning of this existing param has changed for processes that are configured to send security requests to TSS. Formerly, the param determined the interval at which a process raised the warning message indicating that an HSM or TSS process had been marked down. In release 6.0 version 5, the same param determines the interval at which the BASE24 process attempts to reopen a TSS process after it has been marked down. The valid range of 10 -1000 and the default value of 20 are the same as before. For processes that are not configured to use TSS, the meaning is unchanged.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the Base modules enhancements that can be enabled or disabled with release 6.0 version 5, along with the changes, conversion requirements, and implementation requirements. These enhancements are the following:

- Absent ICC Application PAN Sequence Number
- American Express Card Security Code
- Automated Hot Card Updates
- EMV 2000 Session Key Derivation
- Enhanced Second Card Data Processing
- Forward Declined Advices to Host
- Inhibit UI Display of POFST/PVV
- Offline PIN Change Notification
- PBF Refresh Early Notification
- Round Robin Access to TSS
- Stored Value Dormancy and Escheatment
- Visa iCVV Support

BASE24 users can choose whether or not to implement the above enhancements. Users who plan to implement these particular enhancements must address the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements. Users who do not plan to implement these particular enhancements can review the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements for future reference.

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Absent ICC Application PAN Sequence Number

This section describes the implementation requirements associated with the enhancement to provide support for Absent ICC Application PAN Sequence Number.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Absent ICC Application PAN Sequence Number enhancement.

All customers who use the BASE24 EMV product must implement this change.

This enhancement implements support for a value of spaces (instead of zeros) in the Application PAN Sequence Number (APSN) to differentiate between an APSN of “00” and an absent APSN. It includes changes to the BASE24-atm N5XXX EMV device handler, the BASE24-pos ACI Standard POS EMV device handler, the BASE24-pos APACS 30/40 EMV device handler, and the BASE24-pos Hypercom EMV device handler to set the APSN in the EMV Status Token to spaces (instead of zeros) when the APSN is not present on an EMV card.

Changes have also been made to several BASE24 Interchange Interfaces to set the APSN in the EMV Status Token to spaces (instead of zeros) and to ensure that the APSN is not set in external messages when the APSN is not present on an EMV card. The following BASE24 Interchange Interfaces have been modified: AMEX GNS, BankNet, MasterCard Debit ISO, MasterCard Maestro, SPAN2, SWITCH Authorization and VisaNet.

NOTE: When the BASE24-atm Triton EMV device handler, the BASE24-atm Shared (Wincor) NDC+ EMV device handler and the BASE24-atm Diebold 10XX/478X EMV device handler were developed as part of release 6.0, version 5, they included support for this enhancement.

Software Impacts

All BASE24 EMV customers:

- Inline changes to BA60IEMV.CAMUTILS

All BASE24-atm N5XXX EMV device handler customers:

- Inline changes to the EMV extension (AT60IN50.N50EMVS)

All BASE24-pos ACI Standard POS EMV device handler customers:

- Inline changes to PS60ISPD.SPDHEMVS

All BASE24-pos APACS 30/40 EMV device handler customers:

- Inline changes to PS60IAPD.APDHEMVS

All BASE24-pos Hypercom EMV device handler customers:

- Inline changes to PS60IHPD.HPDHEMVS

All BASE24 customers who have licensed one or more of the EMV extensions to the interchange interfaces mentioned below:

- Inline changes to the AMEX GNS Interchange Interface EMV extension (SW60IAEG.AEGNEMVS)
- Inline changes to the BankNet Interchange Interface EMV extension (SW60IBNT.BNETEMVS)
- Inline changes to the MasterCard Debit ISO Interchange Interface EMV extension (SW60IMDS.MDSEMVS)
- Inline changes to the MasterCard Maestro Interchange Interface EMV extension (SW60IMDM.MDSMEMVS)
- Inline changes to the SPAN2 Interchange Interface EMV extension (SW60ISPN.SPN2EMVS)
- Inline changes to the SWITCH Authorization Network Interface EMV extension (SW60ISWN.SWANEMVS)
- Inline changes to the VisaNet Interchange Interface EMV extension (SW60VISA.VISAEMVS).

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNICCA.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

ISO Host Interface Process

An issuer host system must be capable of accepting and processing an APSN of spaces if the EMV Status Token is included in transaction messages.

An acquirer host system must include an APSN of spaces in the EMV Status Token in transaction messages sent to BASE24, if the APSN is not present on the chip.

BIC ISO Interchange Interface Process

An issuer system must be capable of accepting and processing an APSN of spaces if the EMV Status Token is included in transaction messages.

An acquirer system must include an APSN of spaces in the EMV Status Token in transaction messages sent to BASE24, if the APSN is not present on the chip.

Super Extract Process

The host system must be capable of accepting and processing an APSN of spaces if the EMV Status Token is included in PTLF, TLF or ILF Extract records.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

American Express Card Security Code

This section describes the implementation requirements associated with the American Express Card Security Code enhancement.

These changes and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the American Express Card Security Code enhancement.

The American Express Card Security Code enhancement involves changes to BASE24 to detect fraudulently created American Express cards through the implementation and use of Card Security Codes (CSC).

American Express offers three types of CSC:

- Three-digit CSC – located on the signature panel (currently not in use)
- Four-digit CSC – located on the front of the card (currently occupied by four digit batch code (4DBC))
- Five-digit CSC – located on the magnetic stripe

BASE24 requires that American Express Card Verification be performed by the Transaction Security Services (TSS) using hardware encryption.

Additional product specific changes relating to this enhancement can be found in Sections 5 (ATM), 6 (POS) and 12 (Interface Interchanges).

Software Impacts

All BASE24 customers:
- Changes to BA60DDL.DDLGTSS and DDLPSTKN
- Inline changes to BA60SRC.CVUTILS
- Inline changes to BA60SRC.SECUTILS

All BASE24-atm customers:
- Inline changes to AT60AUTH.AUTHG and AUTHLIBS
- Updated event (log) message manual BA60LOGM.ATAUTH

All BASE24-pos customers:
- Inline changes to PS60RTAU.AUTHLIBS, ROUTERS and RTAUG
- Updated event (log) message manuals BA60LOGM.PSAUTH and PSROUTER

All BASE24-pos SIV customers:

- Inline changes to BA60VSRC.SECSIVS

All American Express Global Network (AEGN) Interchange Interface customers:

- Inline changes to SW60AEGN.AEGNS, AEGNG and AEGNRSPS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNACSC.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the existing fields in the files that are required to be modified to support this enhancement.

<u>Card Prefix File (CPF)</u>

Field Name: CV CHECK TYP
Screen: CPF Screen 2
DDL Name: CPF.CPFBASE.CV-CHK-TYP

This field should be set to a value other than 0 if AMEX CSC verification is required.

Field Name: CARD VERIFICATION KEYA GROUP
Screen: CPF Screen 2
DDL Name: CPF.CPFBASE.CV-KEYA-GRP

Field Name: MANUAL CARD VERIF KEYA GROUP
Screen: CPF Screen 2
DDL Name: CPF.CPFBASE.MANUAL-CV-KEYA-GRP

If an electronic and a manually entered CSC value are to be verified in a single BASE24-pos transaction, only the Card Verification KEYA group is used by TSS to perform this verification. If an electronic CSC value only is to be verified, the Card Verification KEYA group is used. If a manually entered CSC value only is to be verified, the Manual Card Verif KEYA Group is used.

Field Name: CV DATE
Screen: CPF Screen 2
DDL Name: CPF.CPFBASE.CV-EFF-DAT

Field Name: MANUAL CV DATE
Screen: CPF Screen 2
DDL Name: CPF.CPFBASE.MANUAL-CV-EFF-DAT

If an electronic and a manually entered CSC value are to be verified in a single BASE24-pos transaction, only the CV Date is used by TSS to perform this verification. If an electronic CSC value only is to be verified, the CV Date is used. If a manually entered CSC value only is to be verified, the Manual CV Date is used.

Field Name: DATE CHECK TYPE
Screen: CPF Screen 2
DDL Name: CPF.CPFBASE.DAT-CHK-TYP

This field should be set to a value of 0 if AMEX CSC verification is required.

For additional information on the usage of these fields, refer to the ***BASE24 Base Files Maintenance Manual.***

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

<u>TSS Considerations</u>

Issuers requiring support for AMEX CSC must implement version 04.4 of TSS.

Issuers not requiring support for AMEX CSC may implement the enhanced BASE24 code, but will not have to implement a new version of TSS.

User/Operational Impacts

<u>Disk Impacts</u>

Customers that do not currently support MasterCard and/or Visa card verification may encounter disk impacts. This is due to the utilization of token data within the Release 5.0, Release 5.1 and Data1 POS tokens as well as the Release 5.0 ATM token.

The TLF and PTLF require additional disk space if these tokens are configured to be included in the transaction records logged to these files. The total impact varies for each BASE24 user depending on the transaction volume in the BASE24 system.

Performance Considerations

If BASE24 is configured to verify AMEX CSC values, then the additional HSM command may have an impact on system performance.

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Customers using the BASE24-pos Standard Device Handler (SPDH) must implement updated firmware from their device vendor in order to be able to pass manually entered CSC values into BASE24.

Reference Documents

American Express – American Express Global Network Prerequisite Requirements and Codes, January 2004.

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Automated Hot Card Updates

This section describes the implementation requirements associated with the Automated Host Card Updates enhancement.

These changes and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Automated Hot Card Updates enhancement.

This enhancement allows the BASE24 From Host Maintenance ISO Host Interface process to send PATH messages to a maximum of three interchange interface processes, with or without updating the CAF or NEG files on BASE24.

The following BASE24 Interchange Interfaces support online hot card updates:

- American Express Global Network
- Alaska Option ISO
- BankNet ISO
- Deluxe ISO Generic Gateway
- JCB ISO
- MAC MASM
- MagicLine ISO
- MasterCard Debit ISO
- NEN/Excel ISO
- Networks ISO
- NYCE ISO
- Publix
- Plus ISO
- Pulse ISO
- Shazam ISO
- Star System ISO
- Tyme ISO
- Visa DPS ISO
- VisaNet ISO

The automated hot card update processing uses the following data elements:

- Data element 101, File Name, contains three codes used by the BASE24 From Host Maintenance process, as follows:
  - CF (Cardholder Authorization File) instructs the From Host Maintenance HISO process to use information in data elements 113, 114, and/or 125 to send PATH messages to the interchange interface

process specified in the destination service field of each data element and update the CAF.

- IC (Interchange Only) instructs the From Host Maintenance HISO process to use information in data elements 113, 114, and/or 125 to send PATH messages to the interchange interface processes specified in the destination service field of each data element without updating the CAF or NEG.
- NF (Negative Card File) instructs the From Host Maintenance HISO process to use information in data elements 113, 114, and/or 125 to send PATH messages to the interchange interface processes specified in the destination service field of each data element and update the NEG.

- Data element 113, Reserved National, contains data used by the From Host Maintenance HISO process to format and send hot card update messages to BASE24 Interchange Interfaces.
- Data element 114, Reserved National, contains data used by the From Host Maintenance HISO process to format and send hot card update messages to BASE24 Interchange Interfaces.
- Data element 125, Reserved Private, contains data used by the From Host Maintenance HISO process to format and send hot card update messages to BASE24 Interchange Interfaces.

The host may use data element 113, 114, or 125 to send a hot card update message to a single BASE24 Interchange Interface or may use any combination of these data elements to send hot card update messages to two or three BASE24 Interchange Interfaces.

The format of the hot card update data used in data elements 113, 114, and /or 125 varies for each BASE24 Interchange Interface. Refer to Appendix G of the ***BASE24 External Messages Manual*** for details on each of the formats.

Software Impacts

All BASE24 customers:

- Changes to BA60DDL.DDLGISO

All BASE24 HISO customers:

- Inline changes to BA60HISO.BAHISOS

All BASE24-fhm HISO customers:

- Inline changes FH60HISO.FHHISOS
- Updated event (log) message manual BA60LOGM.FHMHISO

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNHCRD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

The host may format data elements 113, 114, and/or 125 with the interchange interface specific hot card update PATH message format. The host may configure data element 101, indicating whether to send PATH messages to interchange interface processes with or without updating the CAF or NEG. Refer to appendix G of the ***BASE24 External Messages Manual*** for details on the interchange interface specific PATH message formats and details on data element 101.

User/Operational Impacts

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable.

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EMV 2000 Session Key Derivation

This section describes the implementation requirements associated with EMV 2000 Session Key Derivation enhancement.

These changes and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the EMV 2000 Session Key Derivation enhancement.

The EMV 2000 Session Key Derivation enhancement involves changes to BASE24 to allow issuers to utilize the method of session key derivation defined in the EMV 2000 (version 4.0) specifications in EMV cryptographic processing.

The formatting of TSS Command Message ‘02’ in the CAMUTILS module is based on the EMV Security Scheme supported by the card: Visa, M/Chip 2.1 or M/Chip 4.0. For cards compliant with VIS 1.4.0 (Visa) and M/Chip 4.0 (MasterCard), the Cryptogram Version Number within the Issuer Application Data received in an EMV request message specifies the type of key derivation required:

- For VIS 1.4.0, EMV 2000 key derivation is required if the Cryptogram Version Number is hexadecimal ‘0E’.
- For M/Chip 4.0, EMV 2000 key derivation is required if the Cryptogram Version Number is hexadecimal ‘12’ or hexadecimal ‘13’.

Note that M/Chip 2.1 does not support the EMV 2000 method of key derivation.

If EMV 2000 key derivation is required, then TSS Command Message ‘22’ will be formatted instead of TSS Command Message ‘02’. The formatting of the new message will be based on that for the existing message.

Within the CAMUTILS procedure that formats issuer scripts, logic has been added to use the EMV Security Scheme and Cryptogram Version Number parameters (when passed) to determine whether EMV 2000 key derivation is required. The logic used to determine whether EMV 2000 key derivation is required will be the same as that performed when formatting TSS Command Message ‘22’.

If EMV 2000 key derivation is required, then TSS Command Message ‘23’ will be formatted instead of TSS Command Message ‘03’. The formatting of the new message will be based on that for the existing message.

For backwards-compatibility, the new TSS Command Messages (‘22’ and ‘23’) will not be formatted when EMV 1996 key derivation (i.e. current processing) is required.

Logic has been added to the BASE24-atm authorization EMV extension module and the BASE24-pos router/authorization EMV extension module to set the EMV Security Scheme parameter being passed into the procedure call to CAMUTILS, and to include the Cryptogram Version Number as an additional parameter in the procedure call to CAMUTILS.

In order to minimize the number of messages sent to TSS during transaction processing, the EMV Security Scheme is retrieved when the EMV global structures are first initialized for a transaction. The value is stored in an appropriate EMV global field, and referenced when required.

Software Impacts

All BASE24 customers:

- Changes to BA60DDL.DDLGTSS

All BASE24 EMV customers:

- Inline to BA60IEMV.CAMUTILG, CAMUTILS and EMVG
- Updated event (log) message manual BA60LOGM.BACAMUTL

All BASE24-atm EMV customers:

- Inline to AT60IATH.AUTHEMVG and AUTHEMVS

All BASE24-pos EMV customers:

- Inline changes to PS60IRTA.RTAUEMVG and RTAUEMVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNEMV2.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

TSS Considerations

Issuers requiring support for EMV 2000 Session Key Derivation must implement version 04.4 of TSS.

Issuers not requiring support for EMV 2000 Session Key Derivation may implement the enhanced BASE24 code, but will not have to implement a new version of TSS.

User/Operational Impacts

Performance Considerations

This enhancement will improve system performance in EMV transaction processing for TSS users.

In BASE24-atm, TSS may be invoked on two separate occasions during a single transaction to retrieve the EMV Security Scheme from the TSS database. This occurs on all non-approved EMV transactions and on EMV transactions where a PIN CHANGE script is generated by BASE24.

In BASE24-pos, TSS may be invoked on two separate occasions during a single transaction to retrieve the EMV Security Scheme from the TSS database. This occurs on all non-approved EMV transactions.

This enhancement will ensure that the EMV Security Scheme is retrieved from the TSS database only once during a single transaction.

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Issuers requiring support for EMV 2000 key derivation must implement one of the following HSMs, with the specified minimum software release level:

- Atalla Axx100-VAR version 2.61 or Axx100-AKB version 2.61.
- Thales RG7x10 version 3.0 or RG7x00 version 7.0.

Reference Documents

EMVCo - EMV 2000 Integrated Circuit Card Specification for Payment Systems, Book 2 – Security and Key Management, Version 4.0, December 2000

HP (Atalla) - Axx100 Banking Release 2.36 Functional Specification, version 1.0, 26th November 2003

MasterCard International - M/Chip 4 Card Application Specification, Version 4.0, 16th May 2002

Thales e-Security - 1419HS0B HSM Base Upgrade 2003, 21st August 2003

Visa International - Visa Integrated Circuit Card – Card Specification, Version 1.4.0, 31st October 2001

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Enhanced Second Card Data Processing

This section describes the implementation requirements associated with the Enhanced Second Card Data Processing enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Enhanced Second Card Data Processing enhancement.

Prerequisite: The Enhanced Expiration Date Checking enhancement must be installed prior to implementation of this enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 4)***.

Software Impacts

All BASE24 customers:

- Changes to BA60DDL.DDLFCAF
- Inline changes to BA60AFT.RQCAFS, SCRNCAF and SVCAFS

All BASE24-atm customers:

- Inline changes to AT60AUTH.AUTHLIBS, AUTHG and AUTHEMVD

All BASE24-atm EMV customers:

- Inline changes to AT60IATH.AUTHEMVS

All BASE24-pos customers:

- Inline changes to PS60RTAU.AUTHLIBS, ROUTERS, RTAUG and RTAUEMVD

All BASE24-pos EMV customers:

- Inline changes to PS60IRTA.RTAUEMVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNSCNDC.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the new param record that must be added to the LCONF to support this enhancement. This param is provided in the LCONFBA file located on the BA60MISC subvolume.

CAF-SCND-CRD-PROMOTE

This optional param indicates whether the institution can automatically remove the current primary card from the system and make the second (reissued) card the new primary card. When the first transaction is received from the second card, this feature, if enabled, will automatically replace the primary card's data in the CAF with the second card's data and then clear the second card fields. Valid values are as follows:

Y = Yes, automatically promote the second card to primary when the first transaction is received from the second card.

N = No, do not automatically promote the second card to primary. An operator must do so manually at a time to be determined by the institution. (default)

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Promoting Second Card to Primary

The institution may now automatically promote the second (reissued) card to primary when the first transaction is received from the second card. To do so, set the new CAF-SCND-CRD-PROMOTE LCONF param to Y (Yes). If this param is set to N (No) or if this param is not present in the LCONF, an operator must manually promote the second card to primary at a time to be determined by the institution.

When the second card is promoted automatically to primary, the following occurs:

- CAF.BASE.SCND^CRD^DATA.EXP^DAT^2 is moved to CAF.BASE.EXP^DAT, then CAF.BASE.SCND^CRD^DATA.EXP^DAT^2 is set to ASCII zeroes.
- CAF.BASE.SCND^CRD^DATA.EFFECTIVE^DAT^2 is moved to CAF.BASE.EFFECTIVE^DAT, then CAF.BASE.SCND^CRD^DATA.EFFECTIVE^DAT^2 is set to ASCII zeroes.
- CAF.BASE.SCND^CRD^DATA.CRD^STAT^2 is moved to CAF.BASE.CRD^STAT, then CAF.BASE.SCND^CRD^DATA.CRD^STAT^2 is set to '6' (card not used).
- CAF.EMV.SCND_CRD_DATA.ATC_2 is moved to CAF.EMV.ATC, then CAF.EMV.SCND_CRD_DATA.ATC_2 is set to binary zero.

Similar actions should be performed by an operator should manual processing be chosen.

<u>Application Transaction Counter (ATC) Checking</u>

Prior to this enhancement, EMV Application Transaction Counter (ATC) checking could not be performed on a second (reissued) card while both the primary and second cards were active because only one ATC was stored in the EMV segment of the Cardholder Authorization File (CAF), that being the ATC for the primary card. If the ATC from the second card's microchip were compared with the ATC in the CAF, the transaction would be denied because the two ATCs did not match.

With this enhancement, ATC checking may now be performed while both the primary and second cards are active because a second ATC field has been added to the CAF. The Second Card ATC Number, a read-only field, may be viewed on CAF file maintenance screen 13.

If you wish to enable ATC checking, you may do so by setting the ATC Seq Num Check field on screen 11 of the Card Prefix File (CPF) to Y (Yes).

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Forward Declined Advices to Host

This section describes the implementation requirements associated with the enhancement to Forward Declined Advices to the Host.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Forward Declined Advices to Host enhancement.

The Forward Declined Advices to Host enhancement involves changes to BASE24 to allow all advices to be sent to a Host system, if required.

Logic has been added to the BASE24-atm authorization module to allow all advices (approved and declined) to be sent to a HISO process, if required. Logic has also been added to the ATM HISO module to send an advice to a Host system, even if the advice is for a non-financial or a non-approved transaction.

Similar logic has been added to the BASE24-pos router/authorization modules to allow all advices (including approved, declined, and referred advices) to be sent to a HISO process, if required. Logic has been added to the POS HISO module to send an advice to a Host system, even if the advice is for a non-financial or a non-approved transaction.

The following considerations apply to this enhancement and should be reviewed:

- The ability to forward and log advices will be available for all authorization levels.
- Completions generated by the authorization module will not be logged as per current processing.
- Advice messages received by the authorization module are logged unconditionally and this will continue to be the case.
- Declined requests and advices rejected before the IDF record is successfully retrieved will not be forwarded as declined advices to the Host.
- Declined requests and advices rejected because of an error in retrieving the IPCF record will not be forwarded as declined advices to the Host.
- POS Terminal Totals transactions will continue to generate completions for approved transactions only.
- Electronic follow-ups of previously authorized transactions (e.g. referral overrides generated from a POS terminal) will not be forwarded to the Host.

Software Impacts

All BASE24 customers:

- Changes to comments in BA60DDL.DDLFIPCF, DDLGPSTM and DDLGSTM
- Inline changes to BA60AFT.RQIPCFSS, SVIPCFTG, SVIPCFTS

All BASE24-atm customers:

- Inline changes to AT60AUTH.AUTHG, AUTHS and AUTHLIBS
- Inline changes to AT60HISO.ATHISOG, ATHISOS

All BASE24-pos customers

- Inline changes to PS60AUTH.AUTHS, AUTHLIBS and RTAUG
- Inline changes to PS60HISO.PSHISOG, PSHISOS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNFORW.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the files that are required to support this enhancement.

<u>Issuer Processing Code File (IPCF)</u>

Field Name: COMPLETION REQUIRED TO HOST
Screen: IPCF Screen 2
DDL Name: IPCF.COMPL-REQ

A code used to control the production of 0220 financial transaction advice messages. Valid values are:

N = Do not send 0220 messages
Y = Send 0220 messages for approved transactions
A = Send 0220 messages for all transactions

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

ISO Host Interface Process

If the BASE24 user wishes to enable Forward Declined Advices to Host support, then the host system must be capable of accepting and processing non-approved advices. The host system must also be prepared to handle the increase in message traffic.

User/Operational Impacts

If the BASE24 user wishes to enable Forward Declined Advices to Host support, then the COMPLETION REQUIRED TO HOST field on IPCF screen 2 must be set to ‘A’.

Performance Impacts

If this enhancement is implemented, there will be an increase in the number of messages sent to the Host. The Host must be prepared to handle this increase.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Inhibit UI Display of POFST/PVV

This section describes the implementation requirements associated with the enhancement to Inhibit UI Display of POFST/PVV.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Inhibit UI Display of POFST/PVV enhancement.

The Inhibit UI Display of POFST/PVV enhancement involves CAF AFT changes to prevent the contents of the POFST/PVV field being displayed to a user.

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.RQCAFS, SCRNCAF and SVCAFS

No other files were modified for this enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to the LCONF to support this enhancement. This modification is provided in the LCONFBA file located on the BA60MISC subvolume.

MASK-POFST-PVV

This param is used by the CAF Server. It specifies whether the POFST/PVV field should be masked on CAF Screen 1. Valid values are:

- N – Do not mask POFST/PVV field on CAF Screen 1 (default)
- Y – Mask POFST/PVV field on CAF Screen 1

If the param is not found or is invalid, the default value of “N” is used.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

When the enhanced logic is configured (i.e., MASK-POFST-PVV is set to Y), the impacts described below will apply.

If the user only has READ access to CAF Screen 1:

- When the CAF record is read, the POFST/PVV will display "****************" on the screen.

If the user has ADD or UPDATE access to CAF Screen 1:

- When the CAF record is added or the POFST/PVV is updated, the user must enter clear text (as normal) and once the add or update function key is hit, the POFST/PVV will be replaced with "****************".

Note: If the user performs an UPDATE on CAF Screen 1, and the POFST/PVV field is not changed, the original value will be retained for the update.

If the user has DELETE access to CAF Screen 1:

- The user first reads the CAF record, which will display the POFST/PVV as "****************". Once the DELETE function key is hit, the same record is displayed on screen with the POFST/PVV as "****************".

If the user performs VALIDATE on CAF Screen 1:

- The POFST/PVV will display "****************" and the message "DATA O.K – POFST/PVV VALUE OF ASTERISKS VALID ONLY FOR UPDATE" will be displayed.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Offline PIN Change Notification

This section describes the implementation requirements associated with the Offline PIN Change Notification enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Offline PIN Change Notification enhancement.

Changes have been made to the CPF DDL and AFT to support this enhancement. These changes are described below.

Changes have also been made to BASE24-atm modules for this enhancement. Refer to **Offline PIN Change Notification** in Section 5 for a description of these changes.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLFCPF
- Inline changes to BA60AFT.RQCPFS, SCRNCPF and SVCPFS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD08.SCNOFFP.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new or enhanced fields in CPF that are required to support this enhancement.

<u>Card Prefix File (CPF)</u>

Field Name: OFFLINE PIN TRANSPORT KEY
Screen: CPF Screen 11

DDL Name: CPF.EMV.TRANSPORT-KEY

This field contains the locator to the common PIN block encryption key in the TSS database. Its value must be different from the value in the LCONF parameter MULT-LN-PIN-KEY (i.e., the TSS key locator). The default value is all zeros.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

If the BASE24 user wishes to enable Offline PIN Change Notification, the user must update the new OFFLINE PIN TRANSPORT KEY field in the CPF with the appropriate value. Otherwise, the new CPF field will default to zeros to indicate that Offline PIN Change Notification processing is not being performed.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

PBF Refresh Early Notification

This section describes the implementation requirements associated with the PBF Refresh Early Notification enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PBF Refresh Early Notification enhancement.

The refresh process receives the PBF refresh input tape or disk file from the host and loads the new data into the NSK files. This enhancement will now allow the refresh process to notify the authorization processes immediately after data is loaded on to the new PBF without waiting for impacting to be finished. Refresh will start impacting after notification to the authorization processes completes. The authorization processes will then start to authorize using the new PBF file.

Late notification to the authorization processes (i.e., the current mode of processing of BASE24 Refresh) remains an available option.

Software Impacts

All BASE24 customers:

- Inline changes to BA60REFR.REFRS and REFRG
- Updated event (log) message manual BA60LOGM.BAREFR

All BASE24-atm customers:

- Inline changes to AT60REFR.ATREFRS

All BASE24-pos customers:

- Inline changes to PS60REFR.PSREFRS

All BASE24 Remote Banking customers:

- Inline changes to OC60REFR.OCREFRS

All BASE24-teller customers:

- Inline changes to TR60REFR.TRREFRS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNREFR.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to the LCONF to support this enhancement. This modification is provided in the LCONFBA file located on the BA60MISC subvolume.

REFR-PBF-EARLY-NOTIFICATION

This param indicates whether PBF refresh early notification is enabled or disabled. It is used by the BASE24 Refresh process. The valid values are:

- Y = Yes, PBF refresh early notification is enabled.
- N = No, PBF refresh early notification is disabled. (default)

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Performance Considerations

This enhancement allows the new PBF file to be available at the earliest opportunity, and therefore the possibility of authorization processes working with an outdated PBF file is reduced. Consequently, this improves the performance of refresh by reducing the number of records to be impacted under these circumstances. All transactions authorized after the new PBF delivery will be marked as not requiring impact.

User Considerations

While the impacting is taking place, accounts may not be subjected to the impacting procedure and therefore must be considered out-of-date until the impacting phase completes. All transactions authorized in the window between the release of the PBF and the completion of the impacting may be authorized incorrectly. This is true with or without this enhancement. However, with this enhancement, this window is reduced, and therefore accounts are up-to-date sooner.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Round Robin Access to TSS

This section describes the implementation requirements associated with the enhancement to access TSS processes via a round-robin list.

These changes and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Round Robin Access to TSS enhancement.

Customers who use TSS for security processing can implement this enhancement to define a shared list of TSS processes for the network instead of a primary and secondary device for each BASE24 process.

Software Impacts

All BASE24 customers:

- Changes to BA60DDL.DDLGDEFS
- Inline changes to BA60SRC.SECUTILS
- Updated event (log) message manual BA60LOGM.BASECUTL

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNTSSRR.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this enhancement. This modification is provided in the LCONFBA file located on the BA60MISC subvolume.

<u>SECURE-DEV-xx</u>

These new assigns are defined at the network level, that is, the process symbolic name in the LCONF record must be wildcards. At least one assign of this format is required if any BASE24 process is configured to use the round-robin list. The assigns represent up to 20 TSS processes that will be placed in the list. The xx ending will typically be a number but can be any two characters. If there are more than 20 such assigns, then any assigns past the twentieth is ignored.

SECURE-DEV1, SECURE-DEV2

These existing assigns for the primary and secondary devices are no longer applicable if a process is configured to use the round-robin list.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. This modification is provided in the LCONFBA file located on the BA60MISC subvolume.

SECURE-DEV-ACCESS-TYP

This new optional param can be defined at the process or the network level and is used by all processes that are configured to send security requests to TSS. Valid values are:

0 – Use primary and secondary devices (default)

1 – Use the round-robin list

NOTE: If the process is not configured to use TSS, then primary and secondary devices are used regardless of this param.

SECURE-DEV-REINSTATE-TIME

This new optional param can be defined at the process or the network level and is used by all processes that are configured to use the round-robin list. It specifies the time interval in minutes at which a process should attempt to reopen a security device that is marked down on the list. The default value is 0, indicating that the process should not attempt to reopen a device based on the time interval. The maximum value is 32000. This param provides an additional mechanism for automatically reopening TSS processes besides reopening them based on the SECURE-DEV-FAIL-NOTIFY param as described in the Reopen TSS After Restart section.

SECURE-DEV-MONITOR-THRESHOLD

This existing param, which is used to shift usage from the primary device to the secondary device, is no longer applicable if a process is configured to use the round-robin list since all TSS processes will be used equally.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable.

User/Operational Impacts

This enhancement simplifies configuration and performance management of multiple TSS processes. It eliminates the need to create separate primary and secondary device assigns for each BASE24 process in order to distribute the workload evenly. Instead, it enables the user to specify a common list of TSS processes that will be used equally by all BASE24 processes.

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable.

Stored Value Dormancy and Escheatment

This section describes the implementation requirements associated with the Stored Value Dormancy and Escheatment enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Stored Value Dormancy and Escheatment enhancement.

Changes have been made to the CPF and PBF to support this enhancement. These changes and other Base file changes are described below.

Changes have also been made to BASE24-pos modules for this enhancement. Refer to **Stored Value Dormancy and Escheatment** in Section 6 for a description of these changes.

Software Impacts

All BASE24 customers:

- Changes to BA60DDL.DDLFCPF, DDLFOMF and DDLFPBF
- Inline changes to BA60AFT.RQCPFS, RQPBFS, SCRNCPF, SCRNPBF, SECTBL, SVCPFS, SVMHLPS and SVPBFS
- Inline changes to BA60SRC.COBNAMES

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNSVDE.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the CPF and PBF that are required to support this enhancement. For information on POS files impacted by this enhancement, refer to **Stored Value Dormancy/Escheatment** Database Configuration in Section 6.

<u>Card Prefix File (CPF)</u>

Field Name: ESCHEATMENT
Screen: CPF Screen 10
DDL Name: CPF.SV. SV-ESCHEATMENT-FLG

This field will be used by the Stored Value Escheatment process. Valid values are:

N = Do not perform escheatment. (default)

Y = Perform escheatment when applicable.

<u>Positive Balance File (PBF)</u>

Field Name: DORMANCY DATE
Screen: PBF Screen 1
DDL Name: PBF.BASE. DORMANCY-DAT

The date a Stored Value Dormancy Fee was last assessed. This field is system protected.

Field Name: CARD ACTIVATION STATE
Screen: PBF Screen 1
DDL Name: PBF.BASE. CRD-ACTVT-ST

The geographic location where the Stored Value card was actived. This field is system protected.

Field Name: BALANCE PRIOR TO DORMANCY DATE
Screen: PBF Screen 1
DDL Name: PBF.BASE. BAL-PRIOR-TO-DORMANCY-FEE

The PBF Balance prior to a Dormancy Fee being assessed. This field is system protected.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Visa iCVV Support

This section describes the implementation requirements associated with the Visa iCVV Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Visa iCVV Support enhancement.

The Visa iCVV Support enhancement involves changes to BASE24 to allow issuers to validate the card verification digits (CVD) during EMV authorization processing whenever the CVD on the Track-2 Equivalent Data is different from the CVD on the magnetic stripe.

The iCVV validation can be controlled by the new CPF parameters. NOTE: iCVV validation will be performed only under the following conditions:

- The Track-2 Equivalent Data has been read from the EMV card.
- Card verification has been configured for magnetic stripe transactions on the CPF.
- The card expiration date is present in the Track-2 Equivalent Data.
- The card expiration date is not earlier than the CV DATE on screen 2 of the CPF.
- The length of the Track-2 Equivalent Data is not less than the Minimum Track-2 Length configured on screen 1 of the CPF or greater than the Maximum Track-2 Length configured on screen 1 of the CPF.

It is assumed that all EMV cards with the same prefix have CVD encoded on the magnetic stripe and CVD encoded on the chip. Older cards may have the same CVD on both the magnetic stripe and the chip, whereas newer cards may have different CVDs.

The BASE24-atm authorization EMV extension module and the BASE24-pos router/authorization EMV extension module have been enhanced to utilize new CPF fields to determine whether and how to perform card verification on EMV transactions.

Software Impacts

All BASE24 customers:

- Changes to BA60DDL.DDLFCPF
- Inline changes to BA60AFT.RQCPFS, SCRNCPF, SVCPFS
- Inline changes to BA60OMF.OMFXS

All BASE24-atm customers:

- Inline changes to AT60AUTH.AUTHLIBS and AUTHEMVD

All BASE24-atm EMV customers:

- Inline changes to the AT60IATH.AUTHEMVS

All BASE24-pos customers

- Inline changes to PS60AUTH.AUTHLIBS and RTAUEMVD
- Updated event (log) message manual BA60LOGM.PSAUTH

All BASE24-pos EMV customers

- Inline changes to PS60IRTA.RTAUEMVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNICVV.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the files that are required to support this enhancement.

Card Prefix File (CPF)

Field Name: CARD VERIFY – CHECK TYPE
Screen: CPF Screen 11
DDL Name: CPF.EMV.EMV-CV-CHK-TYP

This field indicates the circumstances under which card verification should be performed on EMV transactions. Valid values are:

0 = Do not perform card verification. (default)

1 = Perform card verification only when an ARQC is not present.

2 = Perform card verification when an ARQC is not present or when the

ARQC has not been verified successfully.

3 = Always perform card verification.

Field Name: CARD VERIFY – CHECK METHOD
Screen: CPF Screen 11
DDL Name: CPF.EMV.EMV-CV-CHK-MTHD

This field indicates the algorithm and data used when card verification is performed on EMV transactions. Valid values are:

0 = Use non-EMV card verification. (default)

1 = Replace the service code with EMV-CV-DATA.

Field Name: EMV CV DATE
Screen: CPF Screen 11
DDL Name: CPF.EMV.EMV-CV-EFF-DAT

This field contains the EMV CVD expiration date. If a card has an expiration date greater than or equal to this date, an EMV CVD value is expected on the card and EMV CVD processing will occur. Otherwise, the card may not have an EMV CVD value and EMV CVD processing will not occur. Valid values are dates in the format YYMM. The default value is 0501.

Field Name: CV DATA
Screen: CPF Screen 11
DDL Name: CPF.EMV.EMV-CV-DATA

This field holds any additional data required when card verification is performed on EMV transactions. Valid values are numeric data only. The default value is all zeros.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

If the BASE24 user wishes to enable Visa iCVV Support, the user must update the new fields in the CPF with the appropriate value. Otherwise, the new fields in the CPF will default to indicate that Visa iCVV processing is not performed.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24 users that utilize the OMF Extract Process to accept and process information in the CPF.

User/Operational Impacts

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Visa Asia Pacific - Member Letter 43/03 – Mandate for New Card Verification Value for Chip Cards, 11th December 2003.

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Section 5

BASE24-atm

Section 5 discusses the changes and enhancements made to BASE24-atm in release 6.0 version 5, along with the resulting conversion and implementation requirements that existing BASE24-atm users must consider when upgrading from release 6.0 version 4.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-atm users, along with the associated conversion and implementation requirements that all BASE24-atm users must address.
- Optional enhancements describe the changes that affect only those BASE24-atm users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-atm users must address to implement the enhancements.

Section 5 must be used in conjunction with Section 4 to identify the changes that BASE24-atm users need to consider when upgrading to release 6.0 version 5.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-atm users must address in order to upgrade to release 6.0 version 5.

General Changes

1. Since all of the available space in the two generic ATM tokens (AT50 Token and AT-FLG1 Token) has been taken, a new generic ATM token was added for future use. No BASE24-atm applications were modified to use the new token at this time. Refer to Section 4 **General Changes** for additional information.

2. BASE24-atm Authorization was revised to not perform CVD validation when an interchange has already performed this validation. If the CVD was successfully checked by an interchange, the card verification will be bypassed. If the CVD was checked by an interchange and failed, then ATM Authorization will check the BAD CV ACTION on the Card Prefix File (CPF) to determine the appropriate authorization action.

   Note: This change was cataloged as a fix to AT60AUTH.AUTHLIBS in late September, 2004 for Clarify case # 353773.

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Optional Enhancements

This section identifies the BASE24-atm enhancements that can be enabled or disabled with release 6.0 version 5, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Absent ICC Application PAN Sequence Number
- American Express Card Security Code
- BASE24-es ATM IFX ATM Manager and BASE24-atm Integration
- Diebold AKDS Capability Check and Load All Keys
- Diebold Cash Source Plus (CSP) 200 Triple DES
- Diebold EMV
- Diebold Text to Speech (ADA) and Agilis 91x (Opteva) Features
- EMV 2000 Session Key Derivation
- Enhanced ATM Status Reporting
- Forward Declined Advices to Host
- Mobile Top-Up Integration
- NCR AKDS Capability Check and Load All Keys
- NCR NDC+ CAM II (EMV) Upgrade
- NCR NDC+ EMV Dip Card Reader
- Offline PIN Change Notification
- PBF Refresh Early Notification
- Shared (Wincor) NDC+ EMV Support
- Triton EMV
- Triton TCP/IP
- Triton Triple DES MACing
- Visa iCVV Support

BASE24-atm users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements for future reference.

Absent ICC Application PAN Sequence Number

For further information, see **Absent ICC Application PAN Sequence Number** in Section 4 where the impacts of this enhancement are described.

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American Express Card Security Code

This section describes the implementation requirements associated with the American Express Card Security Code enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the American Express Card Security Code enhancement.

Logic has been added to the BASE24-atm authorization module to detect fraudulently created American Express cards through the implementation and use of Card Security Codes (CSC).

BASE24-atm provides processing capabilities for the CSC value embedded in the magnetic stripe. In the authorization process, American Express on-us transaction requests will have the CSC value verified via Transaction Security Services (TSS). The results of the electronic CSC verification are placed into the BASE24-atm Release 5.0 Token.

For further information regarding this enhancement see **American Express Card Security Code** in Section 4 where the impacts of this enhancement are described.

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BASE24-es ATM IFX ATM Manager and BASE24-atm Integration

This section describes the implementation requirements associated with integrating the BASE24-es ATM IFX ATM Manager with BASE24-atm.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of this BASE24-atm enhancement.

The BASE24-atm Authorization module has been modified to process transactions sent from the new BASE24-es IFX ATM Manager. The BASE24-es IFX ATM Manager is a new device handler that uses the IFX Business Message Specification as its ATM-to-host message format, and the BASE24-atm Standard Internal Message (STM) format when communicating with BASE24-atm for authorizations. When an STM from the IFX ATM Manager is received by the BASE24-atm Authorization module, the Authorization module moves data from the XPNET message header into a new token (the IFX ATM Manager appends the new token with no data in it prior to sending the STM to BASE24-atm). Before sending a response back to the IFX ATM Manager, the BASE24-atm Authorization module moves the data from the new token back into the XPNET message header. This allows the IFX ATM Manager to correctly determine which ATM the response is destined for. All other STM transaction processing is the same as for STMs received from any BASE24-atm Device Handler.

Note: BASE24-atm does not handle cutover of the IFX ATM Manager, but does accept ATM settlement records from the IFX ATM Manager (in STM format) for logging to the Transaction Log File (TLF). The BASE24-atm Authorization module did not need any changes for this processing. Cutover of the IFX ATM Manager is handled by a BASE24-es EOPP module.

Software Impacts

All BASE24 customers:

- DDL changes to BA60DDL.DDLBATKN
- Inline changes to BA60SRC.BATKNID and BATKNCVS
- Inline changes to BA60AFT.COBTKN

All BASE24-atm customers:

- Inline code changes to AT60AUTH.AUTHG, AUTHS and AUTHLIBS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNIFX.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

The following BASE24 Base token has been added for this enhancement:

- The IFX User Data Token (token ID "09") contains the data sent in the user data area of the header of the XPNET message. The token is actually just a free-format buffer that may contain JMS-specific information required by the IFX device handler in the BASE24-es system.

If the IFX User Data Token is not to be logged to the TLF or ILF, the appropriate token records must be configured in the TKN file.

The appropriate records in the TKN file should not be configured to extract the IFX User Data Token.

The appropriate records in the TKN file should not be configured to send the IFX User Data Token to the host or passed to the BIC ISO Interchange Interface.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The XPNET Node where the BASE24-atm Authorization process resides must be configured with the PAYLOADPAGES attribute set to a value greater than 199. This is also true for the XPNET Node where the BASE24-es ATM IFX Device Handlers reside. Refer to the ***BASE24-es & TSS HP NonStop Install Guide*** for further information.

File Conversions

Not applicable

Host/Co-Network Impacts

The logical network cutover time for the BASE24-es ATM system must be equal to the logical network cutover time for the BASE24-atm system.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Diebold AKDS Capability Check and Load All Keys

This section describes the implementation requirements associated with the Diebold AKDS Capability Check and Load All Keys enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Diebold AKDS Capability Check and Load All Keys enhancement.

The standard BASE24-atm Diebold 10XX/478X device handler module has been modified to support this enhancement.

Prerequisite: The Automated Key Distribution System (AKDS) enhancement must be installed prior to implementation of the Diebold AKDS Capability Check and Load All Keys enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 4)***.

Software Impacts

All BASE24-atm Diebold 10XX/478X device handler customers:

- Hooks to AKDS procedures in the main device handler module (AT601000.C1000S)
- Inline changes to AT601000.C1000G
- Inline changes to the Diebold 10XX DCT screen and requester (AT601000.SCND10XX and RQD10XXS)
- Inline changes to AT601000.C1000D
- Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X device handler Dialup customers:

- Inline changes to AT60P100.C100DLUS

All BASE24-atm Diebold 10XX/478X device handler AKDS customers:

- Inline changes to AT60A100.C100AKDG and C100AKDS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD08.SCNLALLD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

AKDS Capability Check

This enhancement enables the Diebold 10XX/478X device handler to automatically determine the AKDS capability of an ATM before permitting an AKDS load (LOAD PUBLIC KEY, LOAD MASTER KEY, LOAD ALL KEYS) to proceed. When the device handler receives an AKDS load command, it closes the ATM and sends a configuration request message to determine if the ATM has an AKDS-capable Encrypting PIN Pad (EPP) installed. The AKDS load is allowed to proceed if an AKDS-capable EPP is present. If an AKDS-capable EPP is not present, the AKDS load is not allowed to proceed, an informational event message is logged, and the ATM is re-opened. An exception to this case is the LOAD ALL KEYS command – see the Device Control Terminal (DCT) subsection that follows for more information.

Device Control Terminal (DCT)

The following command has been added to the Diebold 10XX Commands Screen (screen 40) of the DCT.

- LOAD ALL KEYS

  When the operator places the cursor on this command and presses function key F1-ENTER, the DCT sends a command to the device handler to initiate a Load All Keys sequence. At the start of this sequence, the device handler closes the specified ATM and checks its AKDS capability.

  If the ATM is AKDS-capable, the device handler downloads the RSA and DES keys and then re-opens the ATM. The keys are downloaded in the following order.

  1. Public Key
  2. Master Key
  3. MAC Encryption Key (if ATD MAC Support Type = ‘3’)
  4. PIN Encryption Key (if ATD PIN Encrypt Type = ‘01’, ‘03’, ‘04’ or ‘05’)

If the ATM is not AKDS-capable, the device handler does not download the RSA keys. It does, however, download the DES keys and then re-opens the ATM. The keys are downloaded in the following order.

1. MAC Encryption Key (if ATD MAC Support Type = ‘3’)
2. PIN Encryption Key (if ATD PIN Encrypt Type = ‘01’, ‘03’, ‘04’ or ‘05’)

NCS Screen / NCPCOM Utility

The Diebold 10XX/478X device handler now supports the following 9500 command. It is entered at the Network Control Supervisor (NCS) screen or the Network Control Point Communications (NCPCOM) utility.

- LOADALLKEYS

  This 9500 command is equivalent to the DCT's LOAD ALL KEYS command. It instructs the device handler to initiate a Load All Keys sequence. For more information refer to the LOAD ALL KEYS command in the preceeding DCT subsection.

  The syntax of this command is as follows.

  DELIVER PROCESS <process name>, "9500LOADALLKEYS <terminal ID>"

  <process name>: The symbolic name assigned to the Diebold 10XX/478X device handler process in the XPNET node.

  <terminal ID>: The terminal ID defined for the ATM in the ATM Terminal Data file (ATD).

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Diebold Cash Source Plus (CSP) 200 Triple DES

This section describes the implementation requirements associated with the BASE24-atm Diebold CSP 200 Triple DES enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Diebold CSP 200 Triple DES enhancement.

The standard Diebold CSP device handler BASE24-atm module has been modified to support triple DES encryption. To support triple DES, the ATM uses double-length keys. Prior to this enhancement, the security keys were stored in BASE24 and at the ATM as 16 hex characters or 8 bytes. For triple DES, double-length keys are needed by BASE24 and the ATM to store the keys as 32 hex characters or 16 bytes. Transaction Security Services (TSS) is required to support the new double-length keys.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Diebold CSP 200 Triple DES enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Software Impacts

All BASE24-atm Diebold CSP 200 device handler customers:

- Inline changes to AT60DCSP.DCSPG and DCSPS
- A new Diebold CSP 200 Manual (AT60DCSP.DCSPMAN)

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

The following records must be configured to support triple DES.

Channel Security Data Source (CSECD)

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Key Length of Exchange Key

The Key Length of the Exchange Key for the Diebold CSP 200 CSECD records must be set to Double to utilize Diebold CSP 200 triple DES using double-length keys.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Exchange Key (Header, Part 1-4, Check Digits)

The Exchange Key for the Diebold CSP 200 CSECD records must be modified to contain the new double-length key value.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Key Length of Inbound Key

The Key Length of the Inbound Key for the Diebold CSP 200 CSECD records must be set to Double to utilize Diebold CSP 200 triple DES using double-length keys.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Inbound Key (Header, Part 1-4, Check Digits)

The Inbound Key for the Diebold CSP 200 CSECD records must be modified to contain the new double-length key value.

Refer to the online help provided with the ACI Desktop user interface for detailed descriptions of this window and its fields.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Diebold CSP 200 devices supporting triple DES encryption must have new software installed and configured to use the triple DES encryption algorithm and double-length keys. Additionally the new double-length Master key must be manually entered at the device and the new working key must be downloaded.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Diebold Cash Source Plus, Cash Source Plus 100, Cash Source Plus 200 Programmer's Reference Manual (Part Number TP-820024-001G 09/2003)

Diebold EMV

This section describes the implementation requirements associated with the Diebold EMV enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Diebold EMV enhancement.

The standard BASE24-atm Diebold 10XX/478X device handler module has been modified to support EMV. In addition a new bindable extension module has been developed to handle the special processing required to support EMV.

The BASE24-atm Diebold Entry Program has also been modified to support EMV. A new ENTIBOLD object, version 6.1.12 (15nov04), is required.

Software Impacts

All BASE24-atm customers:

- Inline changes to AT60SRC.ATMFM and ATMMM

All BASE24-atm Diebold 10XX/478X device handler customers:

- Hooks in the main device handler module to EMV (AT601000.C1000S)
- Inline changes to AT601000.C100DDLS, C1000M and C1000G
- Additional procedures added to AT601000.C1000D
- Inline changes to the Diebold 10XX DCT requester and screen (AT601000.RQD10XXS and SCND10XX)
- Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X device handler customers taking the EMV enhancement:

- A new extension subvolume, AT60I100, with files for the Diebold 10XX/478X device handler EMV extension

All BASE24-atm Diebold Entry Program users:

- A new version of ENTIBOLD (version 6.1.12)

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNI100.

CUSTMACS Changes

The existing atm_dh_c1000_emv_on must be set to true to bind the EMV module to the Diebold 10XX/478X device handler.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes fields, within BASE24-atm files, relevant to EMV support. Field settings depend on desired configuration.

BASE24-atm Terminal Data Files (ATD)

Screen Name:	EMV Terminal Capabilities
Screen:	ATD Screen 21
DDL Name:	ATDS1.[atds1-core].EMV-TERM-CAP
	Indicates the card data input, Card Verification Method (CVM), and security capabilities of the terminal using a bit map to represent possible options.

The ATD has a separate screen 21 to configure the EMV terminal capabilities. For additional information regarding the setting of these fields refer to the ***BASE24-atm EMV Support Manual***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Diebold 10XX/478X Configuration File

The Diebold 10XX/478X Configuration File, AT601000.CNFGIX, has been modified to include five new EMV load groups:

- Load Group 400

  Load group 400 is the basic EMV load group. This load group is loaded on top of base load group 0.

- Load Group 404

  Load group 404 is the EMV Surcharge load group. If surcharging is implemented, this group is loaded on top of EMV load group 400 and load group 304 screens.

- Load Group 405

  Load group 405 is the EMV Fastcash load group. If Fastcash transactions are supported, this load group is loaded on top of EMV load group 400 and load group 5 screens.

- Load Group 406

  Load group 406 is the EMV Non-Currency Dispense (NCD) load group. If NCD has been configured into the system, this load group is loaded on top of EMV load group 400 and load group 500 screens.

- Load Group 424

  Load group 424 is the EMV PIN Change load group. If PIN change transactions are supported, this load group is loaded on top of EMV load group 400 and load group 24 screens.

There are two new configuration tables needed for EMV. These tables are accessed via the Diebold Entry Program (ENTIBOLD):

- EMV Application Data
- EMV Scheme Data

NOTE: Each table must be configured by the customer. The records for these tables are written to the CNFGIX file. Refer to the ***BASE24-atm EMV Support Manual*** for additional information about these tables and their values.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to ATD screen 21 (EMV data screen) must be added through the Security requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Device Control Terminal (DCT)

The following command has been added to the Diebold 10XX Commands Screen (screen 40) of the DCT.

- LOAD EMV DATA

When the operator places the cursor on this command and presses function key F1-ENTER, the DCT sends a command to the device handler to initiate a load of all EMV Application Data and EMV Scheme Data.

Operational Impacts:

New event messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

BASE24-atm EMV related response codes are now possible for transactions performed at Diebold ATMs supporting EMV. Operators should be trained to recognize the following new EMV related response codes on the TLF Perusal screen:

- 81 HSM parameter error
- 82 HSM failure
- 83 KEYI record not found

- 84 ATC check failure
- 85 CVR decline
- 86 TVR decline
- 87 ARQC failure
- 88 Fallback Decline

Vendor Impacts

Not applicable

Reference Documents

Diebold Agilis 91x/TCS Plus EMV Programmer's Guide, Version 1.0.0 (Part Number TP-820874-001A)

Diebold Agilis 91x/TCS Plus EMV User's Guide, (Part Number TP-820847-001, version A).

Diebold Agilis 91X Terminal Programming Manual (Part Number TP-820740-001, version A).

Diebold 91X Terminal Control Software (TCS), TCS Plus, and 91X TCS CSP Terminal Programming Manual (Part Number TP-799387-001, version O).

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Diebold Text to Speech (ADA) and Agilis 91x (Opteva) Features

This section describes the implementation requirements associated with supporting Diebold's Text to Speech feature and various Agilis 91x features.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of Diebold's Text to Speech feature and the various Agilis 91x features being supported in the BASE24-atm Diebold 10XX/478X device handler.

Standard BASE24-atm modules have been modified to support this enhancement. These modules include the Diebold Entry Program, the Diebold 10XX/478X device handler and the Receipt requester/server.

NOTE: ACI has been unable to fully test all of the new features being added due to not having hardware for all features at this time. ACI was unable to test the automatic envelope dispenser and the Text to Speech features. Please check with ACI prior to implementing either of these features.

Software Impacts

All BASE24-atm customers:

- Inline changes to AT60AFT.COBRCPT and SVRCPTS

All BASE24-atm Diebold 10XX/478X device handler customers:

- Inline changes to AT601000.C1000G and C1000S
- Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold Entry Program customers:

- A new ENTIBOLD object, version 6.1.12 (15nov04)

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNDIEB.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

BASE24-atm Receipt File (RCPT)

A new field type code of EC has been introduced with this enhancement. This field type is only allowed for Diebold 1000 devices (i.e., device type D1000).

The EC field type code is used to enable/disable text color printing. Diebold supports two color text printing only on their Opteva devices equipped with a thermal receipt and/or thermal statement printer. Two color text printing also requires special two-color paper.

Additional information on specific field type codes may be found in the ***BASE24-atm Diebold 10xx/478x Device Support Manual***. Information on use of the RCPT file may be found in the ***BASE24-atm Files Maintenance Manual***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

ACI anticipates having to make changes to the Diebold configuration data file (CNFGIX) to support Diebold's automatic envelope dispense feature. Outstanding issues with Diebold hardware prevented ACI from being able to determine these changes at this time. Prior to implementing this feature, please consult with ACI.

Support (via the Diebold Entry Program) was added for the new Text to Speech state (AU). Customers wanting to take advantage of this state will need to make their own configuration changes. Support was also added (via the Diebold Entry Program) to support new voice prompting and new voice keypad feedback commands available in TCS Plus 1.3 and Agilis 91x. Again, customers wanting to take advantage of these new features will need to make their own configuration changes.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

<u>Operational Impacts</u>

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

The appropriate level of TCS Plus or Agilis must be installed on the device to take advantage of the above enhancements. Customers running prior versions of TCS will still be supported but will not be able to take advantage of any of the new enhancements.

Reference Documents

Diebold Agilis 91X Status Reference Manual (Part Number TP-820741-001, version B).

Diebold Agilis 91X Terminal Programming Manual (Part Number TP-820740-001, version A).

Diebold 91X Terminal Control Software (TCS), TCS Plus, and 91X TCS CSP Terminal Programming Manual (Part Number TP-799387-001, version O).

EMV 2000 Session Key Derivation

For further information, refer to **EMV 2000 Session Key Derivation** in Section 4 where the impacts of this enhancement are described.

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Enhanced ATM Status Reporting

This section describes the implementation requirements associated with the Enhanced ATM Status Reporting enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Enhanced ATM Status Message enhancement.

The BASE24-atm Diebold 10XX/478X device handler and the BASE24-atm NCR 5XXX device handler modules have been modified to support the Enhanced ATM Status Reporting enhancement.

Software Impacts

All BASE24-atm Diebold 10XX/478X device handler customers:

- Inline changes to AT601000.C1000G and C1000S
- Updated event (log) message documentation BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X EMV device handler customers:

- Inline changes to AT60I100.C100EMVS

All BASE24-atm Diebold 10XX/478X SSB device handler customers:

- Inline changes to AT60D100.C100SSDS

All BASE24-atm Diebold 10XX/478X Merchant Banking Center (MBC) device handler customers:

- Inline changes to AT60X100.C100MBCS

All BASE24-atm NCR 5XXX device handler customers:

- Inline changes to AT60N50.N50G and N50S
- Updated event (log) message documentation BA60LOGM.AT5070

All BASE24-atm NCR 5XXX EMV device handler customers:

- Inline changes to AT60IN50.N50EMVS

All BASE24-atm NCR 5XXX SSB device handler customers:

- Inline changes to AT60DN50.N50SSDS

A complete list of the files affected by this enhancement can be found in the appropriate SCN file on BA60UD08. SCNEASMD contains the list of files for the Diebold 10XX/478X and SCNEASMN contains the list for the NCR 5XXX.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to the LCONF to support this enhancement. Modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-DISPLAY-NATIVE-STATUS-MSG

This param is used by the Diebold 10XX/478X and the NCR 5XXX device handlers. It indicates for solicited and unsolicited status messages whether or not to log the status message in its native format to EMS. Valid values are Y and N, with a default value of N.

For the NCR 5XXX device handler if this parameter is set to Y, any solicited status message with a Status Descriptor of ‘8’ or ‘F’ will have fields ‘b’ through ‘g’ logged to EMS, and any unsolicited status message will have fields ‘b’ through ‘e’ logged to EMS.

For the Diebold 10XX/478X device handler if this parameter is set to Y, any solicited status message with a Status Descriptor of ‘8’ or ‘J’ and any unsolicited status message will have all fields in the message starting with the ‘Solicited/Unsolicited Message’ field through the ‘Status’ field logged to EMS.

The native status message will be appended to existing event message. The format for the appended message will be:

[NATIVE]NNDCnativemessage	for the NCR 5XXX device handler
[NATIVE]D91Xnativemessage	for the Diebold 10XX/478X device handler

where nativemessage consists of the fields from the status message as described above. All field separators in the native message will be displayed to EMS as an * (an asterisk).

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Users may need updated software for their ATM monitoring systems in order to process the additional native information provided in EMS messages with this enhancement.

Reference Documents

Not applicable

Forward Declined Advices to Host

For further information, refer to **Forward Declined Advices to Host** in Section 4 where the impacts of this enhancement are described.

ACI Worldwide Inc.

Mobile Top-Up Integration

This section describes the implementation requirements associated with the BASE24-atm Mobile Top-Up Integration enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the BASE24-atm Mobile Top-Up enhancement.

Standard BASE24 modules have been modified to support this enhancement. These modules include the Diebold 10XX/478X device handler, the NCR 5XXX device handler and OMF Extract.

A new BASE24-atm module has been added to support this enhancement. This module is the BASE24-atm Transaction Context Manager.

Some parameters/options/fields listed in this section are only required when using the definitions-based model for mobile top-up processing, and these are noted as such. All other parameter/options/fields are required regardless of the model being used.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS and FLGSDOC
- Inline changes to Transaction Based Pricing (TBP) reports (BA60TBP.ATDRPTS, BAMRPTS, DDLFMTAF, DRPTG, RPTLIBS, TBPUPDT)
- Inline changes to BA60DDL.BADDLM, BADDLMM, DDLBATKN, DDLFOMF, DDLGSTM (comments only)
- New files BA60DDL.DDLFMOF and DDLFSTRF
- Inline changes to BA60AFT.BAAFTM, BAAFTMM, CCAFMLM, COBTKN, MEGATBL, SECTBL and SVMHLPS
- A new requester/server (STRF) and its files on BA60AFT
- Inline changes to BA60EXT.OMFXG, OMFXM, OMFXS
- Inline changes to BA60SRC.BACOUTLS, BASEFM, BASEM, BASEMM, BATKNCVS, BATKNID and COBNAMES
- New files and changes to the existing Extended Memory Table Build (EMTB) to accommodate the STRF (BA60EMTB.BEMTDS, BEMTBM, BEMTBMM, BEMTBLM, EMTBRUN, STREMTTG, STREMTTM, STREMTTS)
- Inline changes to BA60DCT.RQDECTS and SVDPCTS

All BASE24-atm customers:

- Inline code changes to AT60AFT.COBRCPT, RQTLFS, SCRNTLF and SVTLFS
- Inline changes to AT60SRC.ATCOUTLS, ATMFM, ATMM, ATMMM and RCPTDEFS

All BASE24-atm BIC ISO Host Interface customers:

- Inline changes to SW60ATBI.ATBICIS

All BASE24-atm ISO Host Interface customers:

- Inline changes to AT60HISO.ATHISOS

All BASE24-atm Diebold 10XX/478X or NCR 5XXX device handler customers:

- Hooks in the main device handler module to support Mobile Top-Up (AT601000.C1000S, AT60N50.N50S)
- Inline changes to AT601000.C1000G and AT60N50.N50G
- Additional procedures in AT601000.C1000D and AT60N50.N50D
- Inline changes to AT601000.C1000M and AT60N50.N50M
- New configuration files AT601000.CNFGIX and AT60N50.CNFG5070
- Updated configuration documentation files AT601000.CN100DOC and AT60N50.CNN50DOC
- Updated event (log) message manuals BA60LOGM.AT1000 and AT5070

All BASE24-atm customers taking the Mobile Top-Up enhancement:

- New subvolume, AT60TCM, with files for the BASE24-atm Transaction Context Manager
- A new event (log) message manual BA60LOGM.ATTCM

All BASE24-atm Diebold 10XX/478X customers taking the Mobile Top-Up enhancement:

- New subvolume, AT60T100, with files for the Diebold 10XX/478X device handler Mobile Top-Up extension

All BASE24-atm NCR 5XXX customers taking the Mobile Top-Up enhancement:

- New subvolume, AT60TN50, with files for the NCR 5XXX device handler Mobile Top-Up extension

All BASE24-atm NCR 5XXX EMV customers taking the Mobile Top-Up enhancement:

- Inline changes to AT60IN50.N50EMVS

All BASE24-atm customers taking the definitions-based model of Mobile Top-Up:

- New subvolume, AT60TTCM, with files for the Transaction Context Manager Definitions-based model extension

- New infrastructure subvolume, BA60TMTU, with files containing the base Definitions-based model for Mobile Top-Up

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD08.SCNMTOP.

CUSTMACS Changes

Various Make flags have been added to the CUSTMACS file for Mobile Top-Up Integration.

Set the following flag to true for the BASE24-atm Transaction Context Manager.

- atm_tcm_on

Set the following flag to true if the Transaction Context Manager will support the definitions-based Mobile Top-Up Integration enhancement.

- atm_tcm_mtu_on

Set the following flag to true to bind the Mobile Top-Up module to the Diebold 10XX/478X device handler.

- atm_dh_c1000_mtu_on

Set the following flag to true to bind the Mobile Top-Up module to the NCR 5XXX device handler.

- atm_dh_n50_mtu_on

The following flag should be set to true only if the definitions-based model of mobile top-up is being used. NOTE: The atm_on, atm_tcm_on and atm_tcm_mtu_on flags must also be set to true.

- mtu_on

LCONF Assign Changes

This section describes the assign records that must be added to or already exist in the LCONF to support this enhancement. Modifications for the new assigns are provided in the LCONFBA and LCONFAT files located on BA60MISC subvolume.

MOF

The fully qualified file name of the Mobile Operator File (MOF). This assign is required for definitions-based mobile top-up processing.

STRF

The fully qualified file name of the Split Transaction Routing File (STRF). This assign is required for the Mobile Top-Up Integration enhancement.

STRFEMT

The fully qualified file name of the Split Transaction Routing File Extended Memory Table (STRFEMT). This assign is required for the Mobile Top-Up Integration enhancement.

TSRF

The fully qualified file name of the Transaction Code/Subtype Relationship File (TSRF). This assign was added in Release 6.0 version 4 and is required for the Mobile Top-Up Integration enhancement.

The following LCONF assigns already exist. They are used by the BASE24-atm Transaction Context Manager and are required for the Mobile Top-Up Integration enhancement.

- LMCF
- TKN
- SPANCTL
- TLF

LCONF Param Changes

This section describes the param records that must be added to or already exist in the LCONF to support this enhancement. Modifications for the new parameters are provided in the LCONFBA and LCONFAT files located on BA60MISC subvolume.

The following params are the same for both device handlers. Different param records can be specified for each device handler process if desired.

ATM-DH-RCPT-PHONE-NUM-FRMT

This param is used by the Diebold 10XX/478X and NCR 5XXX device handlers. It specifies whether the phone number printed on the ATM receipt will be formatted. Valid values are as follows:

0 = Do not format the phone number on the ATM receipt (default)

1 = Format the phone number on the ATM receipt in the following format: xxx-xxx-xxxx.

CARRIER-AMTS-VT*identifier*

This param is used by the Diebold 10XX/478X and NCR 5XXX device handlers and should only be added for the definitions-based model of mobile top-up. It identifies the eight amounts supported for this mobile operator.

The *identifier* is a two-digit numeric value, 00 – 29, representing the voucher type. It is also used to associate the Operator IIN, name, mobile phone operator ID, and valid amounts for each carrier.

Valid values must be formatted with eight amounts and each amount must be six digits separated by semicolons.

Example: 000030;000050;000075;000000;000000;000000;000000;000000
Represents $30, $50, and $75 as valid amounts for this carrier.

CARRIER-VT*identifier*

This param is used by the Diebold 10XX/478X and NCR 5XXX device handlers and should only be added for the definitions-based model of mobile top-up. It identifies the four-digit mobile phone operator ID.

The *identifier* is a two-digit numeric value, 00 – 29, representing the voucher type. It is also used to associate the Operator IIN, name, mobile phone operator ID, and valid amounts for each carrier.

The value of the param links a mobile operator with an entry in the Mobile Operator File (MOF) and enables the mobile operator to be identified for the transaction.

MOBILE-NUM-MAX-LGTH

This param, used by the Diebold 10XX/478X and NCR 5XXX device handlers, specifies the maximum length allowed for the mobile phone number entered by the cardholder. Valid values are 0 - 15. The default value is 11.

MOBILE-NUM-MIN-LGTH

This param, used by the Diebold 10XX/478X and NCR 5XXX device handlers, specifies the minimum length allowed for the mobile phone number entered by the cardholder. Valid values are 0 - 15. The default value is 10.

MOBILE-OPERATOR-ID-VT*identifier*

This param is used by the Diebold 10XX/478X and NCR 5XXX device handlers and should only be added for the definitions-based model of mobile top-up. It specifies the mobile phone operator IIN (7 digits) plus the mobile operator name (up to 20 characters) associated with the mobile phone operator.

The *identifier* is a two-digit numeric value, 00 – 29, representing the voucher type. It is also used to associate the Operator IIN, name, mobile phone operator ID, and valid amounts for each carrier.

Valid values must be formatted with a seven-digit operator IIN followed immediately by a mobile operator name.

Example: 1200991CARRIERNAME

<u>SPLIT-TXN-RTG-DEST</u>

This param, used by the Diebold 10XX/478X and NCR 5XXX device handlers, specifies the destination of the BASE24-atm Transaction Context Manager process. The BASE24-atm Transaction Context Manager is required because of the split transaction routing of mobile top-up transactions (i.e., a Funds part and a mobile I/F part). This parameter is required for the Mobile Top-Up Integration enhancement.

<u>STRFEMT-MAX-TBL-SIZE</u>

The maximum size (in megabytes) of the Split Transaction Routing File Extended Memory Table (STRFEMT). This param is used to set the amount of memory allocated by the extended memory table build utility for the table. Valid values are 1 – 127 MB. The default value is 127 MB.

<u>STRFEMT-WARM*identifier*</u>

A param containing the symbolic name of a process to be warmbooted when warmbooting the STRFEMT. Processes that can be warmbooted for the STRFEMT are the BASE24-atm Transaction Context Manager processes. This param indicates a process that will receive an EMT warmboot command from the Device Control Terminal (DCT). The Device Control Terminal Server process retrieves all STRFEMT-WARM*identifier* params when a network control operator issues a command to reinitialize this specific EMT. A maximum of 500 processes can be EMT warmbooted from the DCT.

Each STRFEMT-WARM param contains the name of one process. When multiple processes must be notified, this param can include an optional identifier to specify params used for different processes in the same logical network. The optional identifier is usually a number, but can be any unique set of characters that has not already been declared by another STRFEMT-WARM*identifier* param. For example, STRFEMT-WARM01 could contain the P1A^ATCM^01 process and STRFEMT-WARM02 could contain the P1A^ATCM^02 process.

Another configuration change required for the Mobile Top-Up Integration enhancement is the adding of the BASE24-atm Transaction Context Manager process name to the Settlement Notify List params in the LCONF.

ATM-SETL-*NOTIFYidentifier*

The following LCONF params already exist. They are used by the BASE24-atm Transaction Context Manager and are required for the Mobile Top-Up Integration enhancement.

- LOGICAL-NET
- ATM-LN-CUTOVER
- ATM-NOTIFY-INTERVAL
- ATM-NOTIFY-PERCENTAGE

Database Configuration

BASE24-atm Receipt File (RCPT)

- Receipt Types

New receipt types of PPTU, VCHR, and VT00 through VT29 have been introduced with this enhancement. These receipt types are only allowed for the Diebold 10XX/478X and NCR 5XXX devices.

The PPTU receipt type is used for real-time mobile top-up transactions that do not involve a voucher. The VCHR receipt type is used for mobile top-up transactions involving a voucher. The receipt types VT00 through VT29 are used for the definitions-based model of mobile top-up transactions to provide a unique voucher receipt record for each mobile carrier supported.

- Field Type Codes

The following existing field type code has been altered to allow mobile top-up functionality.

TT	Print Transaction Type The TT field type code is used for printing the transaction type description on the receipt. If this is a mobile top-up transaction, the value is retrieved from the TN50NAMS file (for the NCR 5XXX device handler) or the T100NAMS file (for the Diebold 10XX/478X device handler). If this is not a mobile top-up transaction, the device handler will search other bindable modules or use the transaction type description found in the vanilla names file (N50NAMS or C100NAMS).

Several new field type codes have been introduced with this enhancement. These field type codes are only allowed for the Diebold 10XX/478X and NCR 5XXX devices.

V0	Activation Number The V0 field type code is used for printing the activation code from the Pre-Pay Top-Up Token.
V1	Stock Number The V1 field type code is used for printing the stock number from the Inventory Voucher Token.
V2	Shelf Date The V2 field type code is used for printing the voucher shelf date. It is populated from the shelf date field in the Pre-Pay Voucher Receipt Token or, if this is an inventory voucher transaction, it is populated from the stock expiration date field in the Inventory Voucher Token. When formatting the date for the receipt, the DATE-DISPLAY-TYPE value from the LCONF is used to determine how to format the data.
V3	Phone Number The V3 field type code is used for printing the phone number from the Pre-Pay Top-Up Token. It will be formatted using the ATM-DH-RCPT-PHONE-NUM-FRMT value in the LCONF.
V4	Unique Transaction Number The V4 field type code is used for printing the reference number from the Pre-Pay Top-Up Token.
V5	Mobile Operator Marketing Message The V5 field type code is used for printing the operator message from the Pre-Pay Receipt Token. The operator message will be printed on the receipt for the length indicated in the length field of the token.
V6	Mobile Operator Name The V6 field type code is used for printing the mobile operator name. If performing definitions-based mobile top-up it is from the LCONF param MOBILE-OPERATOR-ID-VT*identifier* for the operator IIN of the transaction. Otherwise, it is the Carrier Name field within the Pre-Pay Response Token.
V7	Mobile Operator Phone Number The V7 field type code is used for printing the carrier phone number from the Pre-Pay Response Token. It is not formatted by BASE24.
VA	Mobile Phone Balance The VA field type code is used for printing the consumer's new mobile phone balance after a successful real-time mobile top-up transaction.
VB	Batch ID The VB field type code is used for printing the batch id from the Pre-Pay Voucher Receipt Token.
VE	Expiration Days The VE field type code is used for printing the expiration days from the Pre-Pay Voucher Receipt Token.

VI	Control ID Number The VI field type code is used for printing the control number from the Pre-Pay Voucher Receipt Token.
VN	Voucher Number The VN field type code is used for printing the voucher number from the Pre-Pay Voucher Receipt Token.
VS	Expiration Date The VS field type code is used for printing the expiration date for a real-time mobile top-up transaction. It is populated from the expiration date field in the Pre-Pay Online Receipt Token. When formatting the date for the receipt, the DATE-DISPLAY-TYPE value from the LCONF is used to determine how to format the data.
VT	Tax Amount The VT field type code is used for printing the tax amount from the Pre-Pay Response Token.
VX	Requested amount minus tax amount. The VX field type code is used for printing the requested amount minus the tax amount. This value reflects the actual amount added to the consumer's mobile phone.

NOTE: Care should be taken when configuring the receipts as to avoid exceeding the maximum size of the print buffer in the device message. If this should happen on the NCR 5XXX device, the customer should review the use of "large message mode" (refer to the ATD flag on screen 7).

Additional information on specific field type codes can be found in the ***BASE24-atm Diebold 10XX/478X*** or ***BASE24-atm NCR 5XXX Device Support Manual***. Information on use of the RCPT file may be found in the ***BASE24-atm Files Maintenance Manual***.

Surcharge File (SURF)

If the mobile operator authorizer prohibits the surcharging of mobile top-up transactions, it is the BASE24 customer's responsibility to configure the SURF to prevent the generation of surcharges for these transactions.

Surcharging can be prohibited by excluding the surcharge by Subtype (i.e., the Boston Communications Group Interface utilizes a subtype of BBT0). Therefore, if the SURF is configured such that cash withdrawal and/or non-currency dispense transactions are to be surcharged, the SURF will need to be updated to prevent the surcharging of subtype BBT0 (BCGI Mobile Top-Up transactions).

It is the acquiring institution's responsibility to enforce their contractual agreement with the mobile operator authorizer and to ensure the SURF records do not allow the assessment of surcharge fees for these transactions. If the BASE24 customer does not take this action, ACI is not responsible for any fine or other consequences imposed by the mobile operator authorizer.

Token file (TKN)

The following BASE24 base tokens have been added for this enhancement:

- The Split Transaction Routing Token (token ID “BR”) carries data that allows BASE24 to route multiple transaction requests related to a single cardholder request. It also allows BASE24 to identify and merge the multiple responses received into a single response destined for the cardholder.
- The Pre-Pay Switch Token (token ID “BS”) contains data used by an interface for a mobile top-up transaction. The token is variable in length and its content varies based upon the interface.
- The Pre-Pay Response Token (token ID “BT”) contains data used to facilitate the pre-pay notification and selection process required by the ATM / Device Handler and the cardholder.
- The Pre-Pay Selection Token (token ID “BU”) contains data that must be presented to the cardholder for further selection. The content of this token varies based upon the type of response received from the Host.
- The Pre-Pay Voucher Receipt Token (token ID “BV”) contains information associated with a mobile top-up voucher transaction that BASE24 customers may print on the receipt.
- The Pre-Pay Online Receipt Token (token ID “BW”) contains information associated with a mobile top-up online (real-time) transaction that BASE24 customers may print on the receipt.
- The Pre-Pay Original Data Token (token ID “BX”) stores original data from a transaction that may be modified during split routing and restored for processing.
- The Inventory Voucher Token (token ID “N8”) contains information associated with the purchase of a mobile top-up voucher from inventory.

The following existing BASE24 tokens are utilized for this enhancement:

- The Non-Currency Dispense Token (token ID “A5”)
- The Pre-Pay Top-Up Token (token ID “BI”)
- The Transaction Subtype Token (token ID “BM”)

If the tokens for this enhancement should not be logged to the TLF or the ILF, the appropriate token records must be configured in the Token file.

If the tokens for this enhancement are to be extracted with the TLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

Transaction Log File (TLF)

Field Name: MOBILE PHONE NUM
Screen: TLF screen 2
DDL Name: PRE-PAY-TOP-UP-TKN.PHN-NUM

The mobile phone number to receive the top-up. This number is provided by the cardholder at the acquiring device.

Field Name: TAX AMOUNT
Screen: TLF screen 2
DDL Name: PRE-PAY-RESP-TKN.TAX-AMT

The tax amount applied to the top-up transaction.

Field Name: TOP-UP REF NUM
Screen: TLF screen 2
DDL Name: PRE-PAY-TOP-UP-TKN.REF

The reference number issued by the mobile operator in top-up response messages. This number is unique for each transaction.

Field Name: LIS5 CODE
Screen: TLF screen 2
DDL Name: PRE-PAY-TOP-UP-TKN.RESP-CDE

The mobile operator’s response code. Alternatively, this field may contain a generic response code agreed upon between the mobile operator and the service provider. This code is converted by the BASE24 Mobile Operator Interface process into an internal BASE24 response code that is carried in the STM response code field.

Field Name: MNO CODE
Screen: TLF screen 2
DDL Name: PRE-PAY-TOP-UP-TKN.APPRV-CDE

The approval code if supplied by the mobile operator in the response message.

New Data Files

Mobile Operator File (MOF)

The Mobile Operator File (MOF) is an optional file and contains one record for each telecommunications provider supplying mobile top-up services for its customers. The MOF defines details of the mobile operators and the services provided and is currently utilized only in the definitions-based model of mobile top-up.

The MOF contains the following fields:

Field Name: OPERATOR ID
Screen: MOF screen 1
DDL Name: MOF. PRIKEY.MOP-ID

The user assigned identifier for the mobile telephone operator providing application services in the system.

Field Name: FIID
Screen: MOF screen 1
DDL Name: MOF.FIID

The FIID of the financial institution associated with the mobile telephone operator.

Field Name: OPERATOR IIN
Screen: MOF screen 1
DDL Name: MOF.ALTKEY.MOP-IIN

The issuer identification number (IIN) assigned to the mobile telephone operator.

Field Name: OPERATOR NAME
Screen: MOF screen 1
DDL Name: MOF.OPER-NAM

The name of the mobile telephone operator associated with this record. The value placed in this field may appear on customer receipts.

Field Name: CARD TYPE
Screen: MOF screen 1
DDL Name: MOF.ALTKEY.TOP-UP-CRD-TYP

A code identifying the type of card used to initiate the mobile top-up transaction. Codes used in this field are either reserved for BASE24 products or are user-defined. A description of the card type entered is displayed to the right of the CARD TYPE field.

Field Name: CUSTOMER PHONE NUMBER LOOKUP
Screen: MOF screen 1
DDL Name: MOF.MCRF-LOOKUP

A flag indicating whether or not the customer's phone number in the Mobile Customer Registration File (MCRF) is retrieved. Valid values are as follows:

Y – Yes, retrieve the number from the MCRF

N – No, do not retrieve the number from the MCRF

NOTE: This field is not currently being used by BASE24-atm.

Field Name: PAN LENGTH
Screen: MOF screen 1
DDL Name: MOF.PAN-LGTH

A length of the primary account number (PAN) associated with the mobile operator's issuer identification number (IIN).

Field Name: EXPECT RVSL RESPONSE
Screen: MOF screen 1
DDL Name: MOF.REV-RESP-EXPECTED

A flag indicating whether the Transaction Context Manager process expects a response to reversal messages sent to the Mobile Operator Interface process. If this flag is set to Y, the Transaction Context Manager process waits for the reversal response before reversing the funds transaction or logging the 0420 reversal to the TLF. Valid values are as follows:

Y – Yes, expect reversal responses (default)

N – No, do not expect reversal responses

Field Name: REFUND TIMER
Screen: MOF screen 1
DDL Name: MOF.REFUND-TIMER

The maximum number of seconds the Mobile Operator Interface process allows a refund transaction. Valid values are 0 – 99999.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: REVERSE FUNDS ALWAYS
Screen: MOF screen 1
DDL Name: MOF.REV-AUTH-TXN

A flag indicating whether the Transaction Context Manager process sends a reversal message to the Authorization process. If this flag is set to Y, the EXPECT RVSL RESPONSE flag is set to Y. When the Transaction Context Manager receives a declined reversal response message from the Mobile Operator Interface process, the Transaction Context Manager process generates and sends a reversal message to the Authorization process based upon the value in this field. Valid values are as follows:

Y – Yes, send a reversal message (default)

N – No, do not send a reversal message

Field Name: RETAILER ID
Screen: MOF screen 1
DDL Name: MOF.MOB-OPER-RTLR-ID

The mobile operator retailer identifier used to identify the mobile operator when settling top-up transactions with the Card Management System.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: REVERSE ON TIMEOUT
Screen: MOF screen 1
DDL Name: MOF.REV-ON-TIMEOUT

A flag indicating whether the Transaction Context Manager process generates and sends a reversal to the Mobile Operator Interface process in the event of a timeout. If the Mobile Operator Interface timer expires while waiting for a response from the mobile operator, it notifies the Transaction Context Manager process of the timeout condition. The Transaction Context Manager then generates and sends a reversal to the Mobile Operator Interface process based upon the value in this field. Valid values are as follows:

Y – Yes, generate a reversal

N – No, do not generate a reversal (default)

Field Name: MOBILE REVERSALS SUPPORTED
Screen: MOF screen 1
DDL Name: MOF.MOB-REV-SPPT

A flag indicating whether the mobile operator supports receiving reversals. This field is used by the Transaction Context Manager process. Valid values are as follows:

Y – Yes, the mobile operator supports receiving reversals (default)

N – No, the mobile operator does not support receiving reversals

Field Name: CHECK CURRENCY CODE
Screen: MOF screen 2
DDL Name: MOF.CHK-CURR-CDE

A flag indicating whether the currency code in the transaction is checked against the value in the CURRENCY CODE field before the transaction is routed to the Mobile Operator Interface process.

If amount verification is to be performed, this flag must be set to Y in order for the CHECK TOP-UP AMOUNTS, VALID TOP-UP AMOUNTS, TOP-UP AMOUNT MIN, TOP-UP AMOUNT MAX, and TOP-UP AMOUNT MULTIPLE fields to be verified.

If the mobile operator is capable of supporting multiple currencies, this value should be set to N. Valid values are as follows:

Y – Yes, the currency code is checked

N – No, the currency code is not checked (default)

Field Name: CURRENCY CODE
Screen: MOF screen 2
DDL Name: MOF.CURR-CDE

A code indicating the currency of the valid top-up amount fields. Valid values are listed in the ISO 4217:1995 standard, ***Codes for the Representation of Currencies and Funds***. If the CHECK CURRENCY CODE field is set to Y, the currency of the transaction is checked against the value in this field.

A description of the code entered is displayed to the right of the

CURRENCY CODE field.

Field Name: CHECK TOP-UP AMOUNTS
Screen: MOF screen 2
DDL Name: MOF.CHK-TOP-UP-AMT

A code indicating whether the Transaction Context Manager checks the top-up amount and how the amount is validated. If the CHECK CURRENCY CODE flag is set to a Y, the Transaction Context Manager process uses the value in this field to validate amounts. A description of the code entered is displayed to the right of the CHECK TOP-UP AMOUNTS field. Valid values are as follows:

0 – Do not validate the amount (default)

1 – Validate the specific amount
The amount of the transaction must equal one of the VALID TOP-UP AMOUNTS.

2 – Validate the range and multiple
The amount of the top-up transaction must be greater than the value in the TOP-UP AMOUNT MIN field, less than the value in the TOP-UP AMOUNT MAX field, and must be a multiple of the value in the TOP-UP MULITPLE field.

Field Name: VALID TOP-UP AMOUNTS
Screen: MOF screen 2
DDL Name: MOF.VALID-TOP-UP-AMT

The valid top-up amounts supported by the mobile operator. Up to twenty different top-up amounts can be configured. If the CHECK CURRENCY CODE flag is set to a Y, the Transaction Context Manager process validates the transaction amount against the amounts in these fields if the CHECK TOP-UP AMOUNTS flag is set to a value of 1. An amount field set to zero is not checked. Whole amounts must be entered in these fields.

Field Name: TOP-UP AMOUNT MIN
Screen: MOF screen 2
DDL Name: MOF.MIN-TOP-UP-AMT

The minimum amount (in whole currency units) allowed for a top-up transaction. If the CHECK CURRENCY CODE flag is set to a Y and the CHECK TOP-UP AMOUNTS flag is set to the value of 2, the Transaction Context Manager process validates that the top-up transaction amount is greater than the amount in this field.

Field Name: TOP-UP AMOUNT MAX
Screen: MOF screen 2
DDL Name: MOF.MAX-TOP-UP-AMT

The maximum amount (in whole currency units) allowed for a top-up transaction. If the CHECK CURRENCY CODE flag is set to a Y and the CHECK TOP-UP AMOUNTS flag is set to the value 2, the Transaction Context Manager process validates that the top-up transaction amount is less than the amount in this field.

Field Name: TOP-UP AMOUNT MULTIPLE
Screen: MOF screen 2
DDL Name: MOF.TOP-UP-MULTIPLE

The standard increment (in whole currency units) over the minimum amount that can be approved for a top-up transaction. If the CHECK CURRENCY CODE flag is set to a Y and the CHECK TOP-UP AMOUNTS flag is set to the value 2, the Transaction Context Manager process validates that the top-up transaction amount is a multiple of the value in this field.

Field Name: TAX CURRENT START DATE
Screen: MOF screen 2
DDL Name: MOF.VAT-CURRENT.STRT-DAT

The start date (YYMMDD) of the current tax rate.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: TAX NEXT START DATE
Screen: MOF screen 2
DDL Name: MOF.VAT-NEXT.STRT-DAT

The start date (YYMMDD) of the next tax rate.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: TAX CURRENT TIME
Screen: MOF screen 2
DDL Name: MOF.VAT-CURRENT.STRT-TIM

The start time (HH:MM based on a 24 hour clock) of the current tax rate.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: TAX NEXT TIME
Screen: MOF screen 2
DDL Name: MOF.VAT-NEXT.STRT-TIM

The start time (HH:MM based on a 24 hour clock) of the next tax rate.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: TAX CURRENT RATE
Screen: MOF screen 2
DDL Name: MOF.VAT-CURRENT.RATE

The percentage of the current tax rate.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: TAX NEXT RATE
Screen: MOF screen 2
DDL Name: MOF.VAT-NEXT.RATE

The percentage of the next tax rate.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: PRODUCT NAME
Screen: MOF screen 3
DDL Name: MOF.PROD-NAM

The product name of the mobile telephone operator.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: SOLUTION PROVIDER ID
Screen: MOF screen 3
DDL Name: MOF.SOL-PROV-ID

The solution provider identifier used by Mobile Operator Interfaces when building the transaction service ID.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: ALLOWED MANUALLY KEYED
Screen: MOF screen 3

DDL Name: MOF.ALLOW-MANUAL

A flag indicating whether the mobile operator allows manually entered transactions. Valid values are:

Y – Yes, manually keyed transactions are allowed

N – No, manually keyed transactions are not allowed

NOTE: This field is not currently being used by BASE24-atm.

Field Name: TOP-UP AUTHORIZER
Screen: MOF screen 3
DDL Name: MOF.TOP-UP-AUTH-TYP

A flag indicating the type of authorizer used for top-up transactions. A text description is immediately displayed following the field. Valid values are:

0 – Inventory stock manager

1 – Online Telco

NOTE: This field is not currently being used by BASE24-atm.

Field Name: SEND CONFIRMATION
Screen: MOF screen 3
DDL Name: MOF.SND-CONF

A flag indicating whether the Transaction Context Manager process sends a confirmation to the funds authorizer at the end of the transaction. Valid values are:

Y – Yes, a confirmation is sent

N – No, a confirmation is not sent

NOTE: This field is not currently being used by BASE24-atm.

Field Name: LOG FUNDS
Screen: MOF screen 3
DDL Name: MOF.LOG-FUNDS

A flag indicating whether the Transaction Context Manager process logs the 0210 response from the funds authorizer and the 0420 reversal sent to the funds authorizer to the TLF. Valid values are:

Y – Yes, log funds transactions

N – No, do not log funds transactions

NOTE: This field is not currently being used by BASE24-atm.

Field Name: SEND TO NOTIFY
Screen: MOF screen 3
DDL Name: MOF.SND-TO-NTFY

A flag indicating whether the Transaction Context Manager process sends a 0220 confirmation message to the Notify Interface process. If the Notify Interface process receives the 0220 confirmation message, a SMS message is then generated and sent to the customer. Valid values are:

Y – Yes, send the confirmation message

N – No, do not send the confirmation message

NOTE: This field is not currently being used by BASE24-atm.

Field Name: NOTIFY SERVICE
Screen: MOF screen 3
DDL Name: MOF.NTFY-SERVICE

The name of the Notify Interface process or the Notify service. If the SEND TO NOTIFY flag is set to Y, this field is required.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: LINE 1
Screen: MOF screen 4
DDL Name: MOF.DFLT-RCPT-MSG-LINE[1]

The first line of the default receipt message.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: LINE 2
Screen: MOF screen 4
DDL Name: MOF.DFLT-RCPT-MSG-LINE[2]

The second line of the default receipt message.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: LINE 3
Screen: MOF screen 4
DDL Name: MOF.DFLT-RCPT-MSG-LINE[3]

The third line of the default receipt message.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: LINE 4
Screen: MOF screen 4
DDL Name: MOF.DFLT-RCPT-MSG-LINE[4]

The fourth line of the default receipt message.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: LINE 5
Screen: MOF screen 4
DDL Name: MOF.DFLT-RCPT-MSG-LINE[5]

The fifth line of the default receipt message.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: GENERIC MESSAGE USE
Screen: MOF screen 4
DDL Name: MOF.DFLT-RCPT-MSG-FLG

A flag indicating whether the generic message identified in the fields above is printed on the receipt. The generic message can be used alone, or with the mobile operator message contained in the Pre-Pay Receipt Token. A text description is immediately displayed to the right of the field. Valid values are as follows:

0 – Do not use this message

1 – Use this message, regardless of the presence of the mobile operator message

2 – Use this message; do not use the mobile operator message

3 – Use this message only if the mobile operator message is not present

NOTE: This field is not currrently being used by BASE24-atm.

Field Name: GENERIC MESSAGE LAST CHANGED
Screen: MOF screen 4
DDL Name: MOF.LAST-DFLT-RCPT-MSG-TS

A timestamp identifying the date and time the generic message was last updated.

NOTE: This field is not currently being used by BASE24-atm.

Split Transaction Routing File (STRF)

The Split Transaction Routing File (STRF) contains one record for each Transaction Subtype in the BASE24 network that requires unique routing. The STRF enables the routing of transaction requests to two or more destinations based upon information obtained from a device for a single transaction.

The STRF contains the following fields:

Field Name: TRANSACTION SUBTYPE
Screen: STRF screen 1
DDL Name: STRF. PRIKEY.TXN-SUBTYP

The transaction subtype to which this record's information applies. A text description of the transaction subtype is retrieved from the Transaction Code/Subtype Relationship File (TSRF) and is displayed to the right of this field.

Field Name: FIID
Screen: STRF screen 1
DDL Name: STRF. PRIKEY.FIID

The acquiring institution identifier.

Field Name: ROUTING HIERARCHY
Screen: STRF screen 1
DDL Name: STRF. RTE-HRCHY

The routing hierarchy determines how BASE24 communicates with multiple authorization destinations for a single transaction. A text description is displayed to the right of the field. Valid values are:

B0 – Sequential; Funds, Secondary Service

B1 – Sequential; Secondary Service, Funds

B2 – Sequential; Secondary Service, Funds, Secondary Service

B3 – N/A RTE-HRCHY VALUE B3

B4 – N/A RTE-HRCHY VALUE B4

B5 – N/A RTE-HRCHY VALUE B5

B6 – N/A RTE-HRCHY VALUE B6

B7 – N/A RTE-HRCHY VALUE B7

B8 – N/A RTE-HRCHY VALUE B8

B9 – N/A RTE-HRCHY VALUE B9

BF – Funds authorization only

BS – Secondary service only

BZ – Simultaneous

Field Name: AUTH LEVEL
Screen: STRF screen 1
DDL Name: STRF. SCND-SVC.AUTH-LVL

The authorization level pertaining to the secondary request. A text description is displayed to the right of this field. Valid values are:

1 – (Online) Authorize transactions on the host only. If the host is offline, deny transaction. (default)

2 – (Offline) Authorize transactions on BASE24. (Reserved for future use.)

3 – (Online/offline) Authorize transactions on the host if the host is online; if the host is offline, authorize transactions on BASE24 and forward completions to the host when the host is online. (Reserved for future use.)

Field Name: PRIMARY DESTINATION
Screen: STRF screen 1
DDL Name: STRF. SCND-SVC.PRI-DEST

The first destination that BASE24 should attempt to when routing a secondary request.

Field Name: ALTERNATE 1 DESTINATION
Screen: STRF screen 1
DDL Name: STRF. SCND-SVC.ALT1-DEST

The alternate destination that BASE24 should attempt to when routing a secondary request. This destination is used if the primary destination is unavailable.

Field Name: ALTERNATE 2 DESTINATION
Screen: STRF screen 1
DDL Name: STRF. SCND-SVC.ALT2-DEST

The second alternate destination that BASE24 should attempt when routing a secondary request. The second alternate destination is used only in the event that both the primary and alternate 1 destinations are unavailable.

NOTE: This field is not currently being used by BASE24-atm.

Field Name: DEFAULT ACTION
Screen: STRF screen 1
DDL Name: STRF. SCND-SVC.DFLT-ACT

The default action to take if the primary, alternate 1, and alternate 2 destinations are all unavailable and an Offline Authorization File is not specified. A text description of the field is diplayed to the right. Valid values are:

A – Authorize Blind (Reserved for future use.)

C – Issue a Call (Reserved for future use.)

D – Decline (default)

Field Name: OFFLINE AUTHORIZATION FILE

Screen: STRF screen 1
DDL Name: STRF. SCND-SVC.OFFL-AUTH-FNAME

The Offline Authorization filename to be used when providing authorization offline for the secondary destination. This filename may only be used in conjunction with authorization levels of '2' (offline) and '3' (online/offline).

NOTE: This field is not currently being used by BASE24-atm.

Template Changes

Not applicable

Device Configuration

With this enhancement, the following files were modified to include new load groups:

Diebold 10XX/478X Configuration File (CNFGIX)

The following load groups were added for the Mobile Top-Up Integration enhancement:

- Load Group 503

  Load group 503 is the Mobile Top-Up group. It must be loaded on top of load groups 0 (base) and 500 (non-currency dispense).

  Load group 503 contains screens 047, 123, 412, 512, 513, 515, 516, 517, 522, 524, 536, 537, 538, and 560.

  Load group 503 contains states 047, 100, 110, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 536, 537, 538, 560 and 583.

- Load Group 504

  Load group 504 is the Mobile Top-Up group that adds support for the logging of cancelled mobile top-up transactions to the TLF when the cardholder declines the transaction at the tax screen or cancels the transaction at the available carrier or available amounts screen. Load group 504 must be loaded on top of 0 (base), 500 (non-currency dispense), and 503 (mobile top-up).

  Load group 504 contains states 110, 508, 513, 515, and 516.

- Load Group 550

  Load group 550 is the definitions-based Mobile Top-Up group and must be loaded on top of 0 (base), 500 (non-currency dispense), 503 (mobile top-up), and optionally 504 (mobile top-up log cancelled).

  Load group 550 contains screens 517, 519, 520, 521 and 525.

  Load group 550 contains states 506, 514, 517, 519, 520, 521 and 525.

NCR 5XXX Configuration File (CNFG5070)

The following load groups were added for the Mobile Top-Up enhancement:

- Load Group 503

  Load group 503 is the Mobile Top-Up group. It must be loaded on top of load groups 0 (base) and 500 (non-currency dispense).

  Load group 503 contains screens 200, 501, 507, 509, 535, 536, 537, 538, 539, 544, 551, and 560.

  Load group 503 contains states 184, 200, 202, 501, 507, 509, 523, 528, 529, 531, 536, 537, 538, 543, 544, 547, 550, 551, and 560.

- Load Group 504

  Load group 504 is the Mobile Top-Up group that adds support for the logging of cancelled mobile top-up transactions to the TLF when the cardholder declines the transaction at the tax screen or cancels the transaction at the available carrier or available amounts screen. Load group 504 must be loaded on top of 0 (base), 500 (non-currency dispense), and 503 (mobile top-up).

  Load group 504 contains screen 585.

  Load group 504 contains states 501, 507, 509, 513, and 585.

- Load Group 550

  Load group 550 is the definitions-based Mobile Top-Up group and must be loaded on top of 0 (base), 500 (non-currency dispense), 503 (mobile top-up), and optionally 504 (mobile top-up log cancelled).

  Load group 550 contains screens 533, 534, and 544.

  Load group 550 contains states 528, 529, 533, 534, and 535.

- Load Group 655

  Load group 655 is the EMV Mobile Top-Up group and must be loaded on top of 0 (base), 500 (non-currency dispense), 503 (mobile top-up), and optionally 504 (mobile top-up log cancelled), optionally 550 (definitions-based mobile top-up), 600 (EMV) and 605 (EMV non-currency dispense).

Load group 655 contains states 200, 202, and 537.

Mega Table (MEGATBL) Changes

The MOF and STRF requester names were added to the MEGATBL file.

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

SECTBL entries for the MOF and STRF screens have been added. These files will appear on the BASE Product Menu.

PATHCONF Changes

The new MOF server must be defined in the PATHCONF file as SERVER-MOF. This server can be added to the system online through the PATHCOM utility. After adding the server to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server.

The STRF Server object is bound into the Classic Combined AFM Linked Object (i.e., CCAFMLO). No additional PATHCONF changes are required for the STRF server.

CRT Access Security Changes

Access to the new MOF and STRF screens must be added through the SEC requester for the appropriate users.

Routing Changes

In a standard BASE24-atm system, the device handlers always route transactions directly to Authorization. In a Mobile Top-Up environment, the device handlers must accommodate the possibility of multiple authorization destinations. The device handlers will route Mobile Top-Up transactions to the destination specified in the LCONF parameter SPLIT-TXN-RTG-DEST. This destination should be the process name of the BASE24-atm Transaction Context Manager (TCM).

The TCM will utilize the new Split Transaction Routing File (STRF) for purposes of routing.

Network Configuration

A new process for the BASE24-atm Transaction Context Manager must be added to any XPNET Node that processes BASE24-atm mobile top-up traffic. The following is an example of a process under XPNET 3.0. This example is intended to aid BASE24 users in their initial setup and can be altered to fit specific environments.

```
RESET PROCESS
SET PROCESS BCPU 0
SET PROCESS LIBRARY $VOLUME.XPNET.SKELB
SET PROCESS PROGRAM $VOLUME.AT60OBJ.ATCM
SET PROCESS PPD $T1A1
SET PROCESS PRIORITY 145
SET PROCESS CPU 1
SET PROCESS STARTUP AUTOMATIC
SET PROCESS QAT 64
SET PROCESS SERVICE P1A^QUEUE
ADD PROCESS P1A^ATCM1, UNDER SYSNAME \<sys>, UNDER NODE P1A^NODE
```

For more information about adding Processes to the XPNET system, refer to the ***NET24-XPNET Network Control Operations Guide*** or the ***NET24-XPNET Network Control Command Reference Manual***.

File Conversions

Changes to the Transaction Based Pricing (TBP) reports necessitates that all BASE24 TBP customers implement the updated version of the TBP report objects into their system and convert their existing Monthly Transaction Accumulation Files (MTAFs) to the new format. Refer to **General Changes** in Section 4 for additional information.

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the External Interfaces and Super Extract processes.

External Interfaces

Host code changes may be required to support a new reversal reason code of 40 – Split Routing Enabled; Secondary Service Not Approved. This value is supported by the BASE24-atm ISO Host Interface process and the BASE24-atm BIC ISO Interface process.

Super Extract Process

Standard TLF Extract processing is not specifically impacted by the BASE24-atm Mobile Top-Up Integration enhancement but host systems must be aware of the following:

- the TLF.AUTH.RESPONDER field can contain a new value of S – Secondary Service

- the TLF.AUTH.RVSL-RSN field can contain a new value of 40 – Split Routing Enabled; Secondary Service Not Approved

Customers implementating the definitions-based model of the BASE24-atm Mobile Top-Up enhancement must also be aware of the following:

- the TLF.AUTH.TRAN-CDE field may contain new values of 25 – mobile top-up and 26 – mobile top-up with funds

If the new tokens for this enhancement are to be included in the TLF and ILF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

Host code changes are required for all BASE24 users that wish to utilize the OMF Extract Process to accept and process file maintenance made to the MOF and STRF files using the BASE24 AFT subsystem.

User/Operational Impacts

Disk Impacts

The new MOF, STRF and STRFEMT files impact disk requirements. The total impact varies for each BASE24 user depending on the number of MOF and STRF records defined in the BASE24 system.

The TSRF and TSRFEMT files impact disk requirements. The total impact varies for each BASE24 user depending on the number of TSRF records defined in the BASE24 system.

Existing BASE24-atm users that implement this enhancement will experience a change in messaging or logging to the TLF as follows:

- Each successful mobile top-up transaction will be recorded in two TLF records: one for the funds authorization and one for the top-up authorization.

- Declined transactions may have as many as two TLF records.
- Reversed transactions may have an additional one or two TLF records depending on the point of failure.
- If BASE24 is configured to do so, there may be an additional record logged to the TLF to represent that the customer cancelled the transaction.

The TLF and ILF require additional disk space as the new tokens for this enhancement can be included in the transaction records logged to these files.

Operational Impacts

TLF Perusal supports on a new detail screen data defined in the Pre-Pay Top-Up and the Pre-Pay Response tokens. Operators should be trained to recognize the information on the new detail screen.

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Performance Impacts

Performance will be impacted in systems running mobile top-up transactions due to the increase in the transaction path length. The transaction path is longer because mobile top-up transactions are routed to two destinations (i.e., a funds authorizer and a mobile I/F). Additionally, there may be multiple interactions between the BASE24 device handler process, the BASE24-atm Transaction Context Manager process, and the Mobile Operator Interface process prior to the funds authorization and prior to the final top-up authorization.

While each interaction with the device handler process resets the timer, BASE24-atm customers may need to examine the value of the LCONF parameter ATM-DH-AUTH-

TIMER and determine if the increased length of the transaction path necessitates an increase in the value assigned to ATM-DH-AUTH-TIMER.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

NCR AKDS Capability Check and Load All Keys

This section describes the implementation requirements associated with the NCR AKDS Capability Check and Load All Keys enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the NCR AKDS Capability Check and Load All Keys enhancement.

The standard BASE24-atm NCR 5XXX device handler module has been modified to support this enhancement.

Prerequisite: The Automated Key Distribution System (AKDS) enhancement must be installed prior to implementation of the NCR AKDS Capability Check and Load All Keys enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 3)***.

Software Impacts

All BASE24-atm NCR 5XXX device handler customers:

- Hooks to AKDS procedures and inline changes to AT60N50.N50S
- Inline changes to AT60N50.N50G
- Inline changes to the N5XXX DCT screen and requester AT60N50.SCRNN50L and RQN50LS
- Additional procedures in AT60N50.N50D
- Updated event (log) message manual with new events BA60LOGM.AT5070

All BASE24-atm NCR 5XXX Dialup device handler customers:

- Inline changes to AT60PN50.N50DLUS

All BASE24-atm NCR 5XXX SSB device handler customers:

- Inline changes to AT60DN50.N50SSDS

All BASE24-atm NCR 5XXX AKDS device handler customers:

- Inline changes to AT60AN50.N50AKDG, N50AKDM and N50AKDS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD08.SCNLALLN.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

ATM Terminal Data file (ATD)

There are a few communication protocols that NCR has not upgraded to support the longer message lengths required by AKDS. The NCR 5XXX device handler now checks the Terminal Line Protocol field on screen 7 of the ATD to determine if an ATM is configured with an appropriate protocol. Protocols that do not support AKDS are 'B' (Bisync), 'C' (Bisync-ASCII) or 'S' (SDLC). The device handler will not send an AKDS load to an ATM configured with one of these protocols. If an ATM is configured with one of these protocols and the ATM is required to use AKDS, the ATM must be reconfigured to use another protocol.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

AKDS Capability Check

This enhancement enables the NCR 5XXX device handler to automatically determine the AKDS capability of an ATM before permitting an AKDS load (LOAD PUBLIC KEY, LOAD MASTER KEY, LOAD ALL KEYS, LOAD KEY ENTRY MODE) to proceed. When the device handler receives an AKDS load command, it closes the ATM and sends a configuration request message to determine if the ATM has an AKDS-capable Encrypting PIN Pad (EPP) installed. The AKDS load is allowed to proceed if an AKDS-capable EPP is present. If an AKDS-capable EPP is not present, the AKDS load is not allowed to proceed, an informational event message is logged, and the ATM is re-opened. One exception to this case is the LOAD KEY ENTRY

MODE command, which is allowed to proceed because it is used to convert an EPP running in BAPE emulation mode to a fully AKDS-capable EPP. Another exception is the LOAD ALL KEYS command – see the Device Control Terminal (DCT) subsection that follows for more information.

<u>Device Control Terminal (DCT)</u>

The following command has been added to the NCR 5XXX Load Commands Screen (screen 45) of the DCT.

- LOAD ALL KEYS

  When the operator places the cursor on this command and presses function key F6-LOAD, the DCT sends a command to the device handler to initiate a Load All Keys sequence. At the start of this sequence, the device handler closes the specified ATM and checks its AKDS capability.

  If the ATM is AKDS-capable, the device handler downloads the RSA and DES keys and then re-opens the ATM. The keys are downloaded in the following order.

  1. Public Key
  2. Master Key
  3. MAC Encryption Key (if ATD MAC Support Type = ‘3’)
  4. PIN Encryption Key (if ATD PIN Encrypt Type = ‘01’, ‘03’, ‘04’ or ‘05’)

  If the ATM is not AKDS-capable, the device handler does not download the RSA keys. It does, however, download the DES keys and then re-opens the ATM. The keys are downloaded in the following order.

  1. MAC Encryption Key (if ATD MAC Support Type = ‘3’)
  2. PIN Encryption Key (if ATD PIN Encrypt Type = ‘01’, ‘03’, ‘04’ or ‘05’)

<u>NCS Screen / NCPCOM Utility</u>

The NCR 5XXX device handler now supports the following 9500 commands. They are entered at the Network Control Supervisor (NCS) screen or the Network Control Point Communications (NCPCOM) utility.

- LALLKEYS

  This 9500 command is equivalent to the DCT's LOAD ALL KEYS command. It instructs the device handler to initiate a Load All Keys sequence. For more information refer to the LOAD ALL KEYS command in the preceeding DCT subsection.

  The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500LALLKEYS <terminal ID>"

<process name>:	The symbolic name assigned to the NCR 5XXX device handler process in the XPNET node.
<terminal ID>:	The terminal ID defined for the ATM in the ATM Terminal Data file (ATD).

- LOADALLKEYS
- LOADPUBLIC
- LOADKEYMODE
- LOADMASTER

These 9500 commands are equivalent to the NCR 5XXX device handler's LALLKEYS, LPUBLIC, LKEYMODE and LMASTER 9500 commands. They have been added in order to provide command formats that are consistent with those of other BASE24-atm device handlers.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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NCR NDC+ CAM II (EMV) Upgrade

This section describes the implementation requirements associated with the NCR NDC+ CAM II (EMV) Upgrade enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the NCR NDC+ CAM II (EMV) Upgrade enhancement.

Software Impacts

All BASE24 customers:

- A new language file (BA60SRC.COBLANG)
- Inline changes to BA60SRC.BASRCMM

All BASE24-atm NCR 5XXX device handler customers:

- Inline changes to AT60N50.ATN50M, N50DDLM, N50DDLS, RQ50STS, SVN50S and SVLIBS

All BASE24-atm NCR 5XXX EMV device handler customers:

- Inline changes to AT60IN50.ATIN50MM, RQEMAIDS, RQEMCRNS, RQEMTXNS
- A new EMV-specific load table requester AT60IN50.RQEMLANS

A complete list of all files affected by this enhancement can be found in SCN file BA60UD08.SCNIN50.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

EMV N5XXX DCT support has been enhanced with respect to three of the four EMV Load Tables: the Application ID (AID) Table, the Currency Data Objects Table, and the Transaction Data Objects Table. In addition, a fifth table, the Language Support Table, has been added to the EMV Load Table Menu screen.

The following tables have been modified to include:

Application ID Table:

- A Default Application Label field which is stored with the table record in the configuration file (CNFG5070) and downloaded to the ATM. The Default Application Label can have imbedded spaces. For example, “Self Service” is a valid label. Labels are entered and stored as text strings.

- Support for hexadecimal Lowest and Highest Version numbering, allowing hexadecimal values in the range of ‘0000’ thru ‘FFFF’ (spaces not allowed). For example, valid version numbers are: 0001, 0A02, FFFE, 9999. Note that the operator is forced to enter 4 valid digits to eliminate any ambiguity.

- Support for deleting AID table entries at the ATM by a subsequent download of a blank AID (i.e., an AID table record with the same index, but no AID). The requester has been modified to allow entry of a blank Primary AID. If a blank Primary AID is entered, further processing of remaining AID table fields is skipped.

- Support for the Terminal Action Code (TAC) within the AID table. This is a 10 character field stored as ASCII hex in the AID table record as it is entered at the DCT screen (note that this is similar to version numbering processing, except for the length of the field). Valid value range is ‘0000000000’ thru ‘FFFFFFFFFF’ (i.e., each digit has a range of ‘0’ thru ‘9’, ‘A’ thru ‘F’). Blanks are not allowed.

Currency Data Objects Table:

- Support for six distinct Currency Type / Currency Code / Exponent occurrences has been added to the table. The exponent field represents the currency’s decimal place requirement.

Transaction Data Objects Table:

- Support for ten Transaction Category Codes has been added to the table. The new 1 char Category Codes have been paired with the existing Transaction Types on the Transaction Data Objects table screen.

The following table has been added.

Language Support Table:

- This table is used in state ‘-‘ (Automatic Language Selection state) to provide a mapping between the language preferences indicated by the chip and those supported by the terminal. The table shows what languages the terminal supports. State '-' directs the state flow depending on whether the language preference on the chip matches an entry in the table. The table has been added to the EMV table menu and must be downloaded to the ATM during the EMV table load if language selection is desired. A maximum of four languages is supported.

The following EMV state has been modified for the NDC+ CAM II Upgrade enhancement.

EMV State '+' (Begin EMV Initialization)

- This state has been modified to include the 'screen to clear screen number' field (entry #8). This field is used during execution of existing EMV state '/' (Complete EMV Application Selection and Initialization) to clear the screen prior to displaying the selected application name.

Additional information concerning the above can be found in the ***BASE24-atm EMV Support Manual*** and the ***BASE24-atm NCR 5XXX Device Support Manual.***

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

NCR NDC+ EMV Dip Card Reader

This section describes the implementation requirements associated with the NCR NDC+ EMV Dip Card Reader enhancement. This enhancement is required for customers using NCR's EMV Dip card reader due to the different request format used by the EMV Dip card reader. NCR's EMV Dip card reader does not send the track2 data buffer if the track2 equivalent data is first obtained from the chip. This enhancement changes the way requests are processed by the NCR 5XXX EMV device handler to account for this difference.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the NCR NDC+ EMV Dip Card Reader enhancement.

Software Impacts

All BASE24-atm NCR 5XXX device handler customers:

- Inline changes to AT60N50.N50S and N50EMVDS

All BASE24-atm NCR 5XXX EMV device handler customers who want to support ATMs equipped with NCR's EMV Dip Card Reader:

- Inline changes to AT60IN50.N50EMVS

There is no SCN file associated with this enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Offline PIN Change Notification

This section describes the implementation requirements associated with the Offline PIN Change Notification enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Offline PIN Change Notification enhancement.

Changes have been made to the CPF to support this enhancement. Refer to **Offline PIN Change Notification** in Section 4 for a description of the Base changes. The standard BASE24-atm authorization module and the authorization EMV extension module have been modified for this enhancement. These changes are described below.

Some card management systems may require knowledge of a new PIN whenever a PIN change transaction is successfully performed since these systems may have to include the new PIN on the card if/when the card is re-issued. BASE24 allows a new PIN block encryption key field in the Transaction Security Services (TSS) database to act as a transport key between the BASE24-atm system and the card management system.

A key locator to this PIN block encryption key field can be configured in the CPF. Before writing a TLF record, the PIN block containing the new PIN may be translated to encryption under the new key. The card management system also stores the new key on its database. Therefore, when it receives the new PIN block (in a TLF Extract file, for example), the card management system is able to decrypt the new PIN block.

Software Impacts

All BASE24-atm customers:

- Inline changes to AT60AUTH.AUTHLIBS
- Updated event (log) message manual BA60LOGM.ATAUTH

All BASE24-atm EMV customers:

- Inline changes to AT60IATH.AUTHEMVS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD08.SCNOFFP.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

<u>Token File (TKN)</u>

Since the existing PIN Change Token (token ID "06") will now be used to notify a Host system of a new offline PIN, BASE24-atm customers utilizing the Offline PIN Change Notification enhancement should review their Token records.

If the PIN Change Token should not to be logged to the TLF or the ILF, the appropriate record must be configured in the Token file.

If the PIN Change Token is to be extracted with the TLF or ILF or sent to the host, the appropriate Token records must be configured.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

If the PIN Change Token is to be included in the TLF and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

ISO Host Interface Process

If the PIN Change Token is included in the ISO message sent to the host, this change needs to be handled by the host programs.

User/Operational Impacts

Operational Impacts

New event messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Disk Impacts

The TLF and ILF will require additional disk space if PIN Change Token is included in the transaction records logged to these files.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Performance Considerations

This enhancement will impact system performance in EMV transaction processing for users wishing to take advantage of the Offline PIN Change Notification functionality. An additional PIN block translation will be performed (via TSS and an HSM) to ensure that the new PIN block is encrypted under a common transport key for subsequent processing by a card management system.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

PBF Refresh Early Notification

For further information, see **PBF Refresh Early Notification** in Section 4 where the impacts of this enhancement are described.

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Shared (Wincor) NDC+ EMV Support

This section describes the implementation requirements associated with the Shared (Wincor) NDC+ EMV Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared (Wincor) NDC+ EMV Support enhancement.

The Shared (Wincor) NDC+ EMV device handler extension is a separate bindable module, independent from, yet compatible with, the NCR NDC+ EMV device handler extension. It uses the existing BASE24-atm N5XXX EMV DCT requesters and screens from the NCR NDC+ EMV device handler extension subvol (AT60IN50). However, it does not require the BASE24-atm NCR 5XXX EMV device handler extension (AT60IN50.N50EMVS).

Anyone using the Shared (Wincor) NDC+ EMV Support enhancement must have previously applied the NCR NDC+ CAM II Upgrade enhancement. The NCR NDC+ CAM II Upgrade enhancement expanded the shared N5XXX DCT EMV functionality. This enhancement is also part of the release 6.0 version 5 enhancements.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS

All BASE24-atm customers:

- Inline changes to AT60SRC.ATMFM and ATMMM
- DDL changes to AT60DDL.DDLFATD and DDLGATD
- Inline changes to AT60AFT.RQATDS, SCRNATD1, SVATDTBL, SVATDTG and SVATDTS

All BASE24-atm NCR 5XXX device handler customers:

- Hooks in the main device handler module to support Wincor NDC+ EMV (AT60N50.N50S)
- Inline changes to AT60N50.N50DDLS, N50DDLM, N50G and N50M
- Additional procedures added to AT60N50.N50D
- Inline changes to AT60N50.RECIMGCS, RQ50STS, SCRNST, SVLIBS and SVN50S
- Updated event (log) message manual BA60LOGM.AT5070

All BASE24-atm NCR 5XXX EMV device handler customers:

- Inline changes to AT60IN50.ATIN50MM

- Inline changes to AT60IN50.RQEMAIDS, RQEMCRNS, RQEMMENS, RQEMTXNS, SCNEMAID, SCNEMCRN, SCNEMMEN and SCNEMTXN

All BASE24-atm NCR 5XXX device handler customers wanting the Shared (Wincor) NDC+ EMV Support enhancement:

- New subvol, AT60IWN5, with files for the Shared (Wincor) NDC+ EMV extension

A complete list of all files affected by this enhancement can be found in SCN file BA60UD08.SCNIWN5.

CUSTMACS Changes

The following Make flag has been added to the CUSTMACS file.

- atm_dh_n50_win_emv_on

It should be set to true to bind the Shared (Wincor) NDC+ EMV module to the NCR 5XXX device handler.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

The Shared (Wincor) NDC+ EMV Support enhancement uses existing NCR NDC+ EMV (CAM II) support structures, including most EMV-specific CNFG5070 file load groups, DCT functionality (including EMV load tables), and ATD screen 21. However, the Shared (Wincor) NDC+ EMV does not perfectly emulate NCR NDC+ EMV, requiring Shared (Wincor) NDC+ EMV-specific configuration in several areas.

These changes include:

- Since the Shared (Wincor) NDC+ EMV native message is slightly different from the NCR NDC+ EMV native message, ATD screen 21 has been modified to include a new field; the ‘native message format’ field. This field must be set to ‘non-standard’ for all Shared (Wincor) NDC+ EMV ATMs. Customers running NCR NDC+ EMV terminals should set this field to ‘standard’ format. This field makes it possible to run both Shared (Wincor) NDC+ EMV and NCR NDC+ EMV terminals from a single BASE24-ATM NCR 5XXX device handler object. The default for this field is ‘standard’ format.

- As part of the NCR NDC+ CAM II Upgrade enhancement, the TAC field was added to the EMV AID table record. However, Wincor ATMs cannot accept AID table records containing the TAC field. Therefore, a separate load group (group 609) has been designated for non-standard (Wincor) EMV AID table records. The N50 server library code (AT60N50.SVLIBS) was modified to recognize AID table records from group 609 as records for non-standard terminals and to remove the TAC field. Therefore, AID table records for Shared (Wincor) NDC+ EMV ATMs must reside in load group 609.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

ATMs supporting Shared (Wincor) NDC+ EMV must have the proper EMV-compatible software and hardware installed.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to ATD screen 21 is required to configure individual ATMs as either standard or non-standard format.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The ATD conversion program (AT60UC04.CNVATDS) has been modified to initialize the new ‘native message format’ and the ACI user field within the ATDS1-CORE data area to valid values. Note that running the ATD conversion program is required only when converting release 5.3 TDF files to release 6.0 version 5 ATD files.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

New event messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriatae response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Triton EMV

This section describes the implementation requirements associated with the Triton EMV enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Triton EMV enhancement.

A new bindable extension, AT60ITRT.TRITEMVS, has been developed to support EMV in the Triton device handler. The standard BASE24-atm Triton device handler has also been modified with “hooks” (calls) to various TRITEMVS procedures that handle the special EMV processing.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS

All BASE24-atm customers:

- Inline changes to AT60SRC.ATDGX, ATMFM and ATMMM
- Inline changes to AT60DDL.DDLFTDF
- Inline changes to AT60AFT.SVATDTS

All BASE24-atm Triton device handler customers:

- Hooks in the main device handler module to support EMV (AT60TRIT.TRITONS)
- Minor inline changes to AT60TRIT.TRITONG, TRITONM and TRITNAMS
- Additional procedures in AT60TRIT.TRITOND
- Updated event (log) message manual BA60LOGM.ATTRIT

All BASE24-atm Triton device handler customers taking the new EMV functionality:

- A new subvolume, AT60ITRT, with all of the files required to support EMV.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNTEMV.

CUSTMACS Changes

The following Make flag has been added to the CUSTMACS file.

- atm_dh_triton_emv_on

It should be set to true to bind the EMV module to the Triton device handler.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes fields, within BASE24-atm files, relevant to EMV support. Field settings depend on desired configuration.

<u>BASE24-atm Terminal Data Files (ATD/TDF)</u>

Screen Name:	EMV Terminal Capabilities
Screen:	ATD Screen 21
DDL Name:	TDF.ADNL-DATA.EMV-TERM-CAP

Indicates the card data input, Card Verification Method (CVM), and security capabilities of the terminal using a bit map to represent possible options.

The ATD has a separate screen 21 to configure the EMV Terminal Capabilities fields. For additional information regarding the setting of these fields refer to the ***BASE24-atm EMV Support Manual.***

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Triton devices supporting EMV must have new software and hardware configured to read ICC cards and send required EMV data fields to BASE24.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to Screen 21 of the ATD is required to set EMV Terminal Capablities for Triton devices. This existing screen is now accessable for Triton devices (device type B3).

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Since the TDF.SNA field was not used by the Triton device handler it was redefined to allow for the addition of the EMV Terminal Capabilities field in the TDF record. The remainder of the SNA field has been redefined as USER-FLD-ACI. Putting the EMV Terminal Capabilities field here allows the field be used (in the future) by other ATM device handlers that still use the TDF but do not support SNA.

For users already running the 6.0 Triton DH, an inline conversion of the TDF will be performed to initialize the new data field defined for EMV. The conversion also initializes the USER-FLD-ACI.

For users converting a release 5.3 TDF to release 6.0 version 5, the ATD conversion program (AT60UC04.CNVATDS) has been modified to initialize the new data fields in the TDF.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operation Impacts

New event messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Triton Terminal Communications Protocol and Message Format Specification, *Version TSCD5.22, July 19, 2004*

Triton TCP/IP

This section describes the implementation requirements associated with the Triton TCP/IP enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Triton TCP/IP enhancement.

The standard BASE24-atm Triton device handler module has been modified to support the use of TCP/IP. Prior to this enhancement, the only protocol that could be configured for the Triton was VISA II dial-up.

Software Impacts

All BASE24-atm Triton device handler customers:

- Inline changes to AT60TRIT.TRITONG and TRITONS
- Updated Triton manual AT60TRIT.TRITMAN
- Updated event (log) message manual BA60LOGM.ATTRIT

A complete list on the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNTRIT.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

BASE24-atm Terminal Data File

Field Name: Terminal Line Protocol
Screen: ATD screen 7
DDL Name: TERM-PROTO

This field must be set to ‘T’ indicating that TCP/IP is the desired protocol to be used.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Triton devices that are to support TCP/IP must have new terminal software and hardware. In addition, the proper IP addresses must be configured at the ATM. The Triton device can be configured to send or not send a CRC-16 checksum value with each request.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

XPNET configuration changes are needed to use TCP/IP. Refer to the Triton Manual, AT60TRIT.TRITMAN, for detailed DEVICE, LINE, and STATION settings. For additional TCP/IP configuration assistance, refer to the ***XPNET TCP/IP Protocol Implementation Guide***.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

<u>Operation Impacts</u>

New event messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

<u>Compatible Devices and Software</u>

Refer to the Triton vendor documentation to verify device compatibility of the following components to support TCP/IP:

- Eprom Version
- Program Version
- Table Version
- Triton ATM models

Reference Documents

Triton: Terminal Communications Protocol and Message Format Specification*,* Version TSCD 5.22.

Triton: TCP/IP Protocol Specification, Version 1.2.

Triton Triple DES MACing

This section describes the implementation requirements associated with the Triton Triple DES MACing enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Triton Triple DES MACing enhancement.

The standard BASE24-atm Triton device handler module has been modified to support triple DES MACing using double-length keys. Prior to this enhancement, the security keys were stored in BASE24 and in the ATM as 16 hex characters or 8 bytes. However, support of double-length keys requires BASE24 and the ATM to store keys as 32 hex characters of 16 bytes. In order to support the double-length keys, Transaction Security Services (TSS) is required.

An important item to note is that the Triton device uses the MACing algorithm Pseudo 3DES Cipher Block Chaining. The use of this algorithm has been tested and found to be supported by Atalla's AKB (non-variant) security device. It has also been tested with Atalla's variant security device and the Thales security device and found NOT to be supported.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Triton Triple DES MACing enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Software Impacts

All BASE24-atm Triton device handler customers:

- Inline changes to AT60TRIT.TRITONG and TRITONS
- Updated Triton manual AT60TRIT.TRITMAN

A complete list on the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNTRIT.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

The following database changes are needed to support triple DES MACing at Triton ATMs.

Channel Security Data Source (CSECD)

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: MAC Option

The MAC Option for the Triton CSECD records must be set to Pseudo 3DES Cipher Block Chaining. Instead of performing the triple DES algorithm on the entire message, the Triton device performs the single DES algorithm on all but the last 8 bytes of the message. These last 8 bytes have the triple DES algorithm performed on them.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Key Length of Exchange Key

The Key Length of the Exchange Key for the Triton CSECD records must be set to double to utilize Triton triple DES MACing using double-length keys.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Exchange Key

The Exchange Key for the Triton CSECD records must be modified to contain the new double-length MAC key value.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Key Length of Inbound Key

The Key Length of the Inbound Key for the Triton CSECD records must be set to double to utilize Triton triple DES MACing using double-length keys.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Inbound Key

The Inbound Key for the Triton CSECD records must be modified to contain the new double-length MAC key value.

Refer to the online help provided with the ACI Desktop user interface for detailed descriptions of this window and its fields.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Triton devices supporting triple DES MACing must have new software and hardware configured to use the triple DES MACing algorithm and double-length MAC keys. Additionally, the new double-length MAC Master key must be manually entered at the device and the new Working key must be downloaded.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Compatible Devices and Software

Refer to the Triton vendor documentation to verify device compatibility of the following components to support Triton triple DES MACing:

- Eprom Version
- Program Version
- Table Version
- Triton ATM models

Reference Documents

Triton: Terminal Communications Protocol and Message Format Specification, Version TSCD 5.22.

Visa iCVV Support

For further information, see **Visa iCVV Support** in Section 4 where the impacts of this enhancement are described.

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Section 6

BASE24-pos

Section 6 discusses the changes and enhancements made to BASE24-pos in release 6.0 version 5, along with the resulting conversion and implementation requirements that existing BASE24-pos users must consider when upgrading from release 6.0 version 4.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-pos users, along with the associated conversion and implementation requirements that all BASE24-pos users must address.
- Optional enhancements describe the changes that affect only those BASE24-pos users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-pos users must address to implement the enhancements.

Section 6 must be used in conjunction with Section 4 to identify the changes that BASE24-pos users need to consider when upgrading to release 6.0 version 5.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-pos users must address in order to upgrade to release 6.0 version 5.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- Router/Auth/DH Data Stack Relief

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- Changes to support the Standard POS Device Handler (SPDH) Mobile Top-Up Extension Hooks including a new pre-pay generic receipt token and a pre-pay merchant token.
-
- In BASE24-pos the CVD results code are stored in CRD-VRFY-FLG in the PS50 Token for card-read transactions. For manually entered transactions, the CVD2 results code is also stored in CRD-VRFY-FLG in the PS50 Token. If a card-read transaction includes the results code for both CVD and CVD2, then the results code for CVD will be stored in CRD-VRFY-FLG in PS50 Token, and the results code for CVD2 will be stored in CRD-VRFY-FLG2 in the POS Data 1 Token. This standardization ensures Router/Authorization logic remains unaffected, and positions BASE24-pos for future enhancements to perform CVD and CVD2 validation in a single call to the hardware security module.
-
- If the CVD and CVD2 were checked by an interchange and failed, then BASE24-pos Router/Authorization will check the BAD CV ACTION on the Card Prefix File (CPF) to determine the appropriate authorization action.
-
- Note: This change was cataloged as a fix to PS60RTAU.AUTHLIBS in late September, 2004 for Clarify case # 353773.
-

Software Impacts

Note: The following impacts are for the SPDH Mobile Top-Up Hooks changes only.

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS and FLGSDOC
- Changes to the Base Token DDLs and the POS Token DDLs for future support of the new token defines for BASE24-pos Mobile Top-Up hooks (i.e. DDLBATKN, DDLPSTKN)
- Changes to the Base Token Ids, the POS Token Ids, the Base Token Conversion utilities, the POS Token Conversion Utilities and COBOL Token Table for future support of new token defines for the BASE24-pos Mobile Top-Up hooks (i.e., BATKNID, PSTKNID, BATKNCVS, PSTKNCVS, and COBTKN)

All BASE24-pos customers:

- Changes to Make files to include the pos_dh_spdh_mtu_on flag (i.e., PS60SRC.POSFM, PS60SRC.POSMM)
-
- All BASE24-pos customers with the ACI Standard POS Device Handler customers:

- Inline changes to the ACI Standard POS Device Handler to support Mobile Top-Up hooks. (ASPDHG, ASPDHM, ASPDHS, and SPDHD)

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

When a POS Mobile Top-Up solution is available, set the following flag to true to bind the Mobile Top-Up module to the ACI Standard POS Device Handler (ASPDH).

- pos_dh_spdh_mtu_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Router/Auth/DH Data Stack Relief

This section describes the implementation requirements associated with the Router/Auth/DH Data Stack Relief enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Router/Auth/DH Data Stack Relief enhancement.

Standard BASE24-pos Router, Authorization, and Device Handler modules were modified to reduce the total amount of data placed on the data stack, eliminating the incidents of stack overflow failures. The data stack, also known as the lower 32K, is limited to 32768 words. Changes were also made to the standard BASE24-pos Device Handler modules to reduce the amount of data placed in the global primary area of the data stack, allowing more room for future fixes, product enhancements and customer modifications. The global primary area of the data stack is limited to 256 words.

The global files of the ACI Standard POS Device Handler (SPDH) and the Hypercom POS Device Handler (HPDH) have been changed to reduce the amount of data placed in the global primary area of the data stack. This was accomplished by moving the declarations for a number of global variables to a subordinate position within an indirect global structure. The declaration of the indirect global structure itself consumes only one word of the global primary area. This change was made because the TAL compiler will move all of the structure's subordinate variables to the global secondary area, which has no explicit limit, when the variables are used. In the SPDH, this structure is called spdh_g. In the HPDH, this structure is called hpdh_g. **From this point forward, any globals added for fixes, product enhancements or customer modifications should be declared within these structures, if at all possible**.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60RTAU.AUTHLIBS, AUTHS, ROUTERS, RTAUG, RTAUEMVD and RTAUPID
- Updated event (log) message manual BA60LOGM.PSAUTH

All BASE24-pos Address Verification customers:

- Inline changes to PS60DDR.ADDRS

All BASE24-pos Check Authorization customers:

- Inline changes to PS60ERTA.RTAUECKS

All BASE24-pos EMV customers:

- Inline changes to PS60IRTA.RTAUEMVS

All BASE24-pos PRM customers:

- Inline changes to PS60PRTA.RTAUPIS

All BASE24-pos ACI Standard POS device handler customers:

- Inline changes to PS60SPDH.ASPDHG, ASPDHS and SPDHLDRS

All BASE24-pos ACI Standard POS device handler Data Encryption customers:

- Inline changes to PS60DSPD.SPDHDES

All BASE24-pos ACI Standard POS device handler Dynamic Key Management customers:

- Inline changes to PS60KSPD.SPDHDKMS

All BASE24-pos Hypercom device handler customers:

- Inline changes to PS60HPDH.HPDHG and HPDHS

All BASE24-pos Hypercom device handler EMV customers:

- Inline changes to PS60IHPD.HPDHEMVS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD08.SCNPSSTK. This file also contains a list of all modified variables showing their old and new names.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Optional Enhancements

This section identifies the BASE24-pos enhancements that can be enabled or disabled with release 6.0 version 5, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Absent ICC Application PAN Sequence Number
- American Express Card Security Code
- EMV 2000 Session Key Derivation
- Forward Declined Advices to Host
- Hypercom Triple DES Master/Session and DUKPT
- PBF Refresh Early Notification
- Stored Value Dormancy and Escheatment
- Stored Value Offline Card
- Visa iCVV Support

BASE24-pos users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements for future reference.

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Absent ICC Application PAN Sequence Number

For further information, see **Absent ICC Application PAN Sequence Number** in Section 4 where the impacts of this enhancement are described.

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American Express Card Security Code

This section describes the implementation requirements associated with the American Express Card Security Code enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the American Express Card Security Code enhancement.

Logic has been added to the standard BASE24-pos router/authorization module to detect fraudulently created American Express cards through the implementation and use of Card Security Codes (CSC).

American Express offers three types of CSC:

- Three-digit CSC – located on the signature panel (currently not in use)
- Four-digit CSC – located on the front of the card (currently occupied by four digit batch code (4DBC))
- Five-digit CSC – located on the magnetic stripe

BASE24-pos provides processing capabilities for all three types of CSC. BASE24-pos is capable of processing a manual and an electronic CSC value in a single call to the Hardware Security Module (HSM).

If both an electronic and manually entered CSC value are being verified in BASE24-pos, the manual CSC value is stored in the BASE24-pos Release 5.1 Token.

In the BASE24-pos authorization process, American Express on-us transaction requests will have the CSC value(s) verified via Transaction Security Services (TSS).

In BASE24-pos, if a single CSC value is verified, then the results of that verification are placed into the BASE24-pos Release 5.0 Token. If both an electronic and a manually entered CSC value are verified in the same transaction, the results of the electronic CSC verification are placed into the BASE24-pos Release 5.0 Token and the results of the manually entered CSC verification are placed into the BASE24-pos Data1 Token.

For further information regarding this enhancement see **American Express Card Security Code** in Section 4.

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EMV 2000 Session Key Derivation

For further information, refer to **EMV 2000 Session Key Derivation** in Section 4 where the impacts of this enhancement are described.

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Forward Declined Advices to Host

For further information, refer to **Forward Declined Advices to Host** in Section 4 where the impacts of this enhancement are described.

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Hypercom Triple DES Master/Session and DUKPT

This section describes the implementation requirements associated with the Hypercom Triple DES Master/Session and DUKPT enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Hypercom Triple DES Master/Session and DUKPT enhancement.

The standard BASE24-pos Hypercom device handler module has been modified to support this enhancement.

Prerequisite: Single DES Master/Session PIN encryption and Single DES DUKPT do not require the use of Transaction Security Services. However, if triple DES master/session PIN encryption or triple DES DUKPT is required, the Transaction Security Services enhancement must be installed. For more information about Transaction Security Services refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Software Impacts

All BASE24-pos Hypercom device handler customers:

- Inline changes to PS60HPDH.HPDHDDLS, HPDHG, HPDHLDRS and HPDHS
- Updated event (log) message manual BA60LOGM.PSHYPRCM

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCN3DDUK.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign record that must be added to the LCONF to support this enhancement. This assign pertains only to the DUKPT portion of this enhancement. The LCONFBA file located on the BA60MISC subvolume has been modified to include this assign.

<u>KEYD</u>

This optional assign specifies the fully qualified file name of the BASE24 Derivation Key File.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These params pertain only to the DUKPT portion of this enhancement. The LCONFPS file located on the BA60MISC subvolume has been modified to include these params.

POS-DH-DUKPT-UPDATE-METHOD

This optional param indicates whether the device handler can automatically update the PIN ENCRYPTION TYPE field on BASE24-pos Terminal Data files (PTD) screen 7 to a value of 07 (DUKPT) when the Key Serial Number field is received in a message for the first time and a Derivation Key File (KEYD) record exists for the terminal. Valid values are as follows:

Y = Yes, automatically update the PIN encryption method in the BASE24-pos Terminal Data files.

N = No, do not automatically update the PIN encryption method in the BASE24-pos Terminal Data files. (default)

POS-DH-KEYD-READ-FROM-DISK

This optional param indicates whether the device handler reads the Derivation Key File (KEYD) from disk or from memory. Valid values are as follows:

Y = Yes, read the KEYD from disk.

N = No, do not read the KEYD from disk; instead read the file from extended memory. (default)

POS-DH-KSN-APPRV-OPT

This optional param controls the action taken by the device handler if the static data received in the Key Serial Number (KSN) of a request message does not match the KSN static data maintained in PTD device dependent data. Valid values are as follows:

1 = Deny the transaction

2 = Approve the transaction but do not update the PTD key serial number (KSN).

3 = Approve the transaction and update the PTD key serial number (KSN). (default)

POS-DH-KSN-CNTR-APPRV-OPT

This optional param controls the action taken by the device handler if the difference in the transaction counter between two transactions from the same terminal exceeds the limit set in the POS-DH-KSN-MAX-DIFFERENCE param. The previous transaction counter is maintained in PTD device dependent data and the current transaction counter is carried in the Key Serial Number (KSN) field of the request message. Valid values are as follows:

1 = Deny the transaction

2 = Approve the transaction but do not update the PTD key serial number (KSN).

3 = Approve the transaction and update the PTD key serial number (KSN). (default)

POS-DH-KSN-DESCR

This optional param contains the default key serial number (KSN) descriptor required by the Thales e-Security device to perform a translate command when unique keys are derived per transaction at the terminal to encrypt the PIN block. The default KSN descriptor is a three-byte numeric value that contains the sum of the individual digits of the KSN minus the counter. For example, a KSN that is 16 bytes in length would have a KSN descriptor whose individual digits when added together would equal 11 (i.e., 16, the actual length of the KSN – the transaction counter). The KSN descriptor for a 16-byte KSN would then be 803 (8 + 0 + 3 = 11). The default value is 644.

POS-DH-KSN-MAX-DIFFERENCE

This optional param indicates the largest allowable difference in the transaction counter between two transactions from the same terminal. The previous transaction counter is maintained in PTD device dependent data and the current transaction counter is carried in the Key Serial Number (KSN) field of the request message. Valid values are 0 through 4096, where 0 indicates that no difference checking is performed. The default value is 4096.

Database Configuration

BASE24-pos Terminal Data files (PTD)

The following PTD fields are relevant to the DUKPT portion of this enhancement.

Field Name: PIN Encryption Type
Screen: PTD Screen 7
DDL Name: PTDD1.[ptdd1-core].ENCR-PIN.PIN-ENCRYPT-TYP

If PINs are transmitted from the terminal using a DUKPT PIN block and encrypted under keys that are only in the clear within a security module, this field must be set to a value of 07. The DUKPT PIN block is translated into an ANSI standard PIN/PAN

PIN block before it is sent to the router/authorization module for authorization or routing.

Field Name: PIN Master Key
Screen: PTD Screen 7
DDL Name: PTDD1.[ptdd1-core].ENCR-PIN.ENCR-KEYS.M-KEY

If a terminal uses the DUKPT method of PIN encryption, this field must contain an intermediate PIN block encryption key encrypted by or for the specific security module being used in the system. When the PIN block in an incoming transaction is in DUKPT format, the device handler translates the PIN block to encryption under the intermediate PIN block encryption key.

If you are using the Transaction Security Services process, this field must be set to a key locator value used to retrieve the actual intermediate PIN block encryption key from the Transaction Security Services database.

Field Name: Derivation Key Group
Screen: PTD Screen 7
DDL Name: PTDS1.[ptds1-core].KEYD-GRP

Specifies the Derivation Key File (KEYD) group with which this terminal is associated for retrieval of the derivation key. This field is an optional field that allows multiple terminal records to be associated with the same KEYD record.

Derivation Key File (KEYD)

The Derivation Key File (KEYD) can contain 32-byte derivation keys or key locators. Derivation keys are used by the device handler to translate DUKPT PIN blocks received from the terminal into ANSI standard PIN/PAN PIN blocks. Key locators are non-encrypted values used by the Transaction Security Services process to locate hardware-encrypted key information maintained in the Transaction Security Services database.

The 32-byte derivation keys stored in the KEYD or TSS database must be encrypted under a double-length Master File Key (MFK) variant of 8 for Atalla security devices or a double-length Local Master Key (LMK) pair variant of 28–29 for Thales e-Security (Racal) security devices before they are manually entered into the file.

Depending on the value of the POS-DH-KEYD-READ-FROM-DISK LCONF parameter, the device handler retrieves a KEYD record from memory or disk for every DUKPT transaction. The KEYD is read according to the following data hierarchy:

Retailer ID	KEYD Group	Terminal ID
Exact	Exact	Exact
Exact	Exact	****************
Exact	****	****************
****************	****	****************

<u>Token File (TKN)</u>

The following BASE24-pos token is used by this enhancement:

- DUKPT Data token (token ID “CA”)

If this token is not to be logged to the PTLF or ILF, the appropriate token records must be configured in the Token file.

If this token is to be extracted with the PTLF or ILF, sent to the host, or passed through the BIC ISO interchange interface, the appropriate token records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

To utilize this enhancement, Hypercom terminals must run a version of firmware that supports triple DES master/session PIN encryption and DUKPT.

In any given terminal, master/session PIN encryption and DUKPT encryption are mutually exclusive.

Master/Session PIN encryption (single DES or triple DES) is enabled during the initialization parameters download by turning OFF bit 1 of the PIN Pad Type field in the Terminal Configuration Table.

DUKPT encryption is enabled during the initialization parameters download by turning ON bit 1 of the PIN Pad Type field in the Terminal Configuration Table.

When using DUKPT, a unique base derivation key must be manually entered or injected into each terminal.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

DUKPT: Since the KEYD derivation key field is a 32-byte key (double-length) the MFK for the applicable security devices must also be a double-length key.

File Conversions

Not applicable

Host/Co-Network Impacts

BASE24-pos users implementing the DUKPT portion of this enhancement incur host impacts. These impacts are as follows.

<u>External Interfaces</u>

There are no changes required to HISO for passing the KSN between BASE24-pos and host systems. However, the host can make use of the KSN to potentially detect fraud with a given PIN pad. The KSN should be sequential.

Super Extract Process

Standard PTLF Extract processing is not specifically impacted by this enhancement but host systems must be aware that the PTLF.AUTH.ORIGINATOR field may contain a value of “9” (DUKPT driven terminal using a Thales security module).

If the DUKPT Data Token is to be included in the PTLF and ILF extracts, BASE24-pos users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Operational Impacts

If the PIN pad is swapped out at the retailer site, the network operator must update the PTD with the correct KEYD group to insure the correct derivation key is used (if applicable).

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and take the appropriate action for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Performance Impacts

An additional I/O to the security device is required in order to translate the DUKPT PIN block to an ANSI standard PIN/PAN PIN block.

Additional I/O is required to either read the KEYD file into memory on initialization/warmboot or to read the KEYD from disk on each applicable transaction. This can require several disk reads to find the correct KEYD record.

Vendor Impacts

Not applicable

Reference Documents

Hypercom Message Specification "The System", Version 3.42, 08/2003.

Hypercom T7 and ICE System Point of Sale Software Initialization Parameters, Version 1.29, 10/2003.

PBF Refresh Early Notification

For further information, see **PBF Refresh Early Notification** in Section 4 where the impacts of this enhancement are described.

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Stored Value Dormancy and Escheatment

This section describes the implementation requirements associated with the Stored Value Dormancy and Escheatment enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Stored Value Dormancy and Escheatment enhancement.

The standard BASE24-pos authorization and extract modules have been modified to support this enhancement. Changes have also been made to the BASE24-pos stored value module. Section 4 has the Base changes associated with this enhancement.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60DDL.DDLFSVHF, DDLGSVHX, PSDDLM and PSDDLMM
- A new file (PS60DDL.DDLFSVDF)
- Inline changes to PS60EXT.PSOMFXG, PSOMFXM and PSOMFXS
- Inline changes to PS60AUTH.AUTHLIBS and RTAUG
- Updated event (log) message manual BA60LOGM.PSAUTH

All BASE24-pos Stored Value customers:

- Inline changes to PS60SV.CURSVHFS, DORMESCG, DORMESCM, DORMESCR, DORMESCS, PSCUSTTG, PSSVM and PSSVMM
- New SVDF requester/server and its associated files (PS60SV.RQSVDFM, RQSVDFS, SCRNSVDF, SVSVDFS and SVSVDFM)

A list of the Base files affected by this enhancement can be found in Section 4.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD08.SCNSVDE.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this enhancement. The LCONFPS file located on the BA60MISC subvolume has been modified to include these assigns.

POS-SV-CPF-EXTMEM-SWAPVOL

The fully qualified name of the BASE24-pos Stored Value CPF extended memory swap volume. This assign specifies where to temporarily store data when building the CPF stored value extended memory table. If this assign is not present or contains an invalid value, the NSK operating system assigns a swap volume location.

POS-SV-IDF-EXTMEM-SWAPVOL

The fully qualified name of the BASE24-pos Stored Value IDF extended memory swap volume. This assign specifies where to temporarily store data when building the IDF stored value extended memory table. If this assign is not present or contains an invalid value, the NSK operating system assigns a swap volume location.

POS-SVDF

This fully qualified name of the Stored Value Dormancy File. This assign is required for the Stored Value Dormancy and Escheatment enhancement.

POS-SV-SVDF-EXTMEM-SWAPVOL

The fully qualified name of the BASE24-pos Stored Value SVDF extended memory swap volume. This assign specifies where to temporarily store data when building the SVDF extended memory table. If this assign is not present or contains an invalid value, the NSK operating system assigns a swap volume location.

POS-SV-RETAIL-EXTMEM-SWAPVOL

The fully qualified name of the BASE24-pos Stored Value Retailer extended memory swap volume. This assign specifies where to temporarily store data when building the retailer stored value extended memory. If this assign is not present or contains an invalid value, the NSK operating system assigns a swap volume location.

POS-SV-DORMANCY-RPT

The fully qualified name of the BASE24-pos Stored Value Dormancy Report. It specifies the report's print location. If this assign is not present or contains an invalid value, a default value of $S.#DORMRPT is used.

POS-SV-ESCHEATMENT-RPT

The fully qualified name of the BASE24-pos Stored Value Escheatment Report. It specifies the report's print location. If this assign is not present or contains an invalid value, a default value of $S.#ESCHRPT is used.

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

BASE24-pos Stored Value Dormancy File (SVDF)

The BASE24-pos Stored Value Dormancy File (SVDF) is file containing the fees that can be accessed for an account that has been determined to be dormant.

Field Name: RETAILER ID
Screen: SVDF Screen 1
DDL Name: SVDF.PRIKEY.RETAILER-ID

The retailer ID. It must match the stored value retailer ID in the CPF.

Field Name: CARD TYPE
Screen: SVDF Screen 1
DDL Name: SVDF.PRIKEY.CRD-TYP

The stored value card type. It must match one of the entries in the stored value card type table in COBNAMES.

Field Name: ACTIVATION STATE
Screen: SVDF Screen 1
DDL Name: SVDF.PRIKEY.ST

The geographic location where the stored value card was activated.

Field Name: FLAT FEE
Screen: SVDF Screen 1
DDL Name: SVDF.FLAT-FEE

The flat fee to be charged as the dormancy fee. Either the flat fee or the percent fee is used as the fee.

Field Name: PERCENT FEE
Screen: SVDF Screen 1
DDL Name: SVDF.PERCENT-FEE

The percentage of the balance to be charged as the dormancy fee. Either the flat fee or the percent fee is used as the fee.

Field Name: DAYS DORMANT
Screen: SVDF Screen 1
DDL Name: SVDF.DAYS-DORMANT

The number of days an account can be dormant before escheatment.

Field Name: PERIOD
Screen: SVDF Screen 1
DDL Name: SVDF.PERIOD

The period of time before a dormancy fee can be assessed when an account has no activity.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

An entry was added to the SECTBL to support the new SVDF file. This file will appear on the POS Product Menu.

PATHCONF Changes

A new SVDF server must be defined in the PATHCONF file as SERVER-SVDF. This server can be added to the system online through the PATHCOM utility. After adding the server to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server.

CRT Access Security Changes

Access to the new SVDF screen must be added through the SEC requester to the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-pos users that use OMF Extract to accept and process information for the new SVDF.

User/Operational Impacts

Performance Impacts

The Dormancy Process will read the CPF (from memory) to determine which institutions support Stored Value card types. This information will be used to determine which IDF records are to be accessed and to retrieve the correct CAF and PBF file names. The CAF will be read through sequentially and corresponding PBF records will be accessed. Considerable disk I/O will be required for this processing.

Disk Impacts

The new SVDF will impact disk requirements. The total impact varies for each BASE24 user depending upon the number of SVDF records defined in the BASE24 system.

BASE24-pos users utilizing this enhancement must enter the information in the new Stored Value Dormancy File (SVDF).

Dormancy and/or escheatment transactions will be logged to the SVHF. This requires additional disk I/O to the SVHF, and therefore more records will be displayed on the SVHF perusal screens.

Operational Impacts

The Network Operator will be required to start the Dormancy/Escheatment Process from a TACL prompt or via a batch scheduling process.

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Stored Value Offline Card

This section describes the implementation requirements associated with Stored Value Offline Card enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Stored Value Offline Card enhancement.

This enhancement includes changes to the BASE24-pos ACI Standard POS device handler module to support offline stored value transaction processing.

Software Impacts

All BASE24-pos ACI Standard POS device handler customers:

- Inline changes to PS60SPDH.ASPDHG and ASPDHS

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Visa iCVV Support

For further information, see **Visa iCVV Support** in Section 4 where the impacts of this enhancement are described.

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Section 7

BASE24-teller

Section 7 discusses the changes and enhancements made to BASE24-teller in release 6.0 version 5, along with the resulting conversion and implementation requirements that existing BASE24-teller users must consider when upgrading from release 6.0 version 4.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-teller users, along with the associated conversion and implementation requirements that all BASE24-teller users must address.
- Optional enhancements describe the changes that affect only those BASE24-teller users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-teller users must address to implement the enhancements.

Section 7 must be used in conjunction with Section 4 to identify the changes that BASE24-teller users need to consider when upgrading to release 6.0 version 5.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-teller users must address in order to upgrade to release 6.0 version 5.

There are no general enhancements inherent to BASE24-teller.

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Optional Enhancements

This section identifies the BASE24-teller enhancements that can be enabled or disabled with release 6.0 version 5, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- PBF Refresh Early Notification

BASE24-teller users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements for future reference.

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PBF Refresh Early Notification

This enhancement has an additional user impact on the BASE24-teller product. While the other products utilize limits defined in the CAF to restrict the funds withdrawn from an account defined in the PBF, BASE24-teller does not. Thus, once the new PBF file becomes available to the BASE24-teller Authorization process, all of the funds in the account are available for withdrawal. These funds can be restricted based on the customer group defined in the teller segment of the IDF.

For further information, see **PBF Refresh Early Notification** in Section 4 where the impacts of this enhancement are described.

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Section 8

BASE24 Kernel/IEP

Section 8 discusses the changes and enhancements made to BASE24 Kernel/IEP in release 6.0 version 5, along with the resulting conversion and implementation requirements that existing BASE24 Kernel/IEP users must consider when upgrading from release 6.0 version 4.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 Kernel/IEP users, along with the associated conversion and implementation requirements that all BASE24 Kernel/IEP users must address.
- Optional enhancements describe the changes that affect only those BASE24 Kernel/IEP users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 8 must be used in conjunction with Section 4 to identify the changes that BASE24 Kernel/IEP users need to consider when upgrading to release 6.0 version 5.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 Kernel/IEP users must address in order to upgrade to release 6.0 version 5.

There are no general enhancements inherent to BASE24 Kernel/IEP.

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Optional Enhancements

This section identifies the BASE24 Kernel/IEP enhancements that can be enabled or disabled with release 6.0 version 5, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements inherent to BASE24 Kernel/IEP.

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Section 9
BASE24-telebanking

Section 9 discusses the changes and enhancements made to BASE24-telebanking in release 6.0 version 5, along with the resulting conversion and implementation requirements that existing BASE24-telebanking users must consider when upgrading from release 6.0 version 4.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-telebanking users, along with the associated conversion and implementation requirements that all BASE24-telebanking users must address.
- Optional enhancements describe the changes that affect only those BASE24-telebanking users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-telebanking users must address to implement the enhancements.

Section 9 must be used in conjunction with Sections 4 and 8 to identify the changes that BASE24-telebanking users need to consider when upgrading to release 6.0 version 5. Enhanced Telebanking users must also use Section 10, BASE24-billpay, to identify all changes, as Enhanced Telebanking shares modules with BASE24-billpay.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-telebanking users must address in order to upgrade to release 6.0 version 5.

There are no general enhancements inherent to BASE24-telebanking.

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Optional Enhancements

This section identifies the BASE24-telebanking enhancements that can be enabled or disabled with release 6.0 version 5, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- PBF Refresh Early Notification

BASE24-telebanking users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements for future reference.

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PBF Refresh Early Notification

For further information, see **PBF Refresh Early Notification** in Section 4 where the impacts of this enhancement are described.

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Section 10

BASE24-billpay

Section 10 discusses the changes and enhancements made to BASE24-billpay in release 6.0 version 5, along with the resulting conversion and implementation requirements that existing BASE24-billpay users must consider when upgrading from release 6.0 version 4.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-billpay users, along with the associated conversion and implementation requirements that all BASE24-billpay users must address.
- Optional enhancements describe the changes that affect only those BASE24-billpay users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-billpay users must address to implement the enhancements.

Section 10 must be used in conjunction with Sections 4, 8 and 9 to identify the changes that BASE24-billpay users need to consider when upgrading to release 6.0 version 5.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-billpay users must address in order to upgrade to release 6.0 version 5.

There are no general enhancements inherent to BASE24-billpay.

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Optional Enhancements

This section identifies the BASE24-billpay enhancements that can be enabled or disabled with release 6.0 version 5, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements inherent to BASE24-billpay.

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Section 11

BASE24-customer service

Section 11 discusses the changes and enhancements made to BASE24-customer service in release 6.0 version 5, along with the resulting conversion and implementation requirements that existing BASE24-customer service users must consider when upgrading from release 6.0 version 4.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-customer service users, along with the associated conversion and implementation requirements that all BASE24-customer service users must address.
- Optional enhancements describe the changes that affect only those BASE24-customer service users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-customer service users must address to implement the enhancements.

Section 11 must be used in conjunction with Sections 4 and 8 to identify the changes that BASE24-customer service users need to consider when upgrading to release 6.0 version 5.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-customer service users must address in order to upgrade to release 6.0 version 5.

There are no general enhancements inherent to BASE24-customer service.

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Optional Enhancements

This section identifies the BASE24-customer service enhancements that can be enabled or disabled with release 6.0 version 5, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no general enhancements inherent to BASE24-customer service.

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Section 12

BASE24 Interchange Interfaces

Section 12 discusses the changes and enhancements made to BASE24 Interchange Interfaces in release 6.0 version 5, along with the resulting conversion and implementation requirements that existing BASE24 Interchange Interface users must consider when upgrading from release 6.0 version 4.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 Interchange Interface users, along with the associated conversion and implementation requirements that all BASE24 Interchange Interface users must address.
- Optional enhancements describe the changes that affect only those BASE24 Interchange Interface users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 Interchange Interface users must address to implement the enhancements.

Section 12 must be used in conjunction with Section 4 to identify the changes that BASE24 Interchange Interface users need to consider when upgrading to release 6.0 version 5.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 Interchange Interface users must address in order to upgrade to release 6.0 version 5.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

Hooks for the new Boston Communications Group Interface (BCGI) to support BASE24-atm Mobile Top-Up.

Changes for a new reversal code in the BASE24-atm BIC ISO Host Interface and the BASE24-atm ISO Host Interface to support BASE24-atm Mobile Top-Up.

Software fix to HISWUTLS procedure hiswilf^add^tkn to restore the original extended memory segment before returning.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS and FLGSDOC
- Changes to BA60MAKE.ALLMACS to include new defines for: system_credecs, system_taldecs, zspidef_zspital, zspidef_zemstal
- Inline changes BA60SRC.BASEFM and BASEMM
- Software fix to BA60SRC.HISWUTLS

All BASE24 Interchange Interface customers:

- Inline changes to SW60SRC.SWIFM, SWIM and SWIMM

All BASE24-atm BIC ISO Host Interface customers:

- Inline changes to SW60ATBI.ATBICIS

All BASE24-atm ISO Host Interface customers:

- Inline changes to AT60HISO.ATHISOS

Only BASE24-atm customers using the new Mobile Top-Up functionality with the new Boston Communications Group Interface:

- New subvolume, SW60BCGI, containing files for the BCGI Interface and its documentation
- New subvolume, BA60XPT, containg the XML parser files

A complete list of the files affected by the BASE24-atm Mobile Top-Up enhancement can be found in the SCN file BA60UD08.SCNMTOP.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flag to true when implementing the BASE24-atm Mobile Top-Up functionality using the new Boston Communications Group Interface (BCGI).

- bcgi_on

Set the following flag to true for the XML Parser. This flag must be set to true if the BCGI Interface is supported.

- xml_parser_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing the changes to the BASE24-atm BIC ISO Host Interface and the BASE24-atm ISO Host Interface.

<u>External Interfaces</u>

Host code changes may be required to support a new reversal reason code of “40” – Split Routing Enabled; Secondary Service Not Approved. This value is supported by the BASE24-atm ISO Host Interface process and the BASE24-atm BIC ISO Interface process.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable.

Optional Enhancements

This section identifies the BASE24 Interchange Interface enhancements that can be enabled or disabled with release 6.0 version 5, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Absent ICC Application PAN Sequence Number
- American Express Card Security Code

BASE24 Interchange Interface users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements for future reference.

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Absent ICC Application PAN Sequence Number

For further information, see **Absent ICC Application PAN Sequence Number** in Section 4 where the impacts of this enhancement are described.

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American Express Card Security Code

This section describes the implementation requirements associated with the American Express Card Security Code enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the American Express Card Security Code enhancement.

Logic has been added to the American Express Global Network Interchange Interface to handle a transaction response from the Switch that was denied because of an invalid manually entered CSC value.

For further information regarding this enhancement see **American Express Card Security Code** in Section 4.

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Section 13

BASE24 Value-Added Products

Section 13 discusses the changes and enhancements made to the BASE24 value-added products in release 6.0 version 5, along with the resulting conversion and implementation requirements that existing users of these products must consider when upgrading from release 6.0 version 4.

The value-added products discussed in this section include the following:

- BASE24-InfoBase
- BASE24 Retail Products
- BASE24-card
- ACI Host Link
- ACI Plastic Card Control System
- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Settlement Manager
- ACI Payments Manager
- ACI Smart Chip Manager

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 value-added product users, along with the associated conversion and implementation requirements that all BASE24 value-added product users must address.
- Optional enhancements describe the changes that affect only those BASE24 value-added product users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 value-added product users must address to implement the enhancements.

Section 13 must be used in conjunction with Sections 4 through 12 to identify the changes that BASE24 value-added product users need to consider when upgrading to release 6.0 version 5.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 value-added product users must address in order to upgrade to release 6.0 version 5.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the general enhancements described here.

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BASE24-InfoBase

This section describes the enhancements that impact the BASE24-InfoBase value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting BASE24-InfoBase users.

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BASE24 Retail Products

This section describes the enhancements that impact the value-added BASE24 Retail Products and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the users of the BASE24 Retail Products.

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BASE24-card

This section describes the enhancements that impact the BASE24-card value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting BASE24-card users.

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ACI Host Link

This section describes the enhancements that impact the ACI Host Link value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Host Link users.

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ACI Plastic Card Control System

This section describes the enhancements that impact the ACI Plastic Card Control System value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Plastic Card Control System users.

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ACI Claims Manager

This section describes the enhancements that impact the ACI Claims Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Claims Manager users.

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ACI Proactive Risk Manager

This section describes the enhancements that impact the ACI Proactive Risk Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Proactive Risk Manager users.

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ACI Settlement Manager

This section describes the enhancements that impact the ACI Settlement Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Settlement Manager users.

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ACI Payments Manager

This section describes the enhancements that impact the ACI Payments Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Payments Manager users.

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ACI Smart Chip Manager

This section describes the enhancements that impact the ACI Smart Chip Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Smart Chip Manager users.

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Optional Enhancements

This section identifies the BASE24 enhancements that can be enabled or disabled with release 6.0 version 5 that have an impact on the following BASE24 value-added products, along with the changes, conversion requirements, and implementation requirements associated with each.

- BASE24-InfoBase
- BASE24 Retail Products
- BASE24-card
- ACI Host Link
- ACI Plastic Card Control System
- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Payments Manager
- ACI Settlement Manager
- ACI Smart Chip Manager

BASE24 value-added product users can choose whether or not to implement the enhancements described in this section. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 5 changes, conversion requirements, and implementation requirements for future reference.

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BASE24-InfoBase

This section describes the enhancements that impact the BASE24-InfoBase value-added product.

There are no optional enhancements inherent to BASE24-InfoBase.

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BASE24 Retail Products

This section describes the enhancements that impact the value-added BASE24 Retail Products.

There are no optional enhancements inherent to BASE24 Retail Products.

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BASE24-card

This section describes the enhancements that impact the BASE24-card value-added product.

There are no optional enhancements inherent to BASE24-card.

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ACI Host Link

This section describes the enhancements that impact the ACI Host Link value-added product.

There are no optional enhancements inherent to ACI Host Link.

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ACI Plastic Card Control System

This section describes the enhancements that impact the ACI Plastic Card Control System value-added product.

There are no optional enhancements inherent to ACI Plastic Card Control System.

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ACI Claims Manager

This section describes the enhancements that impact the ACI Claims Manager value-added product.

There are no optional enhancements inherent to ACI Claims Manager.

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ACI Proactive Risk Manager

This section describes the enhancements that impact the ACI Proactive Risk Manager value-added product.

There are no optional enhancements inherent to ACI Proactive Risk Manager.

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ACI Payments Manager

This section describes the enhancements that impact the ACI Payments Manager value-added product.

There are no optional enhancements inherent to ACI Payments Manager.

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ACI Settlement Manager

This section describes the enhancements that impact the ACI Settlement Manager value-added product.

There are no optional enhancements inherent to ACI Settlement Manager.

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ACI Smart Chip Manager

This section describes the enhancements that impact the ACI Smart Chip Manager value-added product.

There are no optional enhancements inherent to the ACI Smart Chip Manager.

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Appendix A

Logical Network Configuration File Changes

Appendix A contains the Logical Network Configuration File (LCONF) changes.

This information includes assigns and params that have been added, deleted or modified with release 6.0 version 5 along with their associated page numbers. Refer to the specified page numbers for explanations of how these assigns and params can or must be set.

Refer to the ***BASE24 Logical Network Configuration File (LCONF) Manual*** for instructions on adding them to the LCONF.

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ASSIGNS

The following assigns were added or modified for release 6.0 version 5. Refer to the specified page numbers for explanations of how these assigns can or must be set.

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PARAMS

The following params were added or modified for release 6.0 version 5. Refer to the specified page numbers for explanations of how these params can or must be set.

Enhanced ATM Status Reporting:

Enhanced Second Card Data Processing:

Hypercom Triple DES Master/Session and DUKPT:

Inhibit UI Display of POFST/PVV:

Mobile Top-Up Integration

PBF Refresh Early Notification:

Round Robin Access to TSS:

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Appendix B

Host/CO-Network Impacts

Appendix B contains a list of release 6.0 version 5 enhancements that have host/co-network impacts, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

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Enhancements

The following enhancements have host/co-network impacts. Refer to the specified page numbers for more information about these enhancements.

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Appendix C

CUSTMACS File Changes

Appendix C contains a list of release 6.0 version 5 enhancements that have CUSTMACS changes, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

BASE24 users should review these changes to determine how the CUSTMACS file should be set for their BASE24 environment before running MAKE to compile their system. More information on all MAKE flags used within BASE24 can be found in the FLGSDOC file located on the BA60MAKE subvolume delivered with BASE24 Release 6.0 Version 5 code.

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Enhancements

The following enhancements have CUSTMACS file changes:

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Appendix D

Re-Sequenced Files and New Files

Appendix D consists of the following tables:

- A table illustrating the new files for the optional iCVV Support and Offline PIN Change Notification enhancements.
- A table illustrating the new files for the optional BASE24-atm Mobile Top-Up enhancement.
- A table illustrating the new files for the optional BASE24-atm Diebold EMV enhancement.
- A table illustrating the new files for the optional BASE24-atm NCR NDC+ CAM II (EMV) Upgrade enhancement.
- A table illustrating the new files for the optional BASE24-atm Shared (Wincor) NDC+ EMV enhancement.
- A table illustrating the new files for the optional BASE24-atm Triton EMV enhancement.
- A table illustrating the new files for the optional BASE24-pos Stored Value Dormancy and Escheatment enhancement.

No files were re-sequenced in release 6.0 version 5.

New Files for the Optional iCVV Support and Offline PIN Change Notification Enhancements Table

ICVV Support and Offline PIN Change Notification New File Location and File Name	**File Language Type**
ba60uc08.aaread08	TGAL
ba60uc08.bauc08fm	MAKE
ba60uc08.bauc08m	MAKE
ba60uc08.bauc08mm	MAKE
ba60uc08.cnvcpf	Generated during a compile
ba60uc08.cnvcpfe	Generated during a compile
ba60uc08.cnvcpfm	MAKE
ba60uc08.cnvcpfo	Generated during a compile
ba60uc08.cnvcpfr	TACL
ba60uc08.cnvcpfs	Generated during a compile

New Files for the Optional BASE24-atm Mobile Top-Up Enhancement Table

Mobile Top-Up New File Location and File Name	**File Language Type**
BASE24-atm Transaction Context Manager	
at60tcm.atcm	Generated during a compile
at60tcm.atcmd	TAL
at60tcm.tcme	Generated during a compile
at60tcm.atcmfm	MAKE
at60tcm.atcmg	TAL
at60tcm.atcmloge	Generated during a compile
at60tcm.atcmlogo	Generated during a compile
at60tcm.atcmlogs	TAL
at60tcm.atcmm	MAKE
at60tcm.atcmmm	MAKE
at60tcm.atcmo	Generated during a compile
at60tcm.atcms	TAL
Diebold 10XX/478X Device Handler Mobile Top-Up Extension	
at60t100.att100fm	MAKE
at60t100.att100mm	MAKE
at60t100.c100mtue	Generated during a compile
at60t100.c100mtug	TAL
at60t100.c100mtum	MAKE
At60t100.c100mtuo	Generated during a compile
at60t100.c100mtus	TAL
at60t100.t100nams	TAL
NCR 56XX Device Handler Mobile Top-Up Extension	
at60tn50.attn50fm	MAKE
at60tn50.attn50mm	MAKE
at60tn50.n50mtue	Generated during a compile
at60tn50.n50mtug	TAL
at60tn50.n50mtum	MAKE
At60tn50.n50mtuo	Generated during a compile
at60tn50.n50mtus	TAL
at60tn50.tn50nams	TAL

Mobile Top-Up New File Location and File Name	**File Language Type**
BASE24-atm Transaction Context Manager Mobile Top-Up Extension	
at60ttcm.atcmmtue	Generated during a compile
at60ttcm.atcmmtug	TAL
at60ttcm.atcmmtuo	Generated during a compile
at60ttcm.atcmmtus	TAL
at60ttcm.atttcmfm	MAKE
at60ttcm.atttcmm	MAKE
at60ttcm.atttcmmm	MAKE
Base AFT	
ba60aft.rqstrfss	ACIAID
ba60aft.rqstrfxm	MAKE
ba60aft.rqstrfxo	Generated during a compile
ba60aft.rqstrfxs	COBOL
ba60aft.svstrfte	Generated during a compile
ba60aft.svstrftg	TAL
ba60aft.svstrftm	MAKE
ba60aft.svstrfto	Generated during a compile
ba60aft.svstrfts	TAL
Base DDLS	
ba60ddl.ddlfmof	DDL
ba60ddl.ddlfstrf	DDL
Extended Memory Table Build Utility	
ba60emtb.stremtte	Generated during a compile
ba60emtb.stremttg	TAL
ba60emtb.stremttm	MAKE
Ba60emtb.stremtto	Generated during a compile
ba60emtb.stremtts	TAL
Base Log Message Manuals	
ba60logm.attcm	DOC
Mobile Top-Up Infrastructure	

Mobile Top-Up **New File Location and File Name**	**File Language Type**
ba60tmtu.batmtufm	MAKE
ba60tmtu.batmtug	TAL
ba60tmtu.batmtum	MAKE
ba60tmtu.batmtumm	MAKE
ba60tmtu.moblib	Generated during a compile
ba60tmtu.moflibe	Generated during a compile
ba60tmtu.moflibg	TAL
ba60tmtu.moflibm	MAKE
ba60tmtu.moflibs	TAL
ba60tmtu.rqmofm	MAKE
ba60tmtu.rqmofo	Generated during a compile
ba60tmtu.rqmofs	COBOL
ba60tmtu.scrnmof	ACIAID
ba60tmtu.svmof	Generated during a compile
ba60tmtu.svmofm	MAKE
ba60tmtu.svmofs	COBOL
XML Parser	
ba60xpt.dcdlib	Object
ba60xpt.expat	Object
ba60xpt.xptlib	Object
ba60xpt.xptmm	MAKE
ba60xpt.xpttalh	TAL
Boston Communications Group Interface	
sw60bcgi.bcgi	Generated duing a compile
sw60bcgi.bcgiappe	Generated during a compile
sw60bcgi.bcgiappg	TAL
sw60bcgi.bcgiappm	MAKE
sw60bcgi.bcgiappo	Generated during a compile
sw60bcgi.bcgiapps	TAL
sw60bcgi.bcgic	Generated during a compile
sw60bcgi.bcgicale	Generated during a compile
sw60bcgi.bcgicalg	TAL
sw60bcgi.bcgicalm	MAKE
sw60bcgi.bcgicalo	Generated during a compile
sw60bcgi.bcgicals	TAL

Mobile Top-Up New File Location and File Name	File Language Type
sw60bcgi.bcgicnve	Generated during a compile
sw60bcgi.bcgicnvg	TAL
sw60bcgi.bcgicnvm	MAKE
sw60bcgi.bcgicnvo	Generated during a compile
sw60bcgi.bcgicnvs	TAL
sw60bcgi.bcgicob	Generated during a compile
sw60bcgi.bcgidcd	Document Content Description (DCD) for XML
sw60bcgi.bcgiddlm	MAKE
sw60bcgi.bcgiddls	DDL
sw60bcgi.bcgie	Generated during a compile
sw60bcgi.bcgifm	MAKE
sw60bcgi.bcgifmte	Generated during a compile
sw60bcgi.bcgifmtg	TAL
sw60bcgi.bcgifmtm	MAKE
sw60bcgi.bcgifmto	Generated during a compile
sw60bcgi.bcgifmts	TAL
sw60bcgi.bcgig	TAL
sw60bcgi.bcgim	MAKE
sw60bcgi.bcgimm	MAKE
sw60bcgi.bcgio	Generated during a compile
sw60bcgi.bcgirsps	TAL
sw60bcgi.bcgis	TAL
sw60bcgi.bcgital	Generated during a compile
sw60bcgi.bcgitbls	TAL
sw60bcgi.bcgith	TAL
sw60bcgi.bisoc	Generated during a compile
sw60bcgi.bisoddlm	MAKE
sw60bcgi.bisoddls	DDL
sw60bcgi.bisotal	Generated during a compile
sw60bcgi.emsddlm	MAKE
sw60bcgi.emsddls	DDL
sw60bcgi.emstal	Generated during a compile
sw60bcgi.gxswchd	TAL
sw60bcgi.gxswche	Generated during a compile
sw60bcgi.gxswchg	TAL
sw60bcgi.gxswchm	MAKE

Mobile Top-Up **New File Location and File Name**	**File Language Type**
sw60bcgi.gxswcho	Generated during a compile
sw60bcgi.gxswchs	TAL
sw60bcgi.gxswcob	Generated during a compile
sw60bcgi.gxswddlm	MAKE
sw60bcgi.gxswddls	DDL
sw60bcgi.gxswfup	Generated during a compile
sw60bcgi.gxswseme	Generated during a compile
sw60bcgi.gxswsemg	TAL
sw60bcgi.gxswsemm	MAKE
sw60bcgi.gxswsemo	Generated during a compile
sw60bcgi.gxswsems	TAL
sw60bcgi.gxswtal	Generated during a compile
sw60bcgi.gxtsappd	TAL
sw60bcgi.gxtsappe	Generated during a compile
sw60bcgi.gxtsappg	TAL
sw60bcgi.gxtsappm	MAKE
sw60bcgi.gxtsappo	Generated during a compile
sw60bcgi.gxtsapps	TAL
sw60bcgi.gxtsemse	Generated during a compile
sw60bcgi.gxtsemsg	TAL
sw60bcgi.gxtsemsm	MAKE
sw60bcgi.gxtsemso	Generated during a compile
sw60bcgi.gxtsemss	TAL
sw60bcgi.gxtserrg	TAL
sw60bcgi.gxtsfile	Generated during a compile
sw60bcgi.gxtsfilg	TAL
sw60bcgi.gxtsfilm	MAKE
sw60bcgi.gxtsfilo	Generated during a compile
sw60bcgi.gxtsfils	TAL
sw60bcgi.gxtsmsge	Generated during a compile
sw60bcgi.gxtsmsgg	TAL
sw60bcgi.gxtsmsgm	MAKE
sw60bcgi.gxtsmsgo	Generated during a compile
sw60bcgi.gxtsmsgs	TAL
sw60bcgi.gxtsstae	Generated during a compile
swt60bcgi.gxtsstag	TAL
sw60bcgi.gxtsstam	MAKE

Mobile Top-Up New File Location and File Name	File Language Type
sw60bcgi.gxtsstao	Generated during a compile
sw60bcgi.gxtsstas	TAL
sw60bcgi.gxtstrce	Generated during a compile
swt60bcgi.gxtstrcg	TAL
sw60bcgi.gxtstrcm	MAKE
sw60bcgi.gxtstrco	Generated during a compile
sw60bcgi.gxtstrcs	TAL
sw60bcgi.gxtsudte	Generated during a compile
swt60bcgi.gxtsudtg	TAL
sw60bcgi.gxtsudtm	MAKE
sw60bcgi.gxtsudto	Generated during a compile
sw60bcgi.gxtsudts	TAL
sw60bcgi.gxtsxmle	Generated during a compile
swt60bcgi.gxtsxmlg	TAL
sw60bcgi.gxtsxmlm	MAKE
sw60bcgi.gxtsxmlo	Generated during a compile
sw60bcgi.gxtsxmls	TAL
sw60bcgi.rqbcgim	MAKE
sw60bcgi.rqbcgio	Generated during a compile
sw60bcgi.rqbcgis	COBOL
sw60bcgi.scrnbcgi	ACIAID
sw60bcgi.swbcgim	MAKE

New Files for the Optional BASE24-atm Diebold EMV Enhancement Table

Diebold EMV **New File Location and File Name**	**File Language Type**
at60i100.ati100fm	MAKE
at60i100.ati100mm	MAKE
at60i100.c100emve	Generated during a compile
at60i100.c100emvg	TAL
at60i100.c100emvm	MAKE
at60i100.c100emvo	Generated during a compile
at60i100.c100emvs	TAL

New Files for the Optional BASE24-atm NCR NDC+ CAM II (EMV) Upgrade Table

NCR NDC+ CAM II (EMV) Upgrade New File Location and File Name	**File Language Type**
at60in50.rqemlanm	MAKE
at60in50.rqemlans	TAL
at60in50.scnemlan	Screen File
ba60src.coblang	COBOL

New Files for the Optional BASE24-atm Shared (Wincor) NDC+ EMV Enhancement Table

Wincor NDC+ EMV New File Location and File Name	**File Language Type**
at60iwn5.atiwn5fm	MAKE
at60iwn5.atiwn5mm	MAKE
at60iwn5.n50wemve	Generated during a compile
at60iwn5.n50wemvg	TAL
at60iwn5.n50wemvm	MAKE
at60iwn5.n50wemvo	Generated during a compile
at60iwn5.n50wemvs	TAL

New Files for the Optional BASE24-atm Triton EMV Enhancement Table

Triton EMV **New File Location and File Name**	**File Language Type**
at60itrt.atitrtfm	MAKE
at60itrt.atitrtmm	MAKE
at60itrt.tritemve	Generated during a compile
at60itrt.tritemvg	TAL
at60itrt.tritemvm	MAKE
at60itrt.tritemvo	Generated during a compile
at60itrt.tritemvs	TAL

New Files for the Optional BASE24-pos Stored Value Dormancy and Escheatment Enhancement Table

Stored Value Dormancy and Escheatment New File Location and File Name	**File Language Type**
ps60ddl.ddlfsvdf	DDL
ps60sv.dormesc	Generated during a compile
ps60sv.dormesce	Generated during a compile
ps60sv.dormescg	TAL
ps60sv.dormescm	MAKE
ps60sv.dormesco	Generated during a compile
ps60sv.dormescr	TCAL
ps60sv.dormescs	TAL
ps60sv.pscusttg	TAL
ps60sv.rqsvdfm	MAKE
ps60sv.rqsvdfo	Generated during a compile
ps60sv.rqsvdfs	Scobol
ps60sv.svsvdf	Generated during a compile
ps60sv.svsvdfm	MAKE
ps60sv.svsvdfs	Cobol

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Appendix E

Moved, Renamed, Deleted Files

Appendix E contains a list of files that were Moved, Renamed, or Deleted in BASE24 Release 6.0 Version 5.

No files were moved, renamed or deleted in release 6.0 version 5.

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Appendix F

Regional Specific Information

Appendix F contains the information that is required to implement the changes for release 6.0 version 5 that are specific to each region.

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